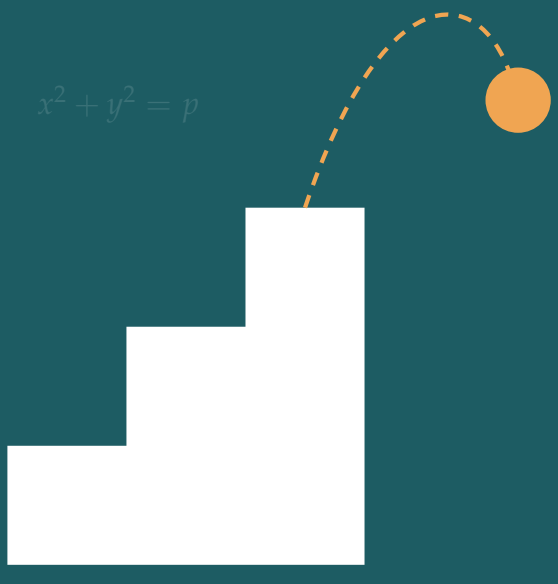
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

Specimen Papers Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

Specimen Papers Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

Specimen Papers Worked Solutions

Worked solutions to all 48 questions of the STEP specimen papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP I Specimen paper, Question 1

Pure • Discriminants and Partial Fractions

STEP I Specimen paper, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s1-q1

The key idea

In part (i) the equation is quadratic in y for each fixed x, and a real y exists only when the discriminant is non-negative; that condition is exactly a restriction on x. Swapping the roles, the equation is also quadratic in x, which gives the restriction on y. Part (ii) is a standard partial-fractions question: split the fraction into pieces whose binomial expansions we know, then add the coefficients of x^n.

Solution

Part (i)

Regard the equation as a quadratic in y with coefficients depending on x:

$$y^2 + 16x\,y + (4x^2 + 24x) = 0.$$

For a given real x, there is a real y satisfying this if and only if the discriminant is non-negative:

$$\begin{aligned}(16x)^2 - 4(4x^2 + 24x) &= 256x^2 - 16x^2 - 96x \\ &= 240x^2 - 96x = 48x(5x - 2) \geq 0.\end{aligned}$$

The product $x(5x-2)$ is non-negative exactly when both factors have the same sign (or one is zero), that is, when $x \leq 0$ or $x \geq \frac{2}{5}$. So any real solution has

$$x \leq 0 \quad \text{or} \quad x \geq \tfrac{2}{5},$$

as required.

Now treat the equation as a quadratic in x:

$$4x^2 + (16y + 24)x + y^2 = 0.$$

A real x exists if and only if

$$\begin{aligned}(16y+24)^2 - 16y^2 &= 256y^2 + 768y + 576 - 16y^2 \\ &= 240y^2 + 768y + 576 \\ &= 48(5y^2 + 16y + 12) = 48(5y+6)(y+2) \geq 0.\end{aligned}$$

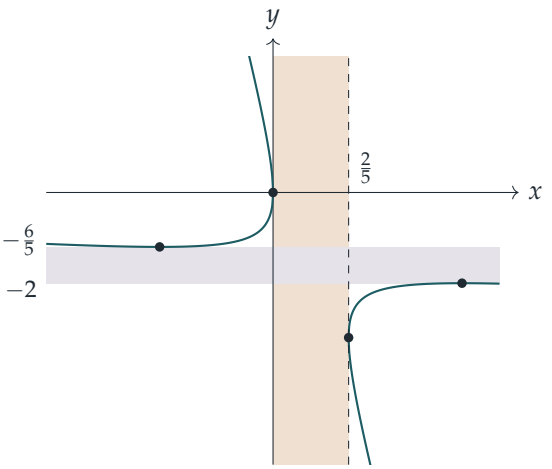
The roots of $(5y+6)(y+2)$ are $y = -2$ and $y = -\frac{6}{5}$, and the quadratic is negative only between them. Hence

$$y \leq -2 \quad \text{or} \quad y \geq -\tfrac{6}{5}.$$

Ask yourself Are these restrictions the best possible, or could the true ranges be smaller?

They are exact. The discriminant argument is an "if and only if": every x with $x \leq 0$ or $x \geq \frac{2}{5}$ gives at least one real y, and likewise for y. The boundary values are attained too, by the double roots: $(x, y) = (0, 0)$ and $(\frac{2}{5}, -\frac{16}{5})$ for the x-bounds, and $(1, -2)$ and $(-\frac{3}{5}, -\frac{6}{5})$ for the y-bounds. You can check each satisfies the original equation.

Geometrically the curve is a hyperbola (since $16^2 > 4 \cdot 4 \cdot 1$), and the two excluded strips are the gaps between its branches.



Part (ii)

Write

$$\frac{9}{(2-x)^2(1+x)} = \frac{A}{1+x} + \frac{B}{2-x} + \frac{C}{(2-x)^2},$$

so that $9 = A(2-x)^2 + B(2-x)(1+x) + C(1+x)$. Putting $x = -1$ gives $9 = 9A$, so $A = 1$; putting $x = 2$ gives $9 = 3C$, so $C = 3$; putting $x = 0$ gives $9 = 4A + 2B + C = 7 + 2B$, so $B = 1$. Thus

$$\frac{9}{(2-x)^2(1+x)} = \frac{1}{1+x} + \frac{1}{2-x} + \frac{3}{(2-x)^2}.$$

Now expand each piece, taking out the factor 2 so that the binomial series is in powers of $\frac{x}{2}$:

$$\frac{1}{1+x} = \sum_{n\geq 0}(-1)^n x^n, \qquad |x|<1,$$

$$\frac{1}{2-x} = \frac{1}{2}\Big(1-\frac{x}{2}\Big)^{-1} = \sum_{n\geq 0}\frac{x^n}{2^{n+1}}, \qquad |x|<2,$$

$$\frac{3}{(2-x)^2} = \frac{3}{4}\Big(1-\frac{x}{2}\Big)^{-2} = \frac{3}{4}\sum_{n\geq 0}(n+1)\frac{x^n}{2^n} = \sum_{n\geq 0}\frac{3(n+1)}{2^{n+2}}x^n, \qquad |x|<2,$$

using $(1-u)^{-2} = \sum_{n\geq 0}(n+1)u^n$ for $|u|<1$. Adding, the coefficient of x^n is

$$(-1)^n + \frac{1}{2^{n+1}} + \frac{3(n+1)}{2^{n+2}} = (-1)^n + \frac{2+3n+3}{2^{n+2}} = (-1)^n + \frac{3n+5}{2^{n+2}}.$$

All three series converge together exactly when $|x|<1$ (the first needs $|x|<1$, the others only $|x|<2$). For $1\leq|x|<2$ the last two still converge but the first does not, so their sum cannot converge either. So the expansion is valid for

$$-1<x<1.$$

Checks

In part (i), the point $(1,-2)$ gives $4-32+4+24=0$, and $(-\frac{3}{5},-\frac{6}{5})$ gives $\frac{36}{25}+\frac{288}{25}+\frac{36}{25}-\frac{360}{25}=0$, confirming that both y-bounds are attained. In part (ii), $n=0$ gives $1+\frac{5}{4}=\frac{9}{4}$, which matches $\frac{9}{4\cdot 1}$ at $x=0$; and $n=1$ gives $-1+\frac{8}{8}=0$, which is right because the denominator $(2-x)^2(1+x)=4-3x^2+x^3$ has no x term. Numerically, summing the series at $x=0.9$ gives $3.914745\ldots$, agreeing with $9/(1.1^2\times 1.9)$.

Common slips

- Getting the inequality the wrong way round and answering $-2\leq y\leq -\frac{6}{5}$. The discriminant is negative *between* its roots, so that interval is the one excluded (and $(0,0)$ lies on the curve with $y=0$ outside it).
- Expanding $(2-x)^{-1}$ as if it were $(1-x)^{-1}$, forgetting to take out the factor $\frac{1}{2}$ (or $\frac{1}{4}$ for the squared term).
- Giving the range of validity as $|x|<2$: the expansion is only valid where every piece converges, so $|x|<1$.

Beyond STEP

The curve in part (i) is a conic, and whether it is an ellipse, parabola or hyperbola depends on the sign of b^2-4ac for its quadratic terms $ax^2+bxy+cy^2$. At university

this is seen through the symmetric matrix of the quadratic form and its eigenvalues. Part (ii) touches the idea of a radius of convergence: the expansion is valid out to the nearest point where the function blows up, here $x = -1$. Strang's *Introduction to Linear Algebra* covers quadratic forms, and Abbott's *Understanding Analysis* covers power series and their convergence.

WORKED SOLUTION

STEP I Specimen paper, Question 2

Pure • Trigonometry and Optimisation

STEP I Specimen paper, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s1-q2

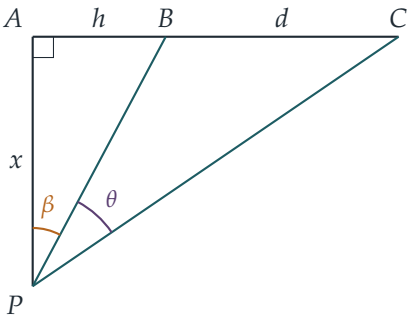
The key idea

The angle θ is the *difference* of two angles at P, each of which sits in a right-angled triangle with the right angle at A. Their tangents can be read straight off the figure, so the subtraction formula for tan does the work. For the maximum, write $\tan\theta$ so that x appears in only one place, as $x + (\text{constant})/x$, and the AM–GM inequality finishes the job without any calculus.

Solution

Finding $\tan\theta$

Let $\alpha = \angle APC$ and $\beta = \angle APB$. Since B lies between A and C, we have $\theta = \alpha - \beta$.



In the right-angled triangles PAC and PAB (right angle at A),

$$\tan\alpha = \frac{AC}{AP} = \frac{h+d}{x}, \qquad \tan\beta = \frac{AB}{AP} = \frac{h}{x}.$$

By the subtraction formula,

$$\tan\theta = \tan(\alpha - \beta) = \frac{\tan\alpha - \tan\beta}{1 + \tan\alpha\tan\beta} = \frac{\dfrac{h+d}{x} - \dfrac{h}{x}}{1 + \dfrac{h(h+d)}{x^2}}.$$

Multiplying top and bottom by x^2,

$$\boxed{\tan\theta = \frac{dx}{x^2 + h(h+d)}}$$

(The denominator $1 + \tan\alpha\tan\beta$ is positive, so the formula never breaks down.)

Maximising θ

Ask yourself Is maximising $\tan\theta$ the same as maximising θ?

Yes, here. Both α and β lie strictly between 0 and $\frac{\pi}{2}$, and $\alpha > \beta$, so $0 < \theta < \frac{\pi}{2}$. On that interval tan is strictly increasing, so the largest θ comes from the largest $\tan\theta$.

Dividing the numerator and denominator by $x > 0$,

$$\tan\theta = \frac{d}{x + \dfrac{h(h+d)}{x}}.$$

Now d is fixed, so $\tan\theta$ is largest when the denominator $x + h(h+d)/x$ is smallest. Both terms are positive, so by the AM–GM inequality

$$x + \frac{h(h+d)}{x} \;\geq\; 2\sqrt{x \cdot \frac{h(h+d)}{x}} = 2\sqrt{h(h+d)},$$

with equality exactly when the two terms are equal, $x = h(h+d)/x$, that is $x^2 = h(h+d)$. So the denominator attains its minimum, and θ its maximum, at

$$\boxed{x = \sqrt{h(h+d)}}$$

and there $\tan\theta = \dfrac{d}{2\sqrt{h(h+d)}}$.

Or by calculus. Writing $k = h(h+d)$ and $f(x) = dx/(x^2 + k)$, the quotient rule gives

$$f'(x) = \frac{d\,(k - x^2)}{(x^2 + k)^2},$$

which is positive for $0 < x < \sqrt{k}$ and negative for $x > \sqrt{k}$, so f rises to a single maximum at $x = \sqrt{k}$ and then falls. AM–GM is quicker and makes it obvious that the maximum is global.

Checks

- Limits: as $x \to 0$ or $x \to \infty$ the formula gives $\tan\theta \to 0$. That is right: from very close to A, or very far away, the segment BC subtends almost no angle at P. So there must be a maximum in between.
- Symmetry: $\tan\theta$ is unchanged when x is replaced by $h(h+d)/x$, so the maximum must sit at the fixed point of that swap, $x = \sqrt{h(h+d)}$. With $h = 2$, $d = 1$, the points $x = 2$ and $x = 3$ give the same angle (about $11.31°$), and the maximum at $x = \sqrt{6}$ gives about $11.54°$.
- With $h = 1$, $d = 3$: $x = 2$, and $\tan\theta = 6/8 = \frac{3}{4}$, a 3–4–5 angle of $36.87°$. Computing θ directly from coordinates for 20 000 random (h, d, x) agreed with the formula to 10^{-14}, and a numerical search for the maximising x matched $\sqrt{h(h+d)}$ every time.

Common slips

- Writing $\tan\theta = d/x$, as if θ were in a right-angled triangle. It isn't: the right angle is at A, not on BC.
- Maximising $\tan\theta$ without saying why that maximises θ; it needs θ to stay in an interval where tan is increasing.

Beyond STEP

There is a lovely geometric reading of the answer. Points from which BC subtends a fixed angle lie on arcs of circles through B and C; the largest angle is reached on the smallest such circle that still meets the line AP, which is the one *tangent* to it at P. The tangent–secant theorem (the power of the point A) then says $AP^2 = AB \cdot AC$, which is exactly $x^2 = h(h+d)$. This is Regiomontanus's problem, often posed as "where should you stand to see a statue on a plinth best?". Optimising a quantity by first changing to one that is easier to handle but rises and falls with it is a theme that runs through first-year analysis; Siklos, *Advanced Problems in Mathematics*, has many more questions in this spirit.

WORKED SOLUTION

STEP I Specimen paper, Question 3

Pure • Second Order Differential Equations

STEP I Specimen paper, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s1-q3

The key idea

A second-order equation is hard; two first-order equations are easy. The suggested substitution $z = y' - y$ (writing y' for $\mathrm{d}y/\mathrm{d}x$) is chosen so that the left-hand side collapses into an expression involving only z and z'. That leaves a first-order linear equation for z, and then a second first-order linear equation for y; each yields to an integrating factor.

Solution

Reducing the order

With $z = y' - y$ we have $z' = y'' - y'$. We split the left-hand side of the equation to match:

$$y'' + 2y' - 3y = (y'' - y') + 3(y' - y) = z' + 3z.$$

So the equation becomes

$$\frac{\mathrm{d}z}{\mathrm{d}x} + 3z = 2\mathrm{e}^x.$$

The initial conditions give $z(0) = y'(0) - y(0) = 2 - 1 = 1$.

? **Ask yourself** How would you have spotted the split $(y'' - y') + 3(y' - y)$?

We need $y'' + 2y' - 3y$ to equal $z' + kz = y'' + (k-1)y' - ky$ for some constant k. Matching the y' terms gives $k = 3$, and then $-ky = -3y$ matches too. (This works because 1 is a root of the auxiliary equation $m^2 + 2m - 3 = (m-1)(m+3) = 0$; the other root, -3, is what appears in $z' + 3z$.)

Solving for z

Multiplying by the integrating factor e^{3x},

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(z\mathrm{e}^{3x}\right) = 2\mathrm{e}^{4x} \quad \Longrightarrow \quad z\mathrm{e}^{3x} = \tfrac{1}{2}\mathrm{e}^{4x} + A.$$

At $x = 0$, $1 = \frac{1}{2} + A$, so $A = \frac{1}{2}$ and

$$z = \tfrac{1}{2}\left(\mathrm{e}^{x} + \mathrm{e}^{-3x}\right).$$

Solving for y

Now $y' - y = z$, that is,

$$\frac{\mathrm{d}y}{\mathrm{d}x} - y = \tfrac{1}{2}\left(\mathrm{e}^{x} + \mathrm{e}^{-3x}\right).$$

The integrating factor is e^{-x}:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\left(y\mathrm{e}^{-x}\right) &= \tfrac{1}{2} + \tfrac{1}{2}\mathrm{e}^{-4x},\\ y\mathrm{e}^{-x} &= \tfrac{1}{2}x - \tfrac{1}{8}\mathrm{e}^{-4x} + B.\end{aligned}$$

At $x = 0$, $1 = -\frac{1}{8} + B$, so $B = \frac{9}{8}$. Multiplying through by e^{x},

$$\boxed{y = \left(\frac{x}{2} + \frac{9}{8}\right)\mathrm{e}^{x} - \frac{1}{8}\mathrm{e}^{-3x} = \frac{(4x+9)\mathrm{e}^{x} - \mathrm{e}^{-3x}}{8}}$$

Notice that we used only $y(0) = 1$ here; the condition $y'(0) = 2$ was already used in finding $z(0)$, so both conditions have been applied.

Checks

- $y(0) = (9-1)/8 = 1$, and $y' = \left((4x+13)\mathrm{e}^{x} + 3\mathrm{e}^{-3x}\right)/8$ gives $y'(0) = 16/8 = 2$.
- Substituting: e^{-3x} solves the homogeneous equation ($9 - 6 - 3 = 0$), and so does e^{x} ($1 + 2 - 3 = 0$). For $x\mathrm{e}^{x}$, $y'' + 2y' - 3y = (x+2) + 2(x+1) - 3x = 4$ times e^{x}, so $\frac{1}{2}x\mathrm{e}^{x}$ contributes exactly $2\mathrm{e}^{x}$.
- Numerically, stepping the original equation from $x = 0$ with a Runge–Kutta method gives $y(1) \approx 4.41098$, agreeing with $(13\mathrm{e} - \mathrm{e}^{-3})/8$.

Common slips

- Applying $y'(0) = 2$ a second time at the end, or forgetting to convert it into $z(0) = 1$.
- Trying a particular integral $c\,\mathrm{e}^{x}$ "otherwise": because e^{x} already solves the homogeneous equation this fails, and one needs $cx\mathrm{e}^{x}$ instead. The substitution handles this automatically.

Beyond STEP

The substitution is really a factorisation of the differential operator: writing D for $\mathrm{d}/\mathrm{d}x$, the equation is $(D+3)(D-1)y = 2\mathrm{e}^x$, and $z = (D-1)y$. Factorising operators in this way explains the whole theory of constant-coefficient linear equations, including why repeated or "resonant" roots produce the extra factor of x. Boyce and DiPrima, *Elementary Differential Equations*, develops this systematically.

WORKED SOLUTION

STEP I Specimen paper, Question 4

Pure • Integration by Substitution

STEP I Specimen paper, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s1-q4

The key idea

Whenever a square root of a quadratic appears, complete the square first. Here the quadratic under the root becomes $(x+1)^2-3$, and the factor outside the root is also $x+1$, so shifting to $u=x+1$ leaves the tidy form $\dfrac{1}{u\sqrt{u^2-3}}$. That shape (a variable times the root of its square minus a constant) is exactly what a secant substitution, or equally a reciprocal substitution $t=1/u$, is built to handle. Either one also turns the infinite range into a finite one.

Solution

Completing the square

We have

$$x^2+2x-2=(x+1)^2-3,$$

which is positive for $x\geq 1$, since then $(x+1)^2\geq 4>3$, so the integrand is defined throughout $[1,\infty)$. Put $u=x+1$, so $\mathrm{d}u=\mathrm{d}x$, $u=2$ when $x=1$, and $u\to\infty$ as $x\to\infty$:

$$I=\int_1^\infty \frac{\mathrm{d}x}{(x+1)\sqrt{x^2+2x-2}}=\int_2^\infty \frac{\mathrm{d}u}{u\sqrt{u^2-3}}.$$

For large u the integrand behaves like $1/u^2$, so the integral converges.

The secant substitution

Let $u=\sqrt{3}\sec\theta$ with $0\leq\theta<\frac{\pi}{2}$. Then

$$\mathrm{d}u=\sqrt{3}\sec\theta\tan\theta\,\mathrm{d}\theta,\qquad \sqrt{u^2-3}=\sqrt{3(\sec^2\theta-1)}=\sqrt{3}\tan\theta,$$

where we have used $\tan\theta\geq 0$ on this range to drop the modulus.

Ask yourself What are the new limits?

At $u = 2$ we need $\sec\theta = \frac{2}{\sqrt{3}}$, that is $\cos\theta = \frac{\sqrt{3}}{2}$, so $\theta = \frac{\pi}{6}$. As $u \to \infty$, $\sec\theta \to \infty$, so $\theta \to \frac{\pi}{2}$ from below. The substitution is increasing on $[\frac{\pi}{6}, \frac{\pi}{2})$, so the order of the limits is preserved.

Everything now cancels:

$$I = \int_{\pi/6}^{\pi/2} \frac{\sqrt{3}\sec\theta\tan\theta}{\sqrt{3}\sec\theta\cdot\sqrt{3}\tan\theta}\,\mathrm{d}\theta$$
$$= \frac{1}{\sqrt{3}}\int_{\pi/6}^{\pi/2}\mathrm{d}\theta = \frac{1}{\sqrt{3}}\left(\frac{\pi}{2}-\frac{\pi}{6}\right) = \frac{\pi}{3\sqrt{3}}.$$

So

$$\int_1^\infty \frac{\mathrm{d}x}{(x+1)\sqrt{x^2+2x-2}} = \frac{\pi}{3\sqrt{3}} = \frac{\sqrt{3}\,\pi}{9}.$$

An alternative: the reciprocal substitution

If secants don't come to mind, $t = 1/u$ works just as well and makes the range finite at once. With $u = 1/t$, $\mathrm{d}u = -\mathrm{d}t/t^2$, and for $t > 0$

$$u\sqrt{u^2-3} = \frac{1}{t}\cdot\frac{\sqrt{1-3t^2}}{t} = \frac{\sqrt{1-3t^2}}{t^2}.$$

The limits $u = 2$ and $u \to \infty$ become $t = \frac{1}{2}$ and $t \to 0$, and the minus sign reverses them:

$$I = \int_0^{1/2}\frac{\mathrm{d}t}{\sqrt{1-3t^2}} = \left[\frac{1}{\sqrt{3}}\arcsin\left(\sqrt{3}\,t\right)\right]_0^{1/2} = \frac{1}{\sqrt{3}}\arcsin\frac{\sqrt{3}}{2} = \frac{1}{\sqrt{3}}\cdot\frac{\pi}{3},$$

in agreement with the first method.

Checks

- Differentiate the antiderivative $F(x) = \frac{1}{\sqrt{3}}\operatorname{arcsec}\left(\frac{x+1}{\sqrt{3}}\right)$ and you recover the integrand; then $F(\infty) - F(1) = \frac{1}{\sqrt{3}}\left(\frac{\pi}{2}-\frac{\pi}{6}\right)$.
- A rough size check: on $[1,\infty)$ the integrand is a little bigger than $1/(x+1)^2$, whose integral is $\frac{1}{2}$. Our answer is $\pi/(3\sqrt{3}) \approx 0.6046$, slightly above $\frac{1}{2}$, as it should be.
- Numerical quadrature of the original integral (splitting the range and removing the infinite tail by $x = 1/w$) gives 0.604600, matching $\pi/(3\sqrt{3}) = 0.6045998\ldots$.

Common slips

- Forgetting to change the limits: the lower limit is $\theta = \frac{\pi}{6}$, not 0; using 0 gives the wrong answer $\pi/(2\sqrt{3})$.

- Writing $\sqrt{\sec^2\theta - 1} = \tan\theta$ without checking the sign of $\tan\theta$ on the range used.
- Leaving the answer as $\frac{1}{\sqrt{3}}\operatorname{arcsec}(\ldots)$ evaluated "at infinity" without taking the limit properly.

Beyond STEP

This is an improper integral: strictly, it means $\lim_{X\to\infty}\int_1^X$, and a first-year analysis course makes precise when such limits exist (comparison with $\int^\infty u^{-2}\,\mathrm{d}u$ is the standard test). The substitutions here are also the start of the general theory of integrating $R\left(x, \sqrt{ax^2+bx+c}\right)$ for rational R, which can always be reduced to integrating a rational function. For improper integrals and the comparison test, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP I Specimen paper, Question 5

Pure • Counting Digits and Factorials

STEP I Specimen paper, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s1-q5

The key idea

Both parts are about counting carefully rather than computing. In (i), first find exactly where the 1000th digit falls by counting digits in blocks (one-digit, two-digit, three-digit numbers), then count the 7s place by place: units, tens, hundreds. In (ii), each trailing zero is a factor of $10 = 2 \times 5$, and fives are scarcer than twos, so we just count the factors of 5 in 365!.

Solution

Part (i)

Where do we stop? The numbers 1 to 9 use 9 digits. The 90 two-digit numbers 10 to 99 use 180 digits, giving 189 so far. That leaves

$$1000 - 189 = 811 = 3 \times 270 + 1$$

digits. The first 270 three-digit numbers are $100, 101, \ldots, 369$, and they use 810 digits. The last digit is the first digit of 370, which is a 3. So we have written all of $1, 2, \ldots, 369$ in full, followed by a single 3.

? **Ask yourself** The number 370 has a 7 in it. Does it count?

No. We stop after writing its first digit, the 3, so its 7 is never written. This is exactly why we need to pin down the stopping point to the digit.

Counting the 7s in 1 to 369. We count by place value. Every 7 is in exactly one place, so nothing is counted twice; a number like 77 correctly contributes two.

- *Units digit.* In each complete block of ten, $10k, \ldots, 10k + 9$, exactly one number ends in 7. The numbers 0 to 359 form 36 complete blocks, giving 36 sevens: $7, 17, \ldots, 357$. The partial block 360 to 369 adds one more, 367. Total: 37.
- *Tens digit.* In each complete hundred, exactly ten numbers have tens digit 7: the

numbers 70–79, 170–179 and 270–279. The block 370–379 is beyond 369. Total: 30.

- *Hundreds digit.* Every number up to 369 has hundreds digit at most 3 (or no hundreds digit at all). Total: 0.

The final lone 3 adds nothing. Hence the digit 7 is used

$$37 + 30 + 0 = 67 \text{ times.}$$

Part (ii)

The number of trailing zeros of N in base 10 is the largest k with $10^k \mid N$. Since $10 = 2 \times 5$, this is the smaller of the power of 2 and the power of 5 in the prime factorisation of N.

Counting the fives. The power of 5 dividing 365! is the sum, over $j = 1, \ldots, 365$, of the number of factors of 5 in j. A multiple of 5 contributes at least one factor, a multiple of 25 contributes a second, a multiple of 125 a third, and so on. Counting each layer separately:

$$\begin{aligned}
\left\lfloor \tfrac{365}{5} \right\rfloor &= 73 && \text{(multiples of 5),} \\
\left\lfloor \tfrac{365}{25} \right\rfloor &= 14 && \text{(multiples of 25),} \\
\left\lfloor \tfrac{365}{125} \right\rfloor &= 2 && \text{(multiples of 125: 125, 250),} \\
\left\lfloor \tfrac{365}{625} \right\rfloor &= 0.
\end{aligned}$$

A number divisible by exactly 5^m is counted once in each of the first m lines, so it contributes exactly m, as it should. The power of 5 in 365! is therefore

$$73 + 14 + 2 = 89.$$

The twos are not the bottleneck. The same argument for 2 gives at least $\lfloor 365/2 \rfloor = 182 > 89$ factors of 2. (In fact there are $182 + 91 + 45 + 22 + 11 + 5 + 2 + 1 = 359$.) So $2^{89} \cdot 5^{89} = 10^{89}$ divides 365!, but 10^{90} does not, since $5^{90} \nmid 365!$.

Hence 365! ends in exactly **89** zeros.

Checks

For (i), a quick check on the stopping point: $9 + 180 + 3 \times 270 = 999$ digits take us to the end of 369, and the next digit is the leading 3 of 370. Writing out the first 1000 digits by computer and counting confirms 67 sevens, with 20 of them in 1–99 and 47 in 100–369. For (ii), compute 365! exactly (it has 779 digits) and count its trailing zeros. There are 89, matching the count of fives. A small case to test the method by hand: 25! should end in $\lfloor 25/5 \rfloor + \lfloor 25/25 \rfloor = 6$ zeros, and indeed $25! = 15\,511\,210\,043\,330\,985\,984\,000\,000$.

Common slips

- Counting 370 (or even 371–379) as written, which gives 68 or more.

- Using only $\lfloor 365/5 \rfloor = 73$ and forgetting that 25, 50, ... contain two fives and 125, 250 contain three.
- Counting 7s by numbers instead of by digits, so that 77, 177 and 277 are counted once rather than twice.

Beyond STEP

The layered count in (ii) is Legendre's formula: the power of a prime p in $n!$ is $\sum_{k \geq 1} \lfloor n/p^k \rfloor$, which also equals $(n - s_p(n))/(p-1)$, where $s_p(n)$ is the sum of the base-p digits of n. Here $365 = 2430_5$, with digit sum 9, and $(365 - 9)/4 = 89$. The formula underlies Kummer's theorem on the prime factors of binomial coefficients. Davenport's *The Higher Arithmetic* is a lovely place to explore this kind of elementary number theory further.

WORKED SOLUTION

STEP I Specimen paper, Question 6

Pure • Areas of Circular Segments

STEP I Specimen paper, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s1-q6

The key idea

The coin is a regular heptagon with a circular "cap" bulging out of each side. So its area is the heptagon plus seven circular segments, and a segment is a sector minus a triangle. Everything then comes down to two angles: the angle each arc subtends at its centre (the inscribed angle theorem gives it at once) and the angle between a long diagonal of the heptagon and the radius of its circumcircle.

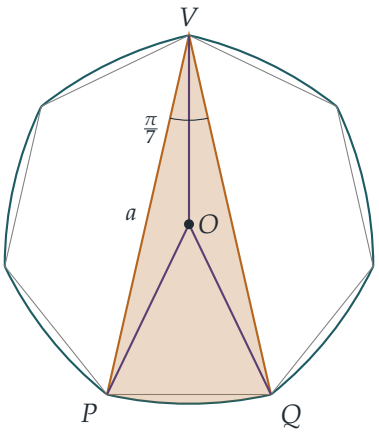
Solution

The shape

Each vertex is at distance a from the two ends of the arc opposite it (both ends lie on the circle of radius a centred there). So the seven "long" diagonals, each joining a vertex to one three places round, all have length a. Equal arcs of equal radius subtend equal angles at their centres, so the seven sides (the chords of the arcs) are also equal.

The long diagonals, taken in turn, form a seven-pointed star with all sides equal to a. Its angle at each point is the angle between the two diagonals from that vertex, and those diagonals, together with the side opposite, form a triangle with sides a, a and the common side length. All seven such triangles are congruent, so the star is equiangular as well as equilateral, hence regular. Therefore the seven vertices are the vertices of a regular heptagon, with centre O and circumradius R, say.

Label one vertex V, and let P and Q be the ends of the arc centred at V; P and Q are adjacent vertices of the heptagon.



The angle of each arc

The side PQ subtends an angle $2\pi/7$ at the centre O. Since V lies on the circumcircle, on the major arc PQ, the inscribed angle theorem gives

$$\angle PVQ = \tfrac{1}{2} \cdot \frac{2\pi}{7} = \frac{\pi}{7}.$$

So each arc is part of a circle of radius a subtending angle $\pi/7$ at its centre, and the sector VPQ (shaded) has area

$$\tfrac{1}{2}a^2 \cdot \frac{\pi}{7} = \frac{\pi a^2}{14}.$$

Adding up the area

The face of the coin consists of the heptagon together with seven segments, one on each side. The heptagon splits into seven triangles like OPQ, and the segment on PQ is the sector VPQ minus the triangle VPQ. Hence

$$\begin{aligned} \text{Area} &= 7\,[OPQ] + 7\big(\text{sector } VPQ - [VPQ]\big) \\ &= \frac{\pi a^2}{2} - 7\big([VPQ] - [OPQ]\big), \end{aligned}$$

where $[\,\cdot\,]$ denotes area.

Ask yourself How can we find $[VPQ] - [OPQ]$ without working out either triangle separately?

The centre O lies inside triangle VPQ, on its axis of symmetry, so removing triangle OPQ from triangle VPQ leaves the two congruent triangles OVP and OVQ:

$$[VPQ] - [OPQ] = [OVP] + [OVQ] = 2 \cdot \tfrac{1}{2}\, R\, a \sin \angle OVP.$$

By symmetry VO bisects $\angle PVQ$, so $\angle OVP = \pi/14$. Triangle OVP is isosceles with $OV = OP = R$, so its base angles are both $\pi/14$, and dropping the perpendicular from

O to VP gives

$$a = VP = 2R\cos\frac{\pi}{14}, \qquad \text{so} \qquad R = \frac{a}{2\cos(\pi/14)}.$$

Therefore

$$[VPQ] - [OPQ] = R\,a\sin\frac{\pi}{14} = \frac{a^2\sin(\pi/14)}{2\cos(\pi/14)} = \frac{a^2}{2}\tan\frac{\pi}{14},$$

and

$$\text{Area} = \frac{\pi a^2}{2} - \frac{7a^2}{2}\tan\frac{\pi}{14} = \frac{a^2}{2}\left(\pi - 7\tan\frac{\pi}{14}\right),$$

as required.

Checks

- Numerically, $\frac{1}{2}(\pi - 7\tan\frac{\pi}{14}) \approx 0.7719$, so the area is about $0.772a^2$. Sampling the seven arcs densely and using the shoelace formula agrees to six decimal places, and so does a Monte Carlo estimate (a point is in the coin exactly when it is within a of all seven vertices).
- The coin has width a in every direction, like a disc of diameter a, whose area is $\pi a^2/4 \approx 0.785a^2$. Of all shapes of constant width a the disc has the greatest area, so we expect a little less, and $0.772a^2 < 0.785a^2$ as it should be.
- The same argument for a Reuleaux n-gon (n odd) gives $\frac{a^2}{2}(\pi - n\tan\frac{\pi}{2n})$; as $n \to \infty$, $n\tan\frac{\pi}{2n} \to \frac{\pi}{2}$ and the area tends to $\pi a^2/4$, the disc. For $n = 3$ it gives $\frac{1}{2}(\pi - \sqrt{3})a^2$, the familiar Reuleaux triangle.

Common slips

- Taking the arc's angle to be $2\pi/7$ (the angle at the centre O) rather than $\pi/7$ (the angle at the opposite vertex, where the arc's centre actually is).
- Confusing a with the circumradius R: a is a long diagonal, $a = 2R\cos(\pi/14)$.

Beyond STEP

The 50p coin is a curve of *constant width*: every pair of parallel tangent lines is the same distance a apart, which is why a coin slot can measure it the same way however it is turned. Barbier's theorem says every such curve has perimeter πa (here $7 \cdot a \cdot \pi/7$), and the Blaschke–Lebesgue theorem says the Reuleaux triangle has the least area. For more on this kind of geometry, Courant and Robbins, *What Is Mathematics?*, has a lovely section on curves of constant width.

WORKED SOLUTION

STEP I Specimen paper, Question 7

Pure • Complex Exponential Series

STEP I Specimen paper, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s1-q7

The key idea

Whenever you see $1+\mathrm{e}^{\mathrm{i}\theta}$, take out the "half-angle" factor $\mathrm{e}^{\mathrm{i}\theta/2}$: what is left is $\mathrm{e}^{-\mathrm{i}\theta/2}+\mathrm{e}^{\mathrm{i}\theta/2}$, which is real. For the series, the binomial coefficients $\frac{n!}{r!(n-r)!}=\binom{n}{r}$ suggest the binomial theorem. So we write each sine as the imaginary part of an exponential, and the sum becomes the imaginary part of a power of $1+\mathrm{e}^{2\mathrm{i}\alpha}$, which the first part has already put into polar form.

Solution

Modulus and argument

Taking out a factor $\mathrm{e}^{\mathrm{i}\alpha}$,

$$\begin{aligned}1+\mathrm{e}^{2\mathrm{i}\alpha} &= \mathrm{e}^{\mathrm{i}\alpha}\left(\mathrm{e}^{-\mathrm{i}\alpha}+\mathrm{e}^{\mathrm{i}\alpha}\right)\\ &= \mathrm{e}^{\mathrm{i}\alpha}\left[(\cos\alpha-\mathrm{i}\sin\alpha)+(\cos\alpha+\mathrm{i}\sin\alpha)\right]\\ &= 2\cos\alpha\,\mathrm{e}^{\mathrm{i}\alpha}.\end{aligned}$$

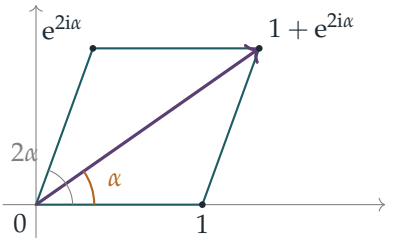
Ask yourself Can we now just read off "modulus $2\cos\alpha$, argument α"? What is the restriction on α for?

The form $r\,\mathrm{e}^{\mathrm{i}\theta}$ gives modulus r and argument θ only when $r>0$. Here $-\frac{1}{2}\pi<\alpha<\frac{1}{2}\pi$, so $\cos\alpha>0$, and this range also lies inside the principal interval $(-\pi,\pi]$. Hence

$$\left|1+\mathrm{e}^{2\mathrm{i}\alpha}\right|=2\cos\alpha,\qquad \arg\left(1+\mathrm{e}^{2\mathrm{i}\alpha}\right)=\alpha.$$

(If $\cos\alpha$ were negative, the modulus would be $-2\cos\alpha$ and the argument would shift by π.)

There is a nice geometric picture too. The points 0, 1 and $1+\mathrm{e}^{2\mathrm{i}\alpha}$ form a rhombus with sides of length 1, together with the point $\mathrm{e}^{2\mathrm{i}\alpha}$. The diagonal from 0 bisects the angle 2α and has length $2\cos\alpha$.



The series

Since $\sin\theta = \operatorname{Im} e^{i\theta}$ for real θ, and the binomial coefficients are real,

$$S = \sum_{r=0}^{n} \binom{n}{r} \sin(2r+1)\alpha = \operatorname{Im} \sum_{r=0}^{n} \binom{n}{r} e^{i(2r+1)\alpha}$$
$$= \operatorname{Im}\left[e^{i\alpha} \sum_{r=0}^{n} \binom{n}{r} \left(e^{2i\alpha}\right)^r\right] = \operatorname{Im}\left[e^{i\alpha}\left(1+e^{2i\alpha}\right)^n\right],$$

using $e^{i(2r+1)\alpha} = e^{i\alpha}\left(e^{2i\alpha}\right)^r$ and then the binomial theorem. Now use the first part, with de Moivre's theorem $\left(e^{i\alpha}\right)^n = e^{in\alpha}$:

$$e^{i\alpha}\left(1+e^{2i\alpha}\right)^n = e^{i\alpha}\, 2^n \cos^n\alpha\, e^{in\alpha} = 2^n \cos^n\alpha\, e^{i(n+1)\alpha}.$$

Because $2^n \cos^n\alpha$ is real, taking imaginary parts gives

$$\boxed{\sum_{r=0}^{n} \frac{n!}{r!\,(n-r)!} \sin(2r+1)\alpha = 2^n \cos^n\alpha \sin(n+1)\alpha.}$$

The factorisation $1 + e^{2i\alpha} = 2\cos\alpha\, e^{i\alpha}$ holds for *every* real α. We needed $\cos\alpha > 0$ only to name the modulus and argument, so the sum formula is valid for all real α, not just the range given.

Checks

- $n = 0$: both sides are $\sin\alpha$. $n = 1$: the left side is $\sin\alpha + \sin 3\alpha = 2\sin 2\alpha\cos\alpha$ by the sum-to-product formula, which matches $2\cos\alpha\sin 2\alpha$.
- $n = 3$, $\alpha = \frac{\pi}{6}$: the left side is $\frac{1}{2} + 3 + \frac{3}{2} - \frac{1}{2} = \frac{9}{2}$, and the right side is $8\left(\frac{\sqrt{3}}{2}\right)^3 \cdot \frac{\sqrt{3}}{2} = \frac{9}{2}$.
- $\alpha = \frac{1}{2}\pi$: every term $\sin(2r+1)\frac{\pi}{2} = (-1)^r$, so the sum is $(1-1)^n = 0$ for $n \geq 1$, matching $\cos\frac{\pi}{2} = 0$.
- We checked numerically that the formula holds for $0 \leq n \leq 24$ at random values of α.

Common slips

- Giving the modulus as $2\cos\alpha$ without noting that the range of α makes it positive. A modulus can't be negative.
- Taking imaginary parts too early. $\mathrm{Im}(zw) \neq \mathrm{Im}\, z \cdot \mathrm{Im}\, w$, so keep everything complex until the very end.
- Writing $\mathrm{e}^{\mathrm{i}(2r+1)\alpha}$ as $\left(\mathrm{e}^{2\mathrm{i}\alpha}\right)^r$ and losing the extra factor $\mathrm{e}^{\mathrm{i}\alpha}$. That turns $\sin(n+1)\alpha$ into $\sin n\alpha$.

Beyond STEP

The same idea, summing $\sum \binom{n}{r}\mathrm{e}^{\mathrm{i}r\theta}$ in closed form, appears in Fourier analysis. There, $\cos^n$-type expressions are expanded into, or collapsed from, sums of multiple angles, and the binomial weights $\binom{n}{r}/2^n$ become the binomial distribution. Factoring out the half-angle is also the standard way to find the modulus of $1-\mathrm{e}^{\mathrm{i}\theta}$, which you meet in geometric series and roots of unity. Needham's *Visual Complex Analysis* develops the geometric view of these manipulations beautifully, rhombus picture included.

WORKED SOLUTION

STEP I Specimen paper, Question 8

Pure • Integral Equations

STEP I Specimen paper, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

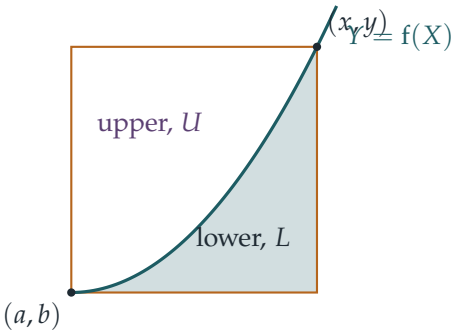
step.maths.org/questions/spec-s1-q8

The key idea

The area condition holds for *every* position of (x, y), so it is really an equation between two functions of x: one side is an integral of f and the other involves $\mathrm{f}(x)$ itself. Differentiating with respect to x removes the integral and leaves a first-order differential equation, which separates (or, more neatly, becomes the derivative of a quotient being zero).

Solution

Since $\mathrm{f}'(x) \geq 0$, the curve rises from the corner (a, b) to the opposite corner (x, y) of the rectangle, so the rectangle has width $x - a$ and height $y - b = \mathrm{f}(x) - b \geq 0$. The lower portion is the region between the curve and the bottom edge $Y = b$ (shaded below); the upper portion is the rest.



Using t as the variable of integration (since x is now the upper limit),

$$L = \int_a^x \big(\mathrm{f}(t) - b\big)\,\mathrm{d}t, \qquad L + U = (x - a)\big(\mathrm{f}(x) - b\big).$$

The condition $L = \frac{1}{2}U$ means $U = 2L$, so the whole rectangle has area $3L$:

$$3\int_a^x \big(\mathrm{f}(t) - b\big)\,\mathrm{d}t = (x-a)\big(\mathrm{f}(x) - b\big) \qquad \text{for all } x \geq a. \tag{$*$}$$

Ask yourself How do we get from an equation involving an integral to something we can solve for f?

Because $(*)$ holds for every $x \geq a$, both sides are the same function of x, so their derivatives agree. The function f is differentiable (we are told about f′), hence continuous, so by the fundamental theorem of calculus the left-hand side has derivative $3(\mathrm{f}(x) - b)$. Differentiating the right-hand side by the product rule,

$$3\big(\mathrm{f}(x) - b\big) = \big(\mathrm{f}(x) - b\big) + (x-a)\,\mathrm{f}'(x),$$

that is,

$$(x-a)\,\mathrm{f}'(x) = 2\big(\mathrm{f}(x) - b\big).$$

To tidy this, write $g(x) = \mathrm{f}(x) - b$ and $u = x - a$, so $g' = \mathrm{f}'$ and the equation is $u\,g' = 2g$. One could separate variables, $g'/g = 2/u$, but that divides by g, which might vanish. It is cleaner to notice that, for $x > a$ (so $u > 0$),

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{g}{u^2}\right) = \frac{u^2 g' - 2u\,g}{u^4} = \frac{u\,g' - 2g}{u^3} = 0.$$

So g/u^2 is a constant, k say, on $x > a$, and

$$\mathrm{f}(x) - b = k(x-a)^2 \qquad (x > a),$$

which also holds at $x = a$ since (a, b) is on the curve. As $\mathrm{f}(x) \geq b$ for $x \geq a$ (the function is non-decreasing from $\mathrm{f}(a) = b$), we have $k \geq 0$, and $k > 0$ unless the curve is the horizontal line $y = b$, for which every rectangle collapses to zero area and the condition is empty.

Thus the curve is $y - b = k(x-a)^2$ with $x \geq a$: half of the parabola with vertical axis $x = a$ and vertex at (a, b). Since $x \geq a$ we only get the right-hand arm, so it is an arc of the parabola, as required.

Conversely, this curve does satisfy the condition: $L = \int_a^x k(t-a)^2\,\mathrm{d}t = \frac{1}{3}k(x-a)^3$, while the rectangle has area $(x-a)\cdot k(x-a)^2 = k(x-a)^3$, so $U = \frac{2}{3}k(x-a)^3 = 2L$.

Checks

Take $a = b = 0$, $k = 1$ and $x = 3$: the rectangle is 3×9, area 27, and $\int_0^3 t^2\,\mathrm{d}t = 9$, so the portions are 9 and 18, ratio $1 : 2$. More generally, for $y - b = k(x-a)^p$ the lower-to-upper ratio is $1/p$ (we checked $p = \frac{1}{2}, 1, \frac{3}{2}, 2, 3$ numerically by Simpson's rule), so the ratio $\frac{1}{2}$ singles out $p = 2$; a straight line ($p = 1$) splits the rectangle equally, as it should.

Common slips

- Using x both as the limit and as the variable of integration, which makes the differentiation step confusing.
- Swapping the portions and getting $U = \frac{1}{2}L$, which leads to $(x-a)\mathrm{f}' = \frac{1}{2}(\mathrm{f}-b)$ and the curve $y - b = k\sqrt{x-a}$, a parabola with a *horizontal* axis and the wrong orientation.
- Dividing by $\mathrm{f}(x) - b$ without comment; the quotient argument above avoids this.

Beyond STEP

Equation $(*)$ is an *integral equation*: the unknown function appears inside an integral. Turning it into a differential equation by differentiating, and remembering that the original equation also fixes the initial value, is a standard first move, and the uniqueness of the answer rests on the fact that a function with zero derivative on an interval is constant (a consequence of the mean value theorem). Abbott's *Understanding Analysis* proves both the fundamental theorem of calculus and the mean value theorem carefully; Boyce and DiPrima, *Elementary Differential Equations*, covers equations like $u\,g' = 2g$.

WORKED SOLUTION

STEP I Specimen paper, Question 9

Pure • Modulus Inequalities

STEP I Specimen paper, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s1-q9

The key idea

Both parts are about getting rid of a modulus sign by finding exactly where the expression inside changes sign or crosses a level. In (i), writing $u = x + 1$ turns the expression into $u - 2/u$, which is increasing on each side of $u = 0$, so the inequality just pins u between two roots of a quadratic on each branch. In (ii), writing $\cos x - \sin x$ as a single cosine shows where it changes sign, and we split the integral there.

Solution

Part (i)

We need $x \neq -1$ for the expression to be defined. Write $u = x + 1$, so $u \neq 0$ and

$$x + \frac{x-1}{x+1} = \frac{x^2+2x-1}{x+1} = \frac{(x+1)^2-2}{x+1} = u - \frac{2}{u}.$$

Call this $f(u)$. The inequality is $-2 < f(u) < 2$.

Ask yourself Why not just multiply through by $x + 1$?

Because $x + 1$ can be negative, and multiplying by a negative number reverses an inequality; doing it carelessly is the classic way to lose half the answer. Instead we use the shape of f. Its derivative is $f'(u) = 1 + 2/u^2 > 0$, so f is strictly increasing on $u < 0$ and, separately, on $u > 0$. On each branch it runs from $-\infty$ to $+\infty$: as $u \to 0^-$, $-2/u \to +\infty$; as $u \to 0^+$, $-2/u \to -\infty$; and $f(u) \sim u$ as $|u| \to \infty$. So on each branch f takes each of the values -2 and 2 exactly once, and $-2 < f < 2$ holds precisely between those two points.

For $u \neq 0$, multiplying by u gives two equivalent equations:

$$f(u) = 2 \iff u^2 - 2u - 2 = 0 \iff u = 1 \pm \sqrt{3},$$
$$f(u) = -2 \iff u^2 + 2u - 2 = 0 \iff u = -1 \pm \sqrt{3}.$$

Branch $u > 0$. The positive roots are $u = -1 + \sqrt{3}$ (where $f = -2$) and $u = 1 + \sqrt{3}$ (where $f = 2$). Since f is increasing, $-2 < f(u) < 2$ exactly when $-1 + \sqrt{3} < u < 1 + \sqrt{3}$, that is, $-2 + \sqrt{3} < x < \sqrt{3}$.

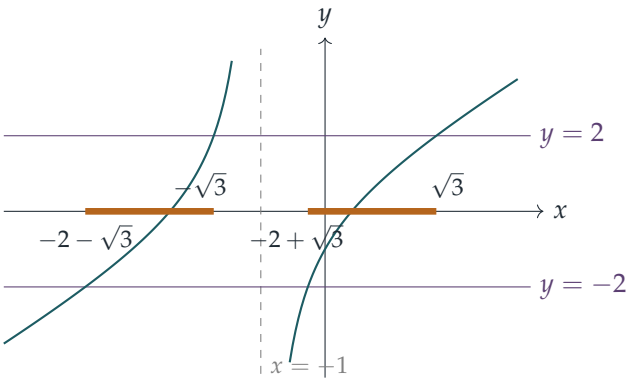
Branch $u < 0$. The negative roots are $u = -1 - \sqrt{3}$ (where $f = -2$) and $u = 1 - \sqrt{3}$ (where $f = 2$). Again by monotonicity, $-2 < f(u) < 2$ exactly when $-1 - \sqrt{3} < u < 1 - \sqrt{3}$, that is, $-2 - \sqrt{3} < x < -\sqrt{3}$.

Neither interval contains $x = -1$ (note $-\sqrt{3} < -1 < -2 + \sqrt{3}$), so the solution set is

$$\boxed{(-2 - \sqrt{3},\ -\sqrt{3}) \cup (-2 + \sqrt{3},\ \sqrt{3})}$$

with every endpoint of the form $a + b\sqrt{3}$ as required. The endpoints are excluded because there the modulus equals 2 exactly.

The graph of $y = x + (x-1)/(x+1)$ against the lines $y = \pm 2$ shows the two intervals (in amber) where the curve lies between the lines.



Part (ii)

Combine the two terms into one cosine:

$$\cos x - \sin x = \sqrt{2}\left(\tfrac{1}{\sqrt{2}}\cos x - \tfrac{1}{\sqrt{2}}\sin x\right) = \sqrt{2}\cos\left(x + \frac{\pi}{4}\right).$$

For $0 \le x \le \pi$ we have $\pi/4 \le x + \pi/4 \le 5\pi/4$, and cosine is positive on $[\pi/4, \pi/2)$ and negative on $(\pi/2, 5\pi/4]$. So $\cos x - \sin x \ge 0$ for $0 \le x \le \pi/4$ and ≤ 0 for $\pi/4 \le x \le \pi$. Splitting the integral at $\pi/4$, and using that an antiderivative of $\cos x - \sin x$ is $\sin x + \cos x$:

$$\begin{aligned}
\int_0^{\pi} |\cos x - \sin x|\,\mathrm{d}x &= \int_0^{\pi/4} (\cos x - \sin x)\,\mathrm{d}x - \int_{\pi/4}^{\pi} (\cos x - \sin x)\,\mathrm{d}x \\
&= \Big[\sin x + \cos x\Big]_0^{\pi/4} - \Big[\sin x + \cos x\Big]_{\pi/4}^{\pi} \\
&= (\sqrt{2} - 1) - (-1 - \sqrt{2}) \\
&= 2\sqrt{2}.
\end{aligned}$$

Checks

- Spot values in (i): at $x = 0$ the expression is -1, inside the modulus bound, and $0 \in (-2+\sqrt{3}, \sqrt{3})$; at $x = -2$ it is $-2+3 = 1$, and $-2 \in (-2-\sqrt{3}, -\sqrt{3})$; at $x = -1.5$ it is $-1.5+5 = 3.5$, outside, and indeed $-1.5 \in (-\sqrt{3}, -1)$ lies in neither interval.
- Testing the inequality on a fine grid and at 2×10^5 random points in $[-50, 50]$ agreed with the two intervals everywhere, and at all four endpoints the modulus equals 2.
- In (ii), Simpson's rule gives $2.828427\ldots$, matching $2\sqrt{2} \approx 2.828427$; the two pieces are $\sqrt{2}-1 \approx 0.4142$ and $\sqrt{2}+1 \approx 2.4142$. As a sanity check, the integrand is at most $\sqrt{2}$, so the integral is at most $\sqrt{2}\,\pi \approx 4.44$.

Common slips

- Multiplying both sides of $x + \frac{x-1}{x+1} < 2$ by $x+1$ without considering its sign, which loses (or wrongly includes) the interval to the left of $x = -1$.
- Solving only $f < 2$, or only $f > -2$: the modulus inequality needs both at once.
- In (ii), integrating $\cos x - \sin x$ straight across $[0, \pi]$, which gives -2 (a negative answer for the integral of a modulus is a red flag).

Beyond STEP

The trick in (i) is to replace an algebraic inequality by a statement about a monotonic function: if f is strictly increasing on an interval, then $a < f(u) < b$ holds exactly between the points where $f = a$ and $f = b$. Making this rigorous uses the intermediate value theorem, one of the first big results of a university analysis course, and inequalities with absolute values, such as $|x - a| < \varepsilon$, are the everyday language of that course. Abbott, *Understanding Analysis*, is a friendly place to see both.

WORKED SOLUTION

STEP I Specimen paper, Question 10

Mechanics • Power and Resistance

STEP I Specimen paper, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

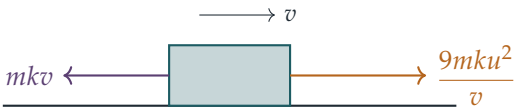
step.maths.org/questions/spec-s1-q10

The key idea

Power is force times speed, so an engine working at constant power P pulls with a driving force P/v. Newton's second law then gives a first-order differential equation for v that separates. Choosing the right form of acceleration is what matters: $\mathrm{d}v/\mathrm{d}t$ when the question asks about *time*, and $v\,\mathrm{d}v/\mathrm{d}x$ when it asks about *distance*.

Solution

Model the engine as a particle of mass m on a level track, so its weight and the normal reaction cancel and only horizontal forces matter. Take the direction of motion as positive.



The maximum speed

At speed v the driving force is $9mku^2/v$, so Newton's second law gives

$$m\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{9mku^2}{v} - mkv \quad\Longrightarrow\quad \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{k\,(9u^2 - v^2)}{v}.$$

While $v < 3u$ the right-hand side is positive, so the engine speeds up. It would stop accelerating only where the driving force balances the resistance, $9u^2 = v^2$. So the maximum possible speed is

$$v_{\max} = 3u.$$

Ask yourself The engine always accelerates while $v < 3u$, so why doesn't it reach $3u$?

The acceleration shrinks to zero as $v \to 3u$. Separating the variables, the time to go from speed v_1 to speed v_2 is

$$t = \frac{1}{k}\int_{v_1}^{v_2} \frac{v}{9u^2 - v^2}\,\mathrm{d}v = \frac{1}{2k}\ln\left(\frac{9u^2 - v_1^2}{9u^2 - v_2^2}\right),$$

and this tends to infinity as $v_2 \to 3u$. So $3u$ is a terminal speed: the engine gets as close to it as we like, but never actually reaches it in a finite time. (Equivalently, $v = 3u$ is an equilibrium of the differential equation, and by uniqueness of solutions a solution starting below it can't reach it.)

From u to $2u$

Put $v_1 = u$ and $v_2 = 2u$ in the formula above:

$$t = \frac{1}{2k}\ln\left(\frac{9u^2 - u^2}{9u^2 - 4u^2}\right) = \frac{1}{2k}\ln\left(\frac{8u^2}{5u^2}\right) = \frac{1}{2k}\ln\left(\frac{8}{5}\right),$$

as required.

Power cut off

Now the only forces are the two resistances, mkv and mkB, both opposing the motion. We want a distance, so write the acceleration as $v\,\mathrm{d}v/\mathrm{d}x$:

$$mv\frac{\mathrm{d}v}{\mathrm{d}x} = -mk(v + B) \quad\Longrightarrow\quad \frac{v}{v+B}\,\mathrm{d}v = -k\,\mathrm{d}x.$$

Writing $\dfrac{v}{v+B} = 1 - \dfrac{B}{v+B}$ and integrating as v falls from $2u$ to 0 while x increases from 0 to D:

$$\begin{aligned} kD &= \int_0^{2u}\left(1 - \frac{B}{v+B}\right)\mathrm{d}v \\ &= \Big[v - B\ln(v+B)\Big]_0^{2u} \\ &= 2u - B\ln\left(\frac{2u+B}{B}\right). \end{aligned}$$

So the distance travelled with the power cut off is

$$D = \frac{1}{k}\left[2u - B\ln\left(1 + \frac{2u}{B}\right)\right].$$

This time the engine does stop in a finite time: $\mathrm{d}v/\mathrm{d}t = -k(v+B) \le -kB < 0$, so the deceleration never drops below kB. That's exactly why the extra constant resistance is needed.

Checks

- As $B \to 0$, $B\ln(1 + 2u/B) \to 0$, so $D \to 2u/k$. That matches the resistance mkv alone, where $v\,\mathrm{d}v/\mathrm{d}x = -kv$ gives v falling linearly with distance, at rate k.

- $D > 0$, since $\ln(1+y) < y$ for $y > 0$ gives $B\ln(1+2u/B) < 2u$. And D decreases as B grows, as it should.
- Numerically, with $k = u = B = 1$, stepping the equations of motion gives a stopping distance of 0.90139, matching $2 - \ln 3 \approx 0.90139$. Stepping the powered phase for time $\frac{1}{2}\ln\frac{8}{5} \approx 0.2350$ from $v = 1$ ends at $v = 2.0000$.

Common slips

- Treating the driving force as constant. At constant power, the force is P/v and falls as the engine speeds up.
- Using $\mathrm{d}v/\mathrm{d}t$ in the last part. That gives the stopping time, not the distance.
- Claiming the engine reaches $3u$ "eventually". The time integral diverges, so it never does.

Beyond STEP

The speed $3u$ is a stable equilibrium of the autonomous equation $\dot{v} = f(v)$. Solutions approach it exponentially but never meet it, because solutions of a well-behaved differential equation can't cross or merge. That's a first taste of the uniqueness theorem and of phase-line analysis in a first course on differential equations. Boyce and DiPrima, *Elementary Differential Equations*, covers both. For more on power and resistive forces, try Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP I Specimen paper, Question 11

Mechanics • Friction and Rolling Contact

STEP I Specimen paper, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s1-q11

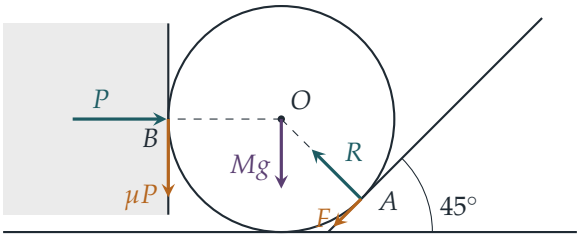
The key idea

At the moment the cylinder is about to move, the floor stops pushing, so only three contacts matter: the weight, the vertical face and the ramp. "Roll" means the cylinder does not slip on the ramp, so it starts to turn about its point of contact with the ramp. Work out how the point touching the vertical face then moves, and you see that the cylinder has to slide against the face. So that is where friction is limiting. A moment equation about the centre then gives the ramp friction with no extra work.

Solution

Model. The cylinder is a uniform rigid body of radius r, and we treat the problem in the vertical cross-section through its centre O. The equipment's face is vertical and moves horizontally. Let B be the point of contact with the face, which is level with O, and A the point of contact with the ramp. Since the ramp is at 45°, OA makes 45° with the downward vertical. Take x to the right (towards the ramp) and y upwards, with O as origin, so

$$B = (-r, 0), \qquad A = \tfrac{r}{\sqrt{2}}(1, -1).$$



Which friction is limiting

When the cylinder is on the point of moving, it is about to leave the floor, so the floor's normal reaction is zero, and with it the floor's friction. To roll up the ramp, the cylinder

must not slip at A, so its motion starts as a rotation about A, with the centre moving up the slope in the direction $(1,1)$. That rotation is clockwise. With angular speed ω, a point X of the cylinder has velocity ω times $X-A$ turned through 90° clockwise, that is $\omega\,(y,\,-x)$ applied to $X-A=(x,\,y)$. For $X=B$,

$$B-A=r\Big(-1-\tfrac{1}{\sqrt2},\,\tfrac{1}{\sqrt2}\Big)\quad\Longrightarrow\quad \mathbf{v}_B=\omega r\Big(\tfrac{1}{\sqrt2},\,1+\tfrac{1}{\sqrt2}\Big).$$

The face moves only horizontally, but B has an upward velocity component, so B slides up the face. The cylinder cannot roll on both surfaces at once, so it must slip on the vertical face, and **the friction is limiting on the vertical face**. Friction opposes the sliding, so it acts *down* on the cylinder with magnitude μP. At the ramp there is no slipping, so the ramp friction F is whatever equilibrium requires, subject only to $F\le \nu R$. (We write ν for the ramp's coefficient of friction, printed as v in the question.)

Ask yourself Why can't friction be limiting at the ramp instead?

If the cylinder slipped at A, it would be sliding up the ramp, not rolling. Rolling means no slip at A. The calculation above shows that rolling forces slip at B, so B is the contact where friction reaches μ times the normal reaction.

Finding P

Let R be the normal reaction of the ramp, along AO (direction $\frac{1}{\sqrt2}(-1,1)$), and let F be the ramp friction, taken as positive *down* the slope (direction $\frac{1}{\sqrt2}(-1,-1)$).

Moments about O. The weight, P and R all pass through O. The friction μP at B acts at distance r and turns the cylinder anticlockwise. The friction F at A is tangential, so it also acts at distance r, and when it points down the slope it turns the cylinder clockwise. Hence

$$\mu P\,r=Fr\quad\Longrightarrow\quad F=\mu P.$$

So the ramp friction really does act down the slope, and it has the same size as the friction at the face.

Resolving horizontally:

$$P-\frac{R}{\sqrt2}-\frac{\mu P}{\sqrt2}=0\quad\Longrightarrow\quad R=\sqrt2\,P-\mu P=P(\sqrt2-\mu).\tag{1}$$

Resolving vertically:

$$\frac{R}{\sqrt2}-\frac{\mu P}{\sqrt2}-\mu P-Mg=0.\tag{2}$$

Substituting $R/\sqrt2=P-\mu P/\sqrt2$ from (1) into (2),

$$\begin{aligned}P-\frac{\mu P}{\sqrt2}-\frac{\mu P}{\sqrt2}-\mu P&=Mg\\ P\big(1-\mu-\sqrt2\,\mu\big)&=Mg,\end{aligned}$$

using $2/\sqrt{2} = \sqrt{2}$. Therefore

$$P = \frac{Mg}{1 - \mu(1 + \sqrt{2})},$$

as required.

The conditions on μ and ν

On μ. P is the magnitude of a push, and $Mg > 0$, so we need $1 - \mu(1 + \sqrt{2}) > 0$, that is

$$\mu < \frac{1}{1 + \sqrt{2}} = \frac{\sqrt{2} - 1}{(\sqrt{2} + 1)(\sqrt{2} - 1)} = \sqrt{2} - 1.$$

If $\mu \geq \sqrt{2} - 1$, no push, however large, will roll the cylinder up the ramp: the extra downward friction at the face grows with P at least as fast as the lift does, and the cylinder jams.

On ν. The cylinder does not slip on the ramp, so $F \leq \nu R$. Using $F = \mu P$ and (1), and noting that $R > 0$ because $\mu < \sqrt{2} - 1 < \sqrt{2}$,

$$\mu P \leq \nu P(\sqrt{2} - \mu) \quad \Longleftrightarrow \quad \nu \geq \frac{\mu}{\sqrt{2} - \mu},$$

where we divided by $P(\sqrt{2} - \mu) > 0$. This completes the question.

Checks

- *Smooth face*, $\mu = 0$: then $P = Mg$ and no ramp friction is needed. This matches a frictionless cylinder wedged between a vertical wall and a 45° ramp: the ramp's reaction has equal horizontal and vertical components, so it must balance P and Mg equally.
- *Numbers*: with $\mu = 0.2$, $P \approx 1.934\,Mg$ and we need $\nu \geq 0.165$. Substituting these values back, the horizontal force, vertical force and moment about O all balance (to rounding error), for random M and μ too.
- As $\mu \to \sqrt{2} - 1 \approx 0.414$, $P \to \infty$, which is consistent with the jamming described above.

Common slips

- Making the friction limiting at the ramp. "Roll" rules out slipping there.
- Leaving the floor reaction in. At the point of rolling up, the cylinder is about to lift off, so the floor's reaction is zero.
- Getting the direction of the face friction wrong. B moves *up* relative to the face, so friction on the cylinder acts *down*. This is why P exceeds Mg.

Beyond STEP

The key step, deciding which contact slips by looking at the instantaneous centre of rotation, is the start of rigid-body kinematics. In a first-year course it leads on to the general rule $\mathbf{v}_X = \mathbf{v}_O + \boldsymbol{\omega} \times (X - O)$ and to problems where several rough contacts compete. The "jamming" at $\mu \geq \sqrt{2} - 1$, where pushing harder never helps, is a self-locking effect that engineers use deliberately in wedges and clamps. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body motion and friction with this kind of care.

WORKED SOLUTION

STEP I Specimen paper, Question 12

Mechanics • Collisions with Friction

STEP I Specimen paper, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s1-q12

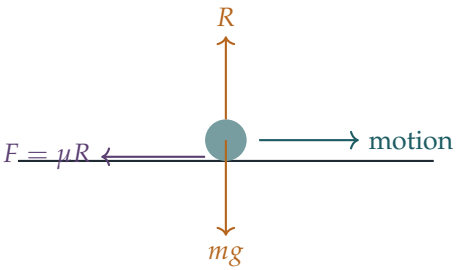
The key idea

Friction does all the work here. A disc sliding on level ice feels a constant retarding force, so it has constant deceleration, and the motion between events is just "constant acceleration" in disguise. We work backwards from the end: B's stopping distance fixes its speed after the impact, the collision laws turn that into A's speed just before the impact, and one more constant-acceleration step takes us back to the start at O.

Solution

The model

Treat each disc as a particle of mass m on a horizontal rough surface with coefficient of friction μ. While a disc slides, the forces on it are its weight mg downwards, the normal reaction R upwards and limiting friction F opposing the motion.



Resolving vertically, there is no vertical acceleration, so $R = mg$. The disc is sliding, so friction is limiting: $F = \mu R = \mu mg$. Resolving horizontally, $m\ddot{x} = -\mu mg$, so every sliding disc decelerates at the constant rate

$$a = \mu g,$$

whatever its mass. We also assume the collision is instantaneous, so friction has no time

to act during the impact itself; only the collision forces between the discs matter then, and momentum along OC is conserved.

Speed of B after the collision

B starts from the point at distance d from O and stops at C, at distance $2d$ from O, so it slides a distance d. Let its speed just after the collision be w. Using $v^2 = u^2 + 2as$ with final speed 0, acceleration $-\mu g$ and displacement d,

$$0 = w^2 - 2\mu g d \quad \Longrightarrow \quad w = \sqrt{2\mu g d},$$

which is the required result.

Speed of A just before the collision

Let A hit B with speed u, and let A's velocity just afterwards be v_A (both measured in the direction $O \to C$). Conservation of momentum, with equal masses m, gives

$$mu = mv_A + mw \quad \Longrightarrow \quad u = v_A + w,$$

and Newton's experimental law gives

$$w - v_A = eu.$$

Adding these eliminates v_A: $2w = (1+e)u$, so

$$u = \frac{2w}{1+e} = \frac{2\sqrt{2\mu g d}}{1+e}.$$

(Subtracting instead gives $v_A = \frac{1}{2}(1-e)u \geq 0$, so A carries on in the same direction more slowly.)

? Ask yourself Could A catch B again and spoil the calculation?

No. After the impact both discs decelerate at the same rate μg, so the gap between their speeds, $w - v_A = eu$, stays constant while both are moving. B is never slower than A, so it stays ahead; and A, starting slower, stops first. So there is exactly one collision, and B's journey from d to $2d$ is undisturbed. (When $e = 0$ the discs move together until they stop, still with no further impact.)

The starting speed

A slides a distance d from O to B, decelerating at μg, so by $v^2 = u^2 + 2as$ again,

$$u^2 = U^2 - 2\mu g d \quad \Longrightarrow \quad U^2 = u^2 + 2\mu g d.$$

Substituting for u,

$$\begin{aligned} U^2 &= \frac{4 \cdot 2\mu g d}{(1+e)^2} + 2\mu g d = \frac{2\mu g d}{(1+e)^2}\left(4 + (1+e)^2\right) \\ &= \frac{2\mu g d}{(1+e)^2}\left(5 + 2e + e^2\right). \end{aligned}$$

Taking the positive square root, as $U > 0$ and $1 + e > 0$,

$$U = \frac{\sqrt{2\mu g d}}{1+e}\,(5 + 2e + e^2)^{\frac{1}{2}},$$

as required.

Checks

- Perfectly elastic, $e = 1$: identical discs swap velocities, so A stops dead and B leaves with A's arrival speed $u = w$. The formula gives $U = \sqrt{2}\,\sqrt{2\mu g d} = \sqrt{2\mu g (2d)}$, exactly the speed needed for a single disc to slide the whole $2d$ to C. That makes sense: the momentum is passed straight on.
- Perfectly inelastic, $e = 0$: the formula gives $U = \sqrt{5}\,\sqrt{2\mu g d}$, larger than for $e = 1$, as it should be, since kinetic energy is lost in the impact. More generally $(5 + 2e + e^2)/(1 + e)^2 = 1 + 4/(1 + e)^2$ decreases as e increases.
- Dimensions: μ and e are pure numbers and $\sqrt{gd}$ is a speed.
- Numerically, with $\mu = 0.1$, $g = 9.8\ \mathrm{m\,s^{-2}}$, $d = 1$ m and $e = 0.5$, the formula gives $U = \frac{7}{3} \approx 2.33\ \mathrm{m\,s^{-1}}$. Stepping the motion forward in small time steps from this U (for many random choices of μ, d and e) brings B to rest within 0.001% of $2d$ from O.

Common slips

- Applying energy conservation across the collision. Unless $e = 1$, kinetic energy is lost in the impact; use momentum and the restitution law there, and the work–energy (or $v^2 = u^2 + 2as$) step only between events.
- Forgetting that A also slides a distance d before the impact, and writing $U = u$.
- Getting the restitution law the wrong way round: it is the speed of separation, $w - v_A$, that equals e times the speed of approach, u.

Beyond STEP

The tidy feature here is that friction gives every disc the same deceleration regardless of its mass, so in a frame decelerating at μg the ice is effectively frictionless while both discs slide, and the collision looks like a textbook one. Choosing a convenient frame is a central technique in undergraduate mechanics, and the friction law $F = \mu R$ itself is an empirical model whose limits are worth knowing. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats both collisions and friction with care.

WORKED SOLUTION

STEP I Specimen paper, Question 13

Mechanics • Circular Motion

STEP I Specimen paper, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s1-q13

The key idea

This is a conical pendulum whose string can stretch. Hooke's law ties the tension to the length of the string, while resolving forces vertically and horizontally ties the tension to g and to ω. Because the string makes a non-zero angle with the vertical, the tension has to be strictly bigger than mg, and that single fact drives both inequalities.

Solution

Model the particle as a point mass on a light elastic string obeying Hooke's law, with no air resistance. Let λ be the modulus of elasticity.

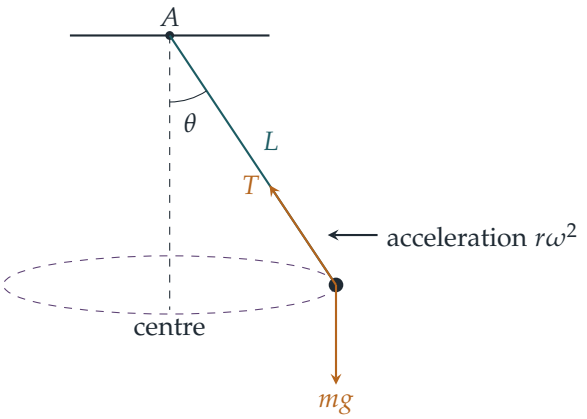
The modulus. Hanging at rest, the string has extension $2a - a = a$ and the tension balances the weight:

$$mg = \frac{\lambda a}{a} \quad \Longrightarrow \quad \lambda = mg.$$

So when the string has length L, its tension is

$$T = \frac{mg(L-a)}{a}.$$

The circular motion. Suppose the particle moves in a horizontal circle below A, with the string of length L making angle θ with the downward vertical. Since the motion is a genuine circle below the level of A, we have $0 < \theta < \frac{\pi}{2}$, so $\sin\theta > 0$ and $0 < \cos\theta < 1$. The centre of the circle is directly below A, and the radius is $r = L\sin\theta$.



The angular speed of the particle about the vertical through A is ω, so its acceleration is $r\omega^2 = L\omega^2 \sin\theta$, directed horizontally towards the centre. There is no vertical acceleration.

Resolving vertically:

$$T\cos\theta = mg. \tag{1}$$

Resolving horizontally, towards the centre:

$$T\sin\theta = mL\omega^2\sin\theta \quad \Longrightarrow \quad T = mL\omega^2, \tag{2}$$

where we may divide by $\sin\theta$ because $\sin\theta \neq 0$.

Ask yourself Where does the fact that the circle is genuine (not just the particle hanging still) get used?

It is used twice: to divide by $\sin\theta$ in (2), and to say $\cos\theta < 1$ below. If $\theta = 0$ the particle simply hangs at rest on the axis, (2) says nothing, and any ω would "work".

The string is longer than $2a$. From (1), since $0 < \cos\theta < 1$,

$$T = \frac{mg}{\cos\theta} > mg.$$

Using Hooke's law, this says

$$\frac{mg(L-a)}{a} > mg \quad \Longrightarrow \quad L - a > a \quad \Longrightarrow \quad L > 2a,$$

so the string is stretched to a length greater than $2a$.

The range of ω. Equating the two expressions for T, from Hooke's law and from (2):

$$mL\omega^2 = \frac{mg(L-a)}{a} \quad \Longrightarrow \quad \omega^2 = \frac{g}{a}\left(1 - \frac{a}{L}\right).$$

Since $L > 2a$, we have $0 < \frac{a}{L} < \frac{1}{2}$, so $\frac{1}{2} < 1 - \frac{a}{L} < 1$. Therefore

$$\frac{g}{2a} < \omega^2 < \frac{g}{a},$$

which is what we had to show.

The converse (not required, but reassuring). If ω lies in this range, set $L = \dfrac{a}{1 - a\omega^2/g}$. This is positive, satisfies $L > 2a$, and gives $\cos\theta = \dfrac{mg}{T} = \dfrac{a}{L-a}$, which lies strictly between 0 and 1. So for every such ω there really is a conical motion.

Checks

- Limits: as $L \to 2a$ the tension tends to mg, $\theta \to 0$ and $\omega^2 \to g/(2a)$, which is the frequency of small swings of a simple pendulum of length $2a$. As $L \to \infty$, $\omega^2 \to g/a$.
- Numerically: with $\omega^2 = \frac{3}{4}\,g/a$ we get $L = 4a$, $T = 3mg$ and $\cos\theta = \frac{1}{3}$. Stepping the full three-dimensional equations of motion from these starting conditions for two revolutions, the particle stayed on the predicted circle to within 10^{-13} of a unit.
- Units: g/a has units of s^{-2}, as ω^2 should.

Common slips

- Taking the radius of the circle to be L rather than $L\sin\theta$. The $\sin\theta$ then fails to cancel in (2).
- Forgetting that $T > mg$ needs $\cos\theta < 1$, that is, $\theta \neq 0$. The strict inequalities depend on it.
- Using the natural length a (or the equilibrium length $2a$) as the radius instead of the stretched length L.

Beyond STEP

The lower limit $\omega^2 = g/(2a)$ is the frequency of small oscillations about the vertical, and this is no accident: steady conical motion near the axis is just two perpendicular small swings, a quarter-period out of phase. The upper limit is where the string's stiffness can no longer supply the centripetal force, however far it stretches. Studying steady motions and their stability, for example by asking what happens if the particle is nudged off its circle, is a standard topic in first-year dynamics. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats the conical pendulum and the analysis of motion in rotating systems with care.

WORKED SOLUTION

STEP I Specimen paper, Question 14

Probability and Statistics • Expected Value

STEP I Specimen paper, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s1-q14

The key idea

Everything depends on one table: the distribution of the total of two fair four-sided dice. Once we have it, the expected gain for any guess x is just "prize times chance of winning, minus penalty times chance of losing". The interesting tension is that large guesses carry large prizes but small chances, so the best guess is not obvious until we compare them all.

Solution

Let S be the total score. Each die shows $1, 2, 3, 4$ with probability $\frac{1}{4}$ each, independently, so each of the 16 ordered pairs is equally likely. Counting the pairs with each total gives

s	2	3	4	5	6	7	8
$\mathrm{P}(S=s)$	$\frac{1}{16}$	$\frac{2}{16}$	$\frac{3}{16}$	$\frac{4}{16}$	$\frac{3}{16}$	$\frac{2}{16}$	$\frac{1}{16}$

(for example, $S=5$ comes from $(1,4),(2,3),(3,2),(4,1)$).

If B guesses x and $p=\mathrm{P}(S=x)$, his gain G in pence is $100x^2$ with probability p and $-50x$ with probability $1-p$, so

$$\begin{aligned}\mathrm{E}(G) &= 100x^2p - 50x(1-p)\\ &= 50x\big[(2x+1)p-1\big].\end{aligned}$$

Writing $p=\frac{k}{16}$, where k is the number of ways of making x,

$$\mathrm{E}(G)=\frac{25x\big[k(2x+1)-16\big]}{8}.$$

Part (i)

Only $(4,4)$ gives a total of 8, so $p = \frac{1}{16}$ and

$$\mathrm{E}(G) = 6400 \cdot \frac{1}{16} - 400 \cdot \frac{15}{16} = 400 - 375 = 25,$$

so B's expected gain when guessing 8 is 25 pence, as required.

Part (ii)

A guess outside $2, 3, \ldots, 8$ can never win, so (taking x positive, as a score must be) it only loses money. For the seven possible totals, the formula above gives

x	2	3	4	5	6	7	8
k	1	2	3	4	3	2	1
$\mathrm{E}(G)$ (pence)	-68.75	-18.75	137.5	437.5	431.25	306.25	25

For instance, for $x = 5$: $\mathrm{E}(G) = \frac{25 \cdot 5\,(4 \cdot 11 - 16)}{8} = \frac{125 \cdot 28}{8} = 437.5$.

Ask yourself The prize grows like x^2, so why isn't a big guess like 7 or 8 best?

The prize does grow quickly, but the chance of winning falls off on either side of 5, and the penalty is paid on the much more likely event of being wrong. At $x = 8$ the chance has dropped to $\frac{1}{16}$, which almost exactly cancels the large prize. The two effects balance best near the middle: $x = 6$ has a bigger prize than $x = 5$ but only three ways to win instead of four, and it comes out just behind.

So we advise B to guess $\boxed{x = 5}$, with expected gain 437.5 pence, i.e. £4.375 (about £4.38) per game.

Checks

- The probabilities in the table sum to $\frac{16}{16} = 1$.
- Part (i) comes out exactly at 25, which confirms the table and formula.
- A simulation of 400 000 games guessing 5 gave an average gain of about 437.2 pence, against the exact 437.5.

Common slips

- Treating the totals 2 to 8 as equally likely: they are not, because the 16 *ordered* pairs are the equally likely outcomes.
- Forgetting that B loses $50x$ with probability $1 - p$, not with probability 1.
- Stopping at $x = 6$ or $x = 7$ because the prize is bigger: 5 and 6 are close, so every value must be computed.

Beyond STEP

This is a small decision problem: choose the action that maximises expected payoff. At university the same idea appears in decision theory and in the study of sums of independent random variables, where the distribution of a sum is found by *convolution* (here, the triangular shape of the table) and, for many dice, approaches a normal curve by the central limit theorem. Grimmett and Welsh, *Probability: An Introduction*, covers both.

WORKED SOLUTION

STEP I Specimen paper, Question 15

Probability and Statistics • Sums of Random Variables

STEP I Specimen paper, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s1-q15

The key idea

The strange-looking distribution of C is exactly the error made when one price is rounded to the nearest 10 pence, provided the last digit of the price is equally likely to be any of $0, 1, \ldots, 9$. The shopper's total error is then a sum of 36 independent copies of C, so its mean and variance follow at once, and the Central Limit Theorem tells us how surprising a discrepancy of 20 pence is.

Solution

Expectation and variance of C

The distribution is symmetric about 0, since $\mathrm{P}(C = i) = \mathrm{P}(C = -i)$ for every i. So the terms $i\,\mathrm{P}(C = i)$ cancel in pairs, and

$$\mathrm{E}(C) = 0.$$

(The probabilities do sum to 1: $9 \cdot \frac{1}{10} + 2 \cdot \frac{1}{20} = 1$.) Since the mean is 0, the variance is just $\mathrm{E}(C^2)$:

$$\begin{aligned}\mathrm{Var}(C) = \mathrm{E}(C^2) &= 2 \cdot \frac{25}{20} + \frac{2}{10}\left(1^2 + 2^2 + 3^2 + 4^2\right) \\ &= \frac{5}{2} + \frac{2 \cdot 30}{10} = \frac{17}{2}.\end{aligned}$$

The shopper's estimate

The model. For one item, let C be the actual price minus the rounded price, in pence. This depends only on the last digit d of the price. We assume that d is equally likely to be any of $0, 1, \ldots, 9$; that is what buying "at random" should mean here, though it is an assumption about real prices.

- $d = 0$: no rounding, so $C = 0$.

- $d = 1, 2, 3, 4$: the price is rounded down by d, so $C = d$.
- $d = 6, 7, 8, 9$: the price is rounded up by $10 - d$, so $C = -(10 - d)$, which is $-4, -3, -2, -1$.
- $d = 5$: the price is rounded down or up with equal probability, so $C = 5$ or -5, each with probability $\frac{1}{2} \cdot \frac{1}{10} = \frac{1}{20}$.

Each of $-4, \ldots, 4$ arises from exactly one digit, with probability $\frac{1}{10}$, and ± 5 each have probability $\frac{1}{20}$. This is precisely the distribution in the first part, which is why it was set up that way.

The total error. Let $C_1, \ldots, C_{36}$ be the errors on the 36 items, and let

$$S = C_1 + C_2 + \cdots + C_{36}$$

be the true bill minus the shopper's estimate. Since the items are chosen at random, we take the C_j to be independent, each distributed like C. Then

$$\mathrm{E}(S) = 36 \cdot 0 = 0, \qquad \mathrm{Var}(S) = 36 \cdot \tfrac{17}{2} = 306,$$

using independence for the variance. So the standard deviation of S is $\sqrt{306} \approx 17.5$ pence.

Ask yourself Is 20 pence a large discrepancy or a small one?

It must be compared with the standard deviation, not with the size of a single rounding error. The cashier's bill is 20 pence above the estimate, which means $S = 20$. That is

$$\frac{20}{\sqrt{306}} \approx 1.14$$

standard deviations above the mean. Because S is a sum of 36 independent, identically distributed variables, the Central Limit Theorem says that S is approximately $\mathrm{N}(0, 306)$. With a continuity correction (as S takes whole-number values),

$$\mathrm{P}(S \geq 20) \approx \mathrm{P}\left(Z \geq \frac{19.5}{\sqrt{306}}\right) = \mathrm{P}(Z \geq 1.11) \approx 0.13,$$

where $Z \sim \mathrm{N}(0, 1)$. An error at least this large (in either direction) happens about 26% of the time.

So the shopper should **not** suspect a mistake. A difference of 20 pence is well within the normal spread of her rounding errors, about one standard deviation, and would happen on roughly one shopping trip in seven or eight. Only a discrepancy of more than about two standard deviations, say 35 pence or more, would give her real grounds for suspicion.

Checks

- Rounding a price to the nearest 10 pence changes it by at most 5 pence, so the worst case for 36 items is 180 pence. A typical error of about 17 pence is far smaller than this, as it should be: positive and negative errors largely cancel.
- The computer confirmed that the rounding error has exactly the distribution of C, with $\mathrm{E}(C) = 0$ and $\mathrm{Var}(C) = \frac{17}{2}$. It then convolved 36 copies exactly using fractions: $\mathrm{Var}(S) = 306$ and $\mathrm{P}(S \geq 20) \approx 0.1330$, against 0.1325 from the normal approximation and 0.132 from 2×10^5 simulated shopping trips.

Common slips

- Taking $\mathrm{Var}(S)$ as $36^2\,\mathrm{Var}(C)$. That would be the variance of $36C$, one error repeated 36 times; for independent errors the variances simply add.
- Treating the probability of S being exactly 20 (about 0.012) as the measure of surprise. Every individual value of S is fairly unlikely; what matters is the probability of an error at least as large as the one observed.
- Forgetting to say what is being assumed: that the last digits of the prices are uniformly distributed and the items' errors are independent.

Beyond STEP

This is the reasoning behind a hypothesis test: assume nothing has gone wrong (here, S is just rounding error), then ask how likely a result at least as extreme as the observed one would be. The same analysis of rounding errors is used in numerical analysis, where the errors from rounding many numbers usually grow like the square root of their count, not linearly. Blitzstein and Hwang, *Introduction to Probability*, covers sums of random variables and the Central Limit Theorem clearly.

WORKED SOLUTION

STEP I Specimen paper, Question 16

Probability and Statistics • Continuous Random Variables

STEP I Specimen paper, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s1-q16

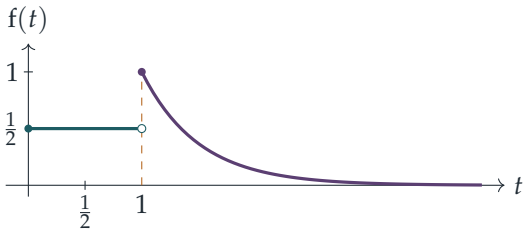
The key idea

The unknown k comes from the fact that the total probability is 1, and everything else follows from integrating the two pieces of the density separately. For the cost, the trick is to turn the continuous variable T into a discrete one: the number of coins depends only on how many complete half-minutes have passed, so we need the probability that T lands in each half-minute interval. A geometric-type series then does the rest.

Solution

Finding k and the mean

The density is a flat piece of height $\frac{1}{2}$ on $[0,1)$ followed by an exponential tail, with a jump at $t = 1$.



The first piece contributes probability $\frac{1}{2}$, so the tail must contribute the other $\frac{1}{2}$:

$$k\int_1^\infty \mathrm{e}^{-2t}\,\mathrm{d}t = k\left[-\tfrac{1}{2}\mathrm{e}^{-2t}\right]_1^\infty = \tfrac{1}{2}k\mathrm{e}^{-2} = \tfrac{1}{2}, \qquad \text{so} \qquad k = \mathrm{e}^2.$$

(The density on $t \geq 1$ is therefore e^{2-2t}, which equals 1 at $t = 1$, as in the sketch.)

For the mean, integrating by parts with $u = t$, $\mathrm{d}v = \mathrm{e}^{-2t}\,\mathrm{d}t$:

$$\int_1^\infty t\,\mathrm{e}^{-2t}\,\mathrm{d}t = \left[-\tfrac{1}{2}t\,\mathrm{e}^{-2t}\right]_1^\infty + \tfrac{1}{2}\int_1^\infty \mathrm{e}^{-2t}\,\mathrm{d}t = \tfrac{1}{2}\mathrm{e}^{-2} + \tfrac{1}{4}\mathrm{e}^{-2} = \tfrac{3}{4}\mathrm{e}^{-2},$$

using $t\,\mathrm{e}^{-2t} \to 0$ as $t \to \infty$. Hence

$$\mathrm{E}(T) = \int_0^1 \tfrac{1}{2}t\,\mathrm{d}t + \mathrm{e}^2 \cdot \tfrac{3}{4}\mathrm{e}^{-2} = \tfrac{1}{4} + \tfrac{3}{4} = 1,$$

so the expected length of a call is one minute, as required.

The cumulative distribution function

Write $\mathrm{F}(t) = \mathrm{P}(T \le t)$. For $t < 0$ it is 0; for $0 \le t < 1$ it is $\int_0^t \tfrac{1}{2}\,\mathrm{d}s = \tfrac{1}{2}t$; and for $t \ge 1$,

$$\mathrm{F}(t) = \tfrac{1}{2} + \int_1^t \mathrm{e}^{2-2s}\,\mathrm{d}s = \tfrac{1}{2} + \tfrac{1}{2}\bigl(1 - \mathrm{e}^{2-2t}\bigr) = 1 - \tfrac{1}{2}\mathrm{e}^{2-2t}.$$

So

$$\mathrm{F}(t) = \begin{cases} 0 & t < 0, \\ \tfrac{1}{2}t & 0 \le t < 1, \\ 1 - \tfrac{1}{2}\mathrm{e}^{2-2t} & t \ge 1. \end{cases}$$

Note that F is continuous at $t = 1$ (both formulas give $\tfrac{1}{2}$) even though f jumps there, and $\mathrm{F}(t) \to 1$ as $t \to \infty$. A form we will use: for $t \ge 1$, $\mathrm{P}(T > t) = \tfrac{1}{2}\mathrm{e}^{2-2t}$.

The expected cost

One coin goes in at the start, and one more each time a half-minute is completed. So if the call lasts T minutes, the number of complete half-minutes is $\lfloor 2T \rfloor$ and

$$C = 10\bigl(1 + \lfloor 2T \rfloor\bigr).$$

(Whether a coin is inserted at the exact instant the call ends at a half-minute mark doesn't matter: T is continuous, so that has probability 0.)

Ask yourself How do we get from a continuous T to the distribution of C?

The event $C = 10(n+1)$ is exactly the event $\tfrac{n}{2} \le T < \tfrac{n+1}{2}$, and we can read its probability off F:

- $n = 0$: $\mathrm{P}(C = 10) = \mathrm{F}(\tfrac{1}{2}) = \tfrac{1}{4}$;
- $n = 1$: $\mathrm{P}(C = 20) = \mathrm{F}(1) - \mathrm{F}(\tfrac{1}{2}) = \tfrac{1}{4}$;
- $n \ge 2$: both endpoints are at least 1, so

$$\mathrm{P}\bigl(C = 10(n+1)\bigr) = \tfrac{1}{2}\mathrm{e}^{2-n} - \tfrac{1}{2}\mathrm{e}^{2-(n+1)} = \tfrac{1}{2}\bigl(1 - \mathrm{e}^{-1}\bigr)\mathrm{e}^{2-n}.$$

Put $r = \mathrm{e}^{-1}$ and $m = n - 2$, so that $\mathrm{e}^{2-n} = r^m$ and $10(n+1) = 10(m+3)$. Then

$$\begin{aligned} \mathrm{E}(C) &= 10 \cdot \tfrac{1}{4} + 20 \cdot \tfrac{1}{4} + \sum_{m=0}^{\infty} 10(m+3) \cdot \tfrac{1}{2}(1-r)\,r^m \\ &= \tfrac{15}{2} + 5(1-r)\left(\sum_{m=0}^{\infty} m r^m + 3\sum_{m=0}^{\infty} r^m\right). \end{aligned}$$

Since $0 < r < 1$, the geometric series gives $\sum r^m = \dfrac{1}{1-r}$, and differentiating it term by term (then multiplying by r) gives $\sum mr^m = \dfrac{r}{(1-r)^2}$. Therefore

$$\mathrm{E}(C) = \tfrac{15}{2} + 5(1-r)\left(\frac{r}{(1-r)^2} + \frac{3}{1-r}\right) = \tfrac{15}{2} + \frac{5r}{1-r} + 15.$$

Finally $\dfrac{r}{1-r} = \dfrac{\mathrm{e}^{-1}}{1-\mathrm{e}^{-1}} = \dfrac{1}{\mathrm{e}-1}$, so

$$\mathrm{E}(C) = 22\tfrac{1}{2} + \frac{5}{\mathrm{e}-1},$$

as required. Numerically this is about 25.41 pence.

The comment

At 20 pence per minute, a call of mean length one minute would cost 20 pence on average, but $\mathrm{E}(C) \approx 25.4$. The difference arises because the man pays *in advance* for each half-minute: every coin buys a full half-minute, but the last one is almost never used up. Indeed $C = 10(1 + \lfloor 2T \rfloor) > 20T$ for every call, since $\lfloor 2T \rfloor > 2T - 1$; the cost rounds the length *up* to the next half-minute, so $\mathrm{E}(C)$ exceeds $20\,\mathrm{E}(T)$ by twenty times the expected unused paid time (about 5.4 pence here).

Checks

- The probabilities of C add to 1: $\frac{1}{4} + \frac{1}{4} + \sum_{m \geq 0} \frac{1}{2}(1-r)r^m = \frac{1}{2} + \frac{1}{2} = 1$.
- The bound $20T < C \leq 20T + 10$ gives $20 < \mathrm{E}(C) \leq 30$, and 25.41 sits comfortably inside. If the unused time in the last half-minute were uniform, we would expect an excess of about 5 pence; the long tail gives slightly more.
- Numerically, Simpson's rule gives total probability 1 and mean 1, and a simulation of 10^6 calls (half uniform on $[0, 1)$, half 1 plus an exponential with rate 2) gave average cost 25.41 pence, matching $22.5 + 5/(\mathrm{e} - 1) = 25.4099$.

Common slips

- Taking the number of coins to be $\lfloor 2T \rfloor$ rather than $1 + \lfloor 2T \rfloor$, forgetting the coin at the start; this loses exactly 10 pence from the answer.
- Using the $t \geq 1$ formula for F at $t = \frac{1}{2}$: the intervals $[0, \frac{1}{2})$ and $[\frac{1}{2}, 1)$ must be handled with the uniform piece.

Beyond STEP

Notice that the tail of T is memoryless: given $T \geq 1$, the remaining time $T - 1$ is exponential with rate 2, which is why the probabilities for $n \geq 2$ form a geometric

sequence. Writing $N = \lfloor 2T \rfloor$, the tail-sum formula $\mathrm{E}(N) = \sum_{n \geq 1} \mathrm{P}(N \geq n)$ gives the same answer even faster. Mixtures of discrete and continuous behaviour like this are treated carefully in Grimmett and Welsh, *Probability: An Introduction*, and Blitzstein and Hwang, *Introduction to Probability*, has many more examples of the memoryless property.

WORKED SOLUTION

STEP II Specimen paper, Question 1

Pure • Partial Fractions

STEP II Specimen paper, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q1

The key idea

Before reaching for the usual partial-fractions machinery, look at the numerator: it is exactly the derivative of the denominator. Since the denominator factorises as a product of linear factors, f is a logarithmic derivative, and the logarithmic derivative of a product is the sum of the logarithmic derivatives of the factors. For the second part, the equation becomes a sum of three unit fractions equal to 1, and size alone forces x to be tiny.

Solution

The partial fractions

The denominator factorises:

$$D(x) = x^3 + (a+b)x^2 + abx = x\big(x^2 + (a+b)x + ab\big) = x(x+a)(x+b),$$

and differentiating gives

$$D'(x) = 3x^2 + 2(a+b)x + ab,$$

which is precisely the numerator. So $\mathrm{f}(x) = D'(x)/D(x)$.

General case, $a \neq b$. The three factors x, $x+a$, $x+b$ are distinct (they differ because $a \neq 0$, $b \neq 0$ and $a \neq b$), so we expect three simple partial fractions. Using the product rule,

$$D'(x) = (x+a)(x+b) + x(x+b) + x(x+a),$$

and dividing by $D(x) = x(x+a)(x+b)$ term by term gives

$$\boxed{\mathrm{f}(x) = \frac{1}{x} + \frac{1}{x+a} + \frac{1}{x+b}} \qquad (a \neq b).$$

(Equivalently, $\mathrm{f}(x) = \dfrac{\mathrm{d}}{\mathrm{d}x}\ln|x(x+a)(x+b)|$, and the logarithm of a product splits into a sum.) Of course the cover-up rule gives the same thing: at $x = 0$ the residue is $ab/(ab) =$

1, and at $x = -a$ it is $\big(3a^2 - 2a^2 - 2ab + ab\big)\,/\,\big((-a)(b-a)\big) = a(a-b)/\big(a(a-b)\big) = 1$, and similarly at $x = -b$.

Ask yourself What goes wrong with this form when $a = b$?

Special case, $a = b$. Now the denominator is $x(x+a)^2$, with a repeated factor, so the standard form would be $\frac{A}{x} + \frac{B}{x+a} + \frac{C}{(x+a)^2}$. But the numerator also has a factor $x + a$:

$$3x^2 + 4ax + a^2 = (3x + a)(x + a),$$

so it cancels, leaving

$$\mathrm{f}(x) = \frac{3x+a}{x(x+a)} = \boxed{\frac{1}{x} + \frac{2}{x+a}} \qquad (a = b),$$

where the last step is the cover-up rule ($x = 0$ gives $a/a = 1$; $x = -a$ gives $-2a/(-a) = 2$). There is no $(x+a)^{-2}$ term at all, so $C = 0$. This agrees with the general formula with b replaced by a, but it is worth noticing that the repeated factor has disappeared rather than producing a squared term.

Since a and b are non-zero, the factor x can never coincide with $x + a$ or $x + b$, so these are the only cases.

When $\mathrm{f}(x) = 1$

Now let x, a, b be positive integers. In both cases above (the $a = b$ formula is just the general one with $b = a$),

$$\mathrm{f}(x) = \frac{1}{x} + \frac{1}{x+a} + \frac{1}{x+b},$$

and all three terms are positive. Because $a, b \geq 1$, the last two terms are each strictly less than $\frac{1}{x}$.

Ask yourself How large can x be if three fractions, each at most $\frac{1}{x}$, add up to 1?

- If $x = 1$, then $\mathrm{f}(x) = 1 + (\text{positive}) > 1$.
- If $x \geq 4$, then $\mathrm{f}(x) < \frac{3}{x} \leq \frac{3}{4} < 1$.
- If $x = 3$, then $x + a \geq 4$ and $x + b \geq 4$, so $\mathrm{f}(x) \leq \frac{1}{3} + \frac{1}{4} + \frac{1}{4} = \frac{5}{6} < 1$.

So $\mathrm{f}(x) = 1$ forces $x = 2$, and then we need

$$\frac{1}{2+a} + \frac{1}{2+b} = \frac{1}{2}.$$

Multiplying by $2(2+a)(2+b)$ gives $2(2+b) + 2(2+a) = (2+a)(2+b)$, that is $8 + 2a + 2b = 4 + 2a + 2b + ab$, so

$$ab = 4.$$

With a, b positive integers the only possibilities are $(a,b) = (1,4)$, $(2,2)$, $(4,1)$. Hence $\mathrm{f}(x) = 1$ exactly for

$$(x,a,b) = (2,1,4), \quad (2,2,2), \quad (2,4,1),$$

which is a finite number (three) of values, as required. Note that $(2,2,2)$ is the special case $a = b$: $\frac{1}{2} + \frac{2}{4} = 1$.

Checks

The three solutions are the familiar unit-fraction identities $\frac{1}{2} + \frac{1}{3} + \frac{1}{6} = 1$ and $\frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1$. Numerically, the partial-fraction forms were compared with the original f in exact rational arithmetic at 2000 random rational points (for both $a \neq b$ and $a = b$), and a brute-force search over all $1 \leq x, a, b < 200$ found exactly the three triples above and no others.

Common slips

- Ploughing into the $a = b$ case with a $\frac{C}{(x+a)^2}$ term and not noticing the cancellation; you should find $C = 0$, and spotting the factor $(x + a)$ in the numerator saves the work.
- Forgetting to rule out $x = 1$ and $x = 3$; the crude bound $\mathrm{f} < 3/x$ only disposes of $x \geq 4$.
- Stopping at "$ab = 4$" without listing the solutions, or listing only $(1,4)$ and forgetting $(4,1)$ and $(2,2)$.

Beyond STEP

The identity $D'/D = \sum 1/(x - r_i)$, where the r_i are the roots of a polynomial D (repeated roots counted with multiplicity), is a workhorse: it underlies Newton's identities for power sums of roots and, in complex analysis, the argument principle, which counts the zeros of a function inside a curve by integrating f'/f around it. Equations like $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1$ in positive integers are Egyptian-fraction problems, a classic corner of Diophantine equations where bounding arguments like ours do all the work. Needham's *Visual Complex Analysis* gives a beautiful geometric account of the argument principle, and Davenport's *The Higher Arithmetic* is a good place to start on Diophantine equations.

WORKED SOLUTION

STEP II Specimen paper, Question 2

Pure • Symmetric Simultaneous Equations

STEP II Specimen paper, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q2

The key idea

The three equations are cyclic, so subtracting them in pairs is natural, and each difference factorises with $x+y+z$ as a common factor. The sum of squared differences is then useful twice over: in terms of x, y, z it is a multiple of $a+b+c$, while the first result expresses it in terms of $a-b$, $b-c$, $c-a$ and $x+y+z$. Once we know $x+y+z$ and the pairwise differences, three linear equations finish the job.

Solution

The equal ratios

Subtract the second equation from the first:

$$\begin{aligned} a-b &= x^2-y^2-yz+zx \\ &= (x-y)(x+y)+z(x-y) \\ &= (x-y)(x+y+z). \end{aligned}$$

In exactly the same way (or by cycling $x \to y \to z \to x$, $a \to b \to c \to a$),

$$b-c=(y-z)(x+y+z), \qquad c-a=(z-x)(x+y+z).$$

Write $S=x+y+z$. Since $a \neq b$, the product $(x-y)S$ is non-zero, so $S \neq 0$ (and likewise x, y, z are distinct). As a, b, c are distinct we may divide by $a-b$, $b-c$, $c-a$ and by S, giving

$$\frac{y-z}{b-c}=\frac{z-x}{c-a}=\frac{x-y}{a-b}=\frac{1}{x+y+z},$$

as required.

Finding $x+y+z$

Expand the suggested sum:

$$\begin{aligned}(y-z)^2+(z-x)^2+(x-y)^2 &= 2\big(x^2+y^2+z^2-xy-yz-zx\big)\\ &= 2\big[(x^2-yz)+(y^2-zx)+(z^2-xy)\big]\\ &= 2(a+b+c).\end{aligned}$$

On the other hand, the first part gives $y-z=(b-c)/S$ and so on, so the same sum equals

$$\frac{(b-c)^2+(c-a)^2+(a-b)^2}{S^2}=\frac{2\big(a^2+b^2+c^2-bc-ca-ab\big)}{S^2}.$$

Equating the two expressions and dividing by $2(a+b+c)$, which is positive because a, b, c are positive,

$$S^2=\frac{a^2+b^2+c^2-bc-ca-ab}{a+b+c}.$$

So $x+y+z=\triangle$ where $\triangle^2$ is this quantity, as required. Note that $\triangle^2>0$, since its numerator is half of $(b-c)^2+(c-a)^2+(a-b)^2$, which is positive for distinct a, b, c. So there are two possibilities, $\triangle=\pm\sqrt{\triangle^2}$, and nothing so far picks one of them.

Solving the equations

We now know

$$x+y+z=\triangle,\qquad x-y=\frac{a-b}{\triangle},\qquad x-z=\frac{a-c}{\triangle}.$$

Adding all three,

$$3x=\triangle+\frac{2a-b-c}{\triangle}=\frac{\triangle^2+2a-b-c}{\triangle}.$$

This simplifies nicely. Put $s=a+b+c$ and $Q=a^2+b^2+c^2-bc-ca-ab$, so $\triangle^2=Q/s$. Then

$$\begin{aligned}s\big(\triangle^2+2a-b-c\big) &= Q+(a+b+c)(2a-b-c)\\ &= \big(a^2+b^2+c^2-bc-ca-ab\big)+\big(2a^2+ab+ca-b^2-c^2-2bc\big)\\ &= 3\big(a^2-bc\big).\end{aligned}$$

Hence $x=\dfrac{a^2-bc}{s\triangle}$, and y, z follow by cycling. Now $(s\triangle)^2=sQ=a^3+b^3+c^3-3abc$, by the standard factorisation

$$a^3+b^3+c^3-3abc=(a+b+c)\big(a^2+b^2+c^2-bc-ca-ab\big).$$

Writing $K=\sqrt{a^3+b^3+c^3-3abc}>0$, we have $s\triangle=\pm K$, and the solutions are

$$(x,y,z)=\pm\frac{1}{K}\big(a^2-bc,\ b^2-ca,\ c^2-ab\big),$$

with the same sign throughout.

Ask yourself We only showed that any solution must be of this form. Are both of these really solutions?

Yes. Everything above was a chain of consequences, so we should substitute back. Since only squares and products of two of the unknowns appear, the overall sign doesn't matter, and

$$\begin{aligned}(a^2-bc)^2-(b^2-ca)(c^2-ab) &= a^4-2a^2bc+b^2c^2-\big(b^2c^2-ab^3-ac^3+a^2bc\big)\\ &= a\big(a^3+b^3+c^3-3abc\big) = aK^2.\end{aligned}$$

Dividing by K^2 gives $x^2-yz=a$; the other two equations follow by cycling. So there are exactly two solutions, one the negative of the other.

Checks

- Take $a=1$, $b=2$, $c=3$. Then $a^3+b^3+c^3-3abc=18$, $K=3\sqrt{2}$, and the solution with the plus sign is $(x,y,z)=\frac{1}{3\sqrt{2}}(-5,1,7)$. For example $x^2-yz=\frac{25-7}{18}=1$, and $x+y+z=\frac{1}{\sqrt{2}}$, whose square is $\frac{1}{2}=\frac{14-11}{6}=\triangle^2$.
- A computer tested 20,000 random triples of distinct positive a, b, c, with both signs: the formulas satisfied all three equations and the equal ratios, with relative error below 10^{-11}.

Common slips

- Taking only the positive square root for $\triangle$: the question's $\triangle$ is defined only through $\triangle^2$, and both signs give genuine solutions.
- Dividing by $x+y+z$ without saying why it is non-zero. Here it follows from $a\neq b$.

Beyond STEP

The identity $a^3+b^3+c^3-3abc=(a+b+c)(a^2+b^2+c^2-ab-bc-ca)$ is the determinant of the circulant matrix with rows (a,b,c), (c,a,b), (b,c,a), and it factorises completely over the complex numbers as $(a+b+c)(a+\omega b+\omega^2 c)(a+\omega^2 b+\omega c)$ with $\omega=e^{2\pi i/3}$. In fact the map $(x,y,z)\mapsto(x^2-yz,\,y^2-zx,\,z^2-xy)$ is the adjugate of a circulant matrix, which explains why it is so easily inverted. Circulant matrices and their eigenvectors are a gentle first meeting with the discrete Fourier transform; Strang's *Introduction to Linear Algebra* is a good place to read more.

WORKED SOLUTION

STEP II Specimen paper, Question 3

Pure • De Moivre's Theorem

STEP II Specimen paper, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q3

The key idea

De Moivre's theorem is a natural induction: multiplying by one more factor of $\cos\theta + \mathrm{i}\sin\theta$ adds θ to the angle, by the compound-angle formulae. For the equations, the pattern $\cos ky$, $\sin ky$ next to x^k shouts "real and imaginary parts". Treat the first equation as the real part and the second as the imaginary part of a single complex equation in $z = x(\cos y + \mathrm{i}\sin y)$, and the pair collapses to a cubic in z with an easy root.

Solution

De Moivre's theorem

We use induction on n. For $n = 1$ both sides are $\cos\theta + \mathrm{i}\sin\theta$, so the result holds.

Suppose it holds for some $n = k \geq 1$. Then

$$\begin{aligned}(\cos\theta + \mathrm{i}\sin\theta)^{k+1} &= (\cos k\theta + \mathrm{i}\sin k\theta)(\cos\theta + \mathrm{i}\sin\theta)\\ &= (\cos k\theta\cos\theta - \sin k\theta\sin\theta) + \mathrm{i}(\sin k\theta\cos\theta + \cos k\theta\sin\theta)\\ &= \cos(k+1)\theta + \mathrm{i}\sin(k+1)\theta,\end{aligned}$$

using the induction hypothesis in the first line, $\mathrm{i}^2 = -1$ in the second, and the compound-angle formulae for $\cos(A+B)$ and $\sin(A+B)$ in the third. So the result holds for $n = k+1$, and by induction it holds for every positive integer n.

Solving the equations

Let x and y be real and put

$$z = x(\cos y + \mathrm{i}\sin y).$$

By de Moivre's theorem, for $n = 1, 2, 3$,

$$z^n = x^n(\cos y + \mathrm{i}\sin y)^n = x^n\cos ny + \mathrm{i}\,x^n\sin ny.$$

(This holds whatever the sign of x: the real factor x simply comes out as x^n.) Since x and y are real, the left-hand sides of the two equations are exactly the real and imaginary parts of z^3+2z^2+2z. So the pair of real equations is equivalent to the single complex equation

$$z^3+2z^2+2z=-1, \qquad \text{that is,} \qquad z^3+2z^2+2z+1=0.$$

Ask yourself Why is one complex equation the same as the two real ones?

Two complex numbers are equal exactly when their real parts agree and their imaginary parts agree. The right-hand side -1 has real part -1 and imaginary part 0, which are the two right-hand sides we were given. So (x,y) satisfies both real equations if and only if z satisfies the cubic.

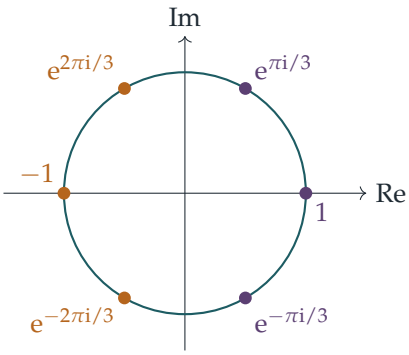
Now $z=-1$ is a root, since $-1+2-2+1=0$, and dividing out gives

$$z^3+2z^2+2z+1=(z+1)(z^2+z+1).$$

The quadratic has roots $z=\frac{1}{2}(-1\pm\mathrm{i}\sqrt{3})$. So the three roots are

$$z=-1=\cos\pi+\mathrm{i}\sin\pi, \qquad z=\cos\tfrac{2\pi}{3}\pm\mathrm{i}\sin\tfrac{2\pi}{3},$$

all on the unit circle.



It remains to turn each root back into real pairs (x,y). Since $|z|=|x|$, every solution has $|x|=1$; in particular $x=0$ is impossible (it would give $0=-1$).

- If $x=1$, then $\cos y+\mathrm{i}\sin y=z$, so y is an argument of one of the three roots: $y=\pi$, $\frac{2\pi}{3}$ or $-\frac{2\pi}{3}$, plus any multiple of 2π.
- If $x=-1$, then $\cos y+\mathrm{i}\sin y=-z$, which is one of 1, $\cos\frac{\pi}{3}\mp\mathrm{i}\sin\frac{\pi}{3}$ (the plum points in the figure). So $y=0$, $-\frac{\pi}{3}$ or $\frac{\pi}{3}$, plus any multiple of 2π.

Each of these gives a root of the cubic, so all of them work. With k any integer, the solutions are

$$x=1,\quad y=\pi+2k\pi \text{ or } y=\pm\tfrac{2\pi}{3}+2k\pi; \qquad x=-1,\quad y=2k\pi \text{ or } y=\pm\tfrac{\pi}{3}+2k\pi.$$

In $[0,2\pi)$ that is six pairs: $(1,\frac{2\pi}{3})$, $(1,\pi)$, $(1,\frac{4\pi}{3})$, $(-1,0)$, $(-1,\frac{\pi}{3})$, $(-1,\frac{5\pi}{3})$.

Checks

Try $x = -1$, $y = 0$: the first equation gives $-1 + 2 - 2 = -1$ and the second is $0 = 0$. Try $x = 1$, $y = \frac{2\pi}{3}$: the cosines are $1, -\frac{1}{2}, -\frac{1}{2}$, giving $1 - 1 - 1 = -1$; the sines are $0, -\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}$, giving $0 - \sqrt{3} + \sqrt{3} = 0$. A computer search over a fine grid of $x \in [-3, 3]$, $y \in [0, 2\pi)$ finds near-solutions only at the six points above (to rounding), so none has been missed.

Common slips

- Assuming x is a modulus and so $x \geq 0$. The question allows any real x, and the $x = -1$ solutions are just as valid; they are the same points z described with y shifted by π.
- Forgetting the $+2k\pi$: y is only determined modulo 2π, so there are infinitely many pairs (x, y).

Beyond STEP

Packaging two real equations as one complex one is a standard move: the real and imaginary parts of a polynomial in $z = re^{i\theta}$ satisfy exactly this kind of trigonometric system, and the factorisation $z^3 + 2z^2 + 2z + 1 = (z + 1)(z^2 + z + 1)$ places the roots on the unit circle. In a first course you will meet $e^{i\theta}$ defined by a power series, which makes de Moivre's theorem hold for every integer (and gives a meaning to non-integer powers, with care over branches). Needham's *Visual Complex Analysis* is a beautiful geometric account of all this.

WORKED SOLUTION

STEP II Specimen paper, Question 4

Pure • Hyperbolic Functions

STEP II Specimen paper, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q4

The key idea

Every one of these identities is a statement about a single angle (or a single hyperbolic "angle"). Give the right-hand side a name, read off one function of it, and use the Pythagorean identities $1 + \tan^2\theta = \sec^2\theta$ and $1 + \sinh^2 u = \cosh^2 u$ to get the other. The only real care needed is with signs of square roots, which is exactly where the ranges of principal values come in — and, in the final part, where the identity stops being true.

Solution

Part (i)

Let $\theta = \tan^{-1}(\sinh x)$, so that $\tan\theta = \sinh x$ and, being a principal value, $-\frac{1}{2}\pi < \theta < \frac{1}{2}\pi$. Then

$$\sec^2\theta = 1 + \tan^2\theta = 1 + \sinh^2 x = \cosh^2 x.$$

On $(-\frac{1}{2}\pi, \frac{1}{2}\pi)$ we have $\cos\theta > 0$, so $\sec\theta > 0$; and $\cosh x > 0$ always. Taking the positive square roots, $\sec\theta = \cosh x$, and therefore

$$\sin\theta = \tan\theta\cos\theta = \frac{\sinh x}{\cosh x} = \tanh x.$$

Since θ lies in $[-\frac{1}{2}\pi, \frac{1}{2}\pi]$, the range of principal values of $\sin^{-1}$, this says precisely that $\theta = \sin^{-1}(\tanh x)$. Hence $\sin^{-1}(\tanh x) = \tan^{-1}(\sinh x)$ for every real x, as required.

? **Ask yourself** Why isn't $\sin\theta = \tanh x$ already the end of the argument?

Because $\sin\theta = \tanh x$ has infinitely many solutions θ. It is only because our θ lies in $[-\frac{1}{2}\pi, \frac{1}{2}\pi]$, where sin is one-to-one, that we may conclude θ *is* the principal value $\sin^{-1}(\tanh x)$. The same range also fixed the sign of $\sec\theta$.

Part (ii)

Let $u = \sinh^{-1}(\tan y)$, which is defined because $\tan y$ is finite for $-\frac{1}{2}\pi < y < \frac{1}{2}\pi$. Then $\sinh u = \tan y$, and part (i) with $x = u$ gives

$$\sin^{-1}(\tanh u) = \tan^{-1}(\sinh u) = \tan^{-1}(\tan y) = y,$$

where the last step uses $-\frac{1}{2}\pi < y < \frac{1}{2}\pi$, the range of principal values of $\tan^{-1}$. Taking sin of both sides, $\tanh u = \sin y$. As $|\sin y| < 1$ on this interval, $\tanh^{-1}(\sin y)$ is defined, and tanh is one-to-one, so $u = \tanh^{-1}(\sin y)$. That is,

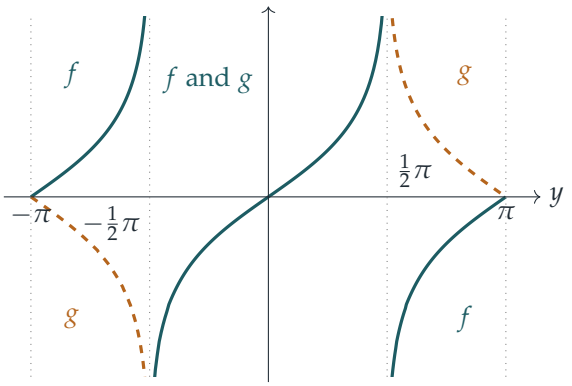
$$\sinh^{-1}(\tan y) = \tanh^{-1}(\sin y) \qquad \left(-\tfrac{1}{2}\pi < y < \tfrac{1}{2}\pi\right).$$

(Directly: $\cosh u = \sqrt{1+\tan^2 y} = \sec y$, positive because $\cos y > 0$, so $\tanh u = \tan y \cos y = \sin y$.)

The sketch Write $f(y) = \sinh^{-1}(\tan y)$ and $g(y) = \tanh^{-1}(\sin y)$. Both are undefined at $y = \pm\frac{1}{2}\pi$, where $\tan y$ is infinite and $\sin y = \pm 1$. Their behaviour elsewhere follows from the symmetries of tan and sin.

- Both are odd, and both vanish at $y = 0$ and $y = \pm\pi$.
- f has period π, because tan does. So on $(\frac{1}{2}\pi, \pi)$ it repeats its values on $(-\frac{1}{2}\pi, 0)$: it rises from $-\infty$ to 0.
- $\sin(\pi - y) = \sin y$, so g is symmetric about $y = \frac{1}{2}\pi$. On $(\frac{1}{2}\pi, \pi)$ it repeats its values on $(0, \frac{1}{2}\pi)$ in mirror image: it falls from $+\infty$ to 0.
- Differentiating, $f'(y) = \dfrac{\sec^2 y}{\sqrt{1+\tan^2 y}} = |\sec y|$ and $g'(y) = \dfrac{\cos y}{1-\sin^2 y} = \sec y$. So f is increasing on every interval where it is defined, while g increases on $(-\frac{1}{2}\pi, \frac{1}{2}\pi)$ and decreases outside it. Both have gradient 1 at the origin; at $y = \pi$, f has gradient 1 and g has gradient -1.

On $(-\frac{1}{2}\pi, \frac{1}{2}\pi)$ the two graphs are the same curve; on the outer intervals each is the reflection of the other in the horizontal axis.



Solid: $f(y) = \sinh^{-1}(\tan y)$. Dashed: $g(y) = \tanh^{-1}(\sin y)$ where it differs from f. The dotted lines $y = \pm\frac{1}{2}\pi$ are asymptotes.

The relationship for $\frac{1}{2}\pi < y < \pi$ The sketch suggests $f = -g$ there, and we can prove it. If $\frac{1}{2}\pi < y < \pi$ then $y - \pi$ lies in $(-\frac{1}{2}\pi, 0)$, so the identity just proved applies to $y - \pi$:

$$\begin{aligned}\sinh^{-1}(\tan y) &= \sinh^{-1}\big(\tan(y-\pi)\big)\\ &= \tanh^{-1}\big(\sin(y-\pi)\big)\\ &= \tanh^{-1}(-\sin y) = -\tanh^{-1}(\sin y),\end{aligned}$$

using $\tan(y-\pi) = \tan y$, $\sin(y-\pi) = -\sin y$ and the fact that $\tanh^{-1}$ is odd. So for $\frac{1}{2}\pi < y < \pi$,

$$\sinh^{-1}(\tan y) = -\tanh^{-1}(\sin y).$$

(By oddness the same relation holds on $-\pi < y < -\frac{1}{2}\pi$.)

Ask yourself Where does the argument of part (ii) break down for $\frac{1}{2}\pi < y < \pi$?

At the step $\tan^{-1}(\tan y) = y$: for y outside $(-\frac{1}{2}\pi, \frac{1}{2}\pi)$ the principal value is $y - \pi$, not y. Equivalently, in the direct argument $\sqrt{1+\tan^2 y} = |\sec y| = -\sec y$ there, because $\cos y < 0$, and that minus sign is exactly the one in the answer.

Checks

- At $y = 1$: $\sinh^{-1}(\tan 1) = \tanh^{-1}(\sin 1) \approx 1.2262$. At $y = 2$: $\sinh^{-1}(\tan 2) \approx -1.5235$ while $\tanh^{-1}(\sin 2) \approx 1.5235$, confirming the change of sign.
- Signs: for $\frac{1}{2}\pi < y < \pi$, $\tan y < 0$ but $\sin y > 0$, so the two expressions must have opposite signs; they cannot be equal.
- Part (i) at large x: both sides tend to $\frac{1}{2}\pi$, as they should.

We tested both identities at 10^5 random points, and the relation $f = -g$ on $\frac{1}{2}\pi < |y| < \pi$, numerically.

Common slips

- Writing $\sqrt{1+\tan^2 y} = \sec y$ without checking the sign of $\sec y$. This is the whole point of the final part.
- Concluding $\theta = \sin^{-1}(\tanh x)$ from $\sin\theta = \tanh x$ without saying why θ is the principal value.
- Drawing g with the same shape as f beyond $\frac{1}{2}\pi$: g inherits the symmetry of sin about $\frac{1}{2}\pi$, not the periodicity of tan.

Beyond STEP

The function $\mathrm{gd}(x) = \tan^{-1}(\sinh x) = \sin^{-1}(\tanh x)$ is the *Gudermannian* function. It links circular and hyperbolic functions without complex numbers, and part (ii) shows its inverse is $\sinh^{-1}(\tan y)$. It appears in the Mercator map projection, where it converts the vertical map coordinate into latitude. Inverse functions and their branches, which tripped up the identity here, become central when you study $\log z$ and $z^{1/2}$ for complex z. Needham's *Visual Complex Analysis* is a beautiful place to see this.

WORKED SOLUTION

STEP II Specimen paper, Question 5

Pure • Series and Integral Bounds

STEP II Specimen paper, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q5

The key idea

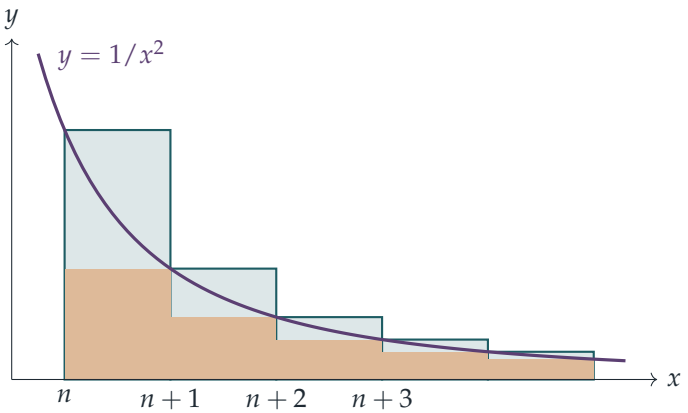
A sum of values of a decreasing function can be pictured as a row of unit-width rectangles, and the integral as the area under the curve. Putting the rectangles to the right of each sample point makes them stick out above the curve; putting them to the left tucks them underneath. That gives the sandwich, and everything after it is bookkeeping: the tail of the series is trapped between two simple fractions, so adding a guess for the tail improves the estimate enormously.

Solution

The sandwich

Let $f(x) = 1/x^2$, which is positive and strictly decreasing for $x > 0$. We also need

$$\int_n^\infty \frac{\mathrm{d}x}{x^2} = \lim_{M\to\infty}\left[-\frac{1}{x}\right]_n^M = \frac{1}{n}.$$



Upper rectangles (teal outlines). On $[r, r+1]$ put a rectangle of height $f(r) = 1/r^2$. Since

f is decreasing, $f(x) < f(r)$ for $r < x \leq r+1$, so this rectangle has area $1/r^2$ strictly greater than $\int_r^{r+1} f(x)\,\mathrm{d}x$. Adding these for $r = n, n+1, \ldots$ gives

$$\sum_{r=n}^{\infty} \frac{1}{r^2} > \int_n^{\infty} \frac{\mathrm{d}x}{x^2}.$$

Lower rectangles (amber). On $[r-1, r]$ put a rectangle of height $f(r) = 1/r^2$. Now $f(x) > f(r)$ for $r-1 \leq x < r$, so this rectangle lies under the curve, and its area $1/r^2$ is strictly less than $\int_{r-1}^{r} f(x)\,\mathrm{d}x$. Adding for $r = n+1, n+2, \ldots$, whose intervals fill $[n, \infty)$,

$$\int_n^{\infty} \frac{\mathrm{d}x}{x^2} > \sum_{r=n+1}^{\infty} \frac{1}{r^2}.$$

(All three quantities are finite: the integral is $1/n$ and the sums are bounded by it.) This is the required pair of inequalities, and for the record they say

$$\sum_{r=n}^{\infty} \frac{1}{r^2} > \frac{1}{n} > \sum_{r=n+1}^{\infty} \frac{1}{r^2}. \qquad (*)$$

Trapping the tail

Write $S_n = \sum_{r=1}^{n} 1/r^2$. Then $A - S_n$ is exactly the tail $T = \sum_{r=n+1}^{\infty} 1/r^2$.

The right-hand inequality of $(*)$ gives $T < 1/n$ directly.

> **Ask yourself** The left-hand inequality of $(*)$ starts its sum at $r = n$, but the tail starts at $r = n+1$. How do we get a lower bound on T?

The inequality $(*)$ holds for every positive integer, so apply its left-hand half with $n+1$ in place of n:

$$T = \sum_{r=n+1}^{\infty} \frac{1}{r^2} > \int_{n+1}^{\infty} \frac{\mathrm{d}x}{x^2} = \frac{1}{n+1}.$$

Putting the two together,

$$\frac{1}{n} > A - \sum_{r=1}^{n} \frac{1}{r^2} > \frac{1}{n+1},$$

as required.

The smallest n

All terms are positive, so $S_n < A$ and the error is $A - S_n$.

- If $n \leq 9999$, then $n+1 \leq 10^4$, so $A - S_n > \dfrac{1}{n+1} \geq 10^{-4}$: the error is too big.
- If $n = 10^4$, then $A - S_n < \dfrac{1}{n} = 10^{-4}$: the error is small enough.

So the smallest value is $n = 10\,000$. Notice that the strictness of both inequalities matters here: at $n = 9999$ the lower bound is exactly 10^{-4}, and at $n = 10^4$ the upper bound is exactly 10^{-4}.

The improved estimate

Subtracting $\frac{1}{n+1}$ throughout the trapping inequality,

$$\frac{1}{n} - \frac{1}{n+1} > A - \left(\frac{1}{n+1} + \sum_{r=1}^{n} \frac{1}{r^2}\right) > 0.$$

So $\frac{1}{n+1} + S_n$ still underestimates A, by less than

$$\frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)}.$$

For $n = 10^4$ this is $\dfrac{1}{10^4 \times 10\,001} < \dfrac{1}{10^4 \times 10^4} = 10^{-8}$, so the improved estimate approximates A with an error of less than 10^{-8}, as required.

The extra term $1/(n+1)$ is a guess at the tail, and since the tail is pinned into an interval of width only $1/(n(n+1))$, the guess cannot be far wrong.

Checks

- Small case: with $n = 1$, $S_1 = 1$ and $A = \pi^2/6 \approx 1.6449$, so $A - S_1 \approx 0.645$, which does lie between $\frac{1}{2}$ and 1.
- Using $A = \pi^2/6$ and summing numerically: $A - S_{9999} \approx 1.00005 \times 10^{-4}$ (just too big) and $A - S_{10000} \approx 0.99995 \times 10^{-4}$ (just small enough), confirming $n = 10\,000$.
- For $n = 10^4$ the improved estimate is out by about 5.0×10^{-9}, below the bound $1/(n(n+1)) \approx 1.0 \times 10^{-8}$; it is roughly half the bound, as the picture suggests (the curve cuts each rectangle's overhang roughly in half).

Common slips

- Using the left-hand inequality at n rather than $n+1$, which gives a lower bound on $\sum_{r=n}^{\infty}$, not on the tail $A - S_n$.
- Answering $n = 9999$ or $n = 10\,001$ by treating a strict inequality as non-strict, or by using only one of the two bounds. You need the lower bound to rule out $n \leq 9999$ and the upper bound to accept $n = 10^4$.
- Drawing the rectangles on the wrong side of each sample point, which reverses the inequalities.

Beyond STEP

This is the integral test for convergence, together with its quantitative bonus: the same rectangles that prove a series converges also estimate its tail. Refining the idea, by comparing the sum with the integral and correcting with derivatives at the ends, leads to the Euler–Maclaurin formula, which here predicts $A - S_n \approx \frac{1}{n} - \frac{1}{2n^2} + \frac{1}{6n^3}$ and was how Euler computed $\sum 1/r^2$ to many decimal places before he proved it equals $\pi^2/6$. Abbott's *Understanding Analysis* treats series and their convergence tests carefully.

WORKED SOLUTION

STEP II Specimen paper, Question 6

Pure • Addition Formulae from ODEs

STEP II Specimen paper, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q6

The key idea

We are not allowed to say "f is e^x", so everything has to come from the differential equations themselves. The one tool we may use is that a function whose derivative is zero everywhere is constant. So the trick is to build an expression whose derivative cancels to zero, and then pin down the constant by putting $x = 0$. Pairing the value at $a - x$ with the value at x is natural because the two arguments always add up to a, which is exactly the shape of an addition formula.

Solution

The exponential addition law

Fix a and let $p(x) = \mathrm{f}(a - x)\,\mathrm{f}(x)$. By the chain rule, $\frac{\mathrm{d}}{\mathrm{d}x}[\mathrm{f}(a - x)] = -\mathrm{f}'(a - x) = -\mathrm{f}(a - x)$, using $\mathrm{f}' = \mathrm{f}$. So by the product rule

$$\begin{aligned} p'(x) &= -\mathrm{f}(a - x)\,\mathrm{f}(x) + \mathrm{f}(a - x)\,\mathrm{f}'(x) \\ &= -\mathrm{f}(a - x)\,\mathrm{f}(x) + \mathrm{f}(a - x)\,\mathrm{f}(x) = 0, \end{aligned}$$

which is the first result.

Since $p'(x) = 0$ for every x, the function p is constant. Its value at $x = 0$ is $\mathrm{f}(a)\,\mathrm{f}(0) = \mathrm{f}(a)$, since $\mathrm{f}(0) = 1$. Hence

$$\mathrm{f}(a - x)\,\mathrm{f}(x) = \mathrm{f}(a) \qquad \text{for all } x \text{ and all } a.$$

Now take any x and y and choose $a = x + y$. Then $a - x = y$, and the identity becomes

$$\mathrm{f}(y)\,\mathrm{f}(x) = \mathrm{f}(x + y),$$

as required.

The cosine addition formula

We copy the same strategy, so we need a combination of c and s, evaluated at $a - x$ and at x, whose derivative vanishes.

> **Ask yourself** Which combination should we differentiate?

Let the answer guide us: with $a = x + y$ we want $\mathrm{c}(x)\mathrm{c}(y) - \mathrm{s}(x)\mathrm{s}(y)$, that is, $\mathrm{c}(a - x)\mathrm{c}(x) - \mathrm{s}(a - x)\mathrm{s}(x)$. So that is the function to try.

Fix a and let

$$g(x) = \mathrm{c}(a - x)\,\mathrm{c}(x) - \mathrm{s}(a - x)\,\mathrm{s}(x).$$

By the chain rule and the given derivatives,

$$\frac{\mathrm{d}}{\mathrm{d}x}[\mathrm{c}(a - x)] = -\mathrm{c}'(a - x) = \mathrm{s}(a - x), \qquad \frac{\mathrm{d}}{\mathrm{d}x}[\mathrm{s}(a - x)] = -\mathrm{s}'(a - x) = -\mathrm{c}(a - x).$$

Hence, by the product rule,

$$\begin{aligned} g'(x) &= \big[\mathrm{s}(a - x)\,\mathrm{c}(x) - \mathrm{c}(a - x)\,\mathrm{s}(x)\big] - \big[-\mathrm{c}(a - x)\,\mathrm{s}(x) + \mathrm{s}(a - x)\,\mathrm{c}(x)\big] \\ &= 0. \end{aligned}$$

So g is constant, and its value is

$$g(0) = \mathrm{c}(a)\,\mathrm{c}(0) - \mathrm{s}(a)\,\mathrm{s}(0) = \mathrm{c}(a) \cdot 1 - \mathrm{s}(a) \cdot 0 = \mathrm{c}(a).$$

Therefore $\mathrm{c}(a - x)\mathrm{c}(x) - \mathrm{s}(a - x)\mathrm{s}(x) = \mathrm{c}(a)$ for all x and a. Given any x and y, put $a = x + y$, so that $a - x = y$:

$$\mathrm{c}(x + y) = \mathrm{c}(x)\,\mathrm{c}(y) - \mathrm{s}(x)\,\mathrm{s}(y),$$

which is the required result.

The same method with $h(x) = \mathrm{s}(a - x)\,\mathrm{c}(x) + \mathrm{c}(a - x)\,\mathrm{s}(x)$ gives $h' = 0$ and $h(0) = \mathrm{s}(a)$, so $\mathrm{s}(x + y) = \mathrm{s}(x)\mathrm{c}(y) + \mathrm{c}(x)\mathrm{s}(y)$ too.

Checks

- Put $y = 0$: the results reduce to $\mathrm{f}(x) = \mathrm{f}(x)$ and $\mathrm{c}(x) = \mathrm{c}(x)$, using $\mathrm{f}(0) = \mathrm{c}(0) = 1$ and $\mathrm{s}(0) = 0$.
- Put $y = -x$ in the cosine result: $1 = \mathrm{c}(0) = \mathrm{c}(x)\mathrm{c}(-x) - \mathrm{s}(x)\mathrm{s}(-x)$, which is $\cos^2 x + \sin^2 x = 1$ in disguise.
- We solved the differential equations numerically (a Runge–Kutta method, with no built-in exponential or trigonometric functions) and tested both identities at 60 random pairs in $[-3, 3]$; for example, with $x = 0.7$ and $y = 1.1$ both sides of the cosine result are -0.2272 to four decimal places. All differences were below 10^{-11}.

Common slips

- Forgetting the minus sign from the chain rule when differentiating $\mathrm{f}(a-x)$, $\mathrm{c}(a-x)$ or $\mathrm{s}(a-x)$.
- Stopping at "the derivative is zero" without evaluating the constant at $x=0$ and then choosing $a=x+y$.
- Quoting $\cos(x+y)$ or $\mathrm{e}^{x+y}=\mathrm{e}^{x}\mathrm{e}^{y}$, which the question explicitly forbids.

Beyond STEP

At university this is the honest way to define the exponential and trigonometric functions: as solutions of differential equations (or as power series), with every familiar property deduced from that definition. The fact we used, that zero derivative everywhere forces a function to be constant, is a consequence of the Mean Value Theorem, and it also shows these solutions are unique. Abbott's *Understanding Analysis* builds the theory carefully.

WORKED SOLUTION

STEP II Specimen paper, Question 7

Pure • Integrals of Step Functions

STEP II Specimen paper, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q7

The key idea

The integrand $[\mathrm{e}^x]$ is a step function: it is constant on each interval where e^x stays between two consecutive integers. So we find those intervals, integrate a constant over each one, and add up. The sum that comes out looks awkward, but most of its logarithms cancel in pairs once we line up equal terms.

Solution

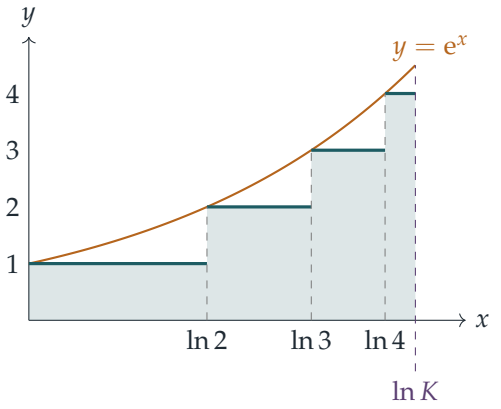
Where the integrand is constant

We take $K \geq 1$, so that $\ln K \geq 0$ and $N = [K] \geq 1$. (If $0 < K < 1$, then for x between $\ln K$ and 0 we have $0 < \mathrm{e}^x \leq 1$, and $[\mathrm{e}^x] = 0$ except at the single point $x = 0$, so $I = 0$; also $N = 0$ and $N \ln K - \ln 0! = 0$. The result holds there too.)

For an integer $k \geq 1$,

$$k \leq \mathrm{e}^x < k+1 \iff \ln k \leq x < \ln(k+1),$$

because ln is increasing. So $[\mathrm{e}^x] = k$ on $[\ln k, \ln(k+1))$. On the range of integration $0 \leq x \leq \ln K$ we have $1 \leq \mathrm{e}^x \leq K$, so $[\mathrm{e}^x]$ takes the values $1, 2, \ldots, N$, and the last value N holds on $[\ln N, \ln K]$, since $N \leq \mathrm{e}^x \leq K < N+1$ there.



The shaded area is I (drawn with $K = 4.5$, so $N = 4$).

Adding up the pieces

Integrating the constant on each interval,

$$I = \sum_{k=1}^{N-1} k\big(\ln(k+1) - \ln k\big) + N\big(\ln K - \ln N\big).$$

(When $N = 1$ the sum is empty; when $K = N$ the last term is zero. Neither case needs separate treatment.)

> **Ask yourself** How do we simplify $\sum k\big(\ln(k+1) - \ln k\big)$? It doesn't telescope as it stands.

Split it into two sums and shift the index in the first, putting $j = k + 1$:

$$\sum_{k=1}^{N-1} k\ln(k+1) = \sum_{j=2}^{N}(j-1)\ln j.$$

Now both sums are sums of multiples of $\ln j$, and we can collect the coefficient of each $\ln j$. Using $\ln 1 = 0$,

$$\begin{aligned}\sum_{k=1}^{N-1} k\big(\ln(k+1) - \ln k\big) &= \sum_{j=2}^{N}(j-1)\ln j - \sum_{j=2}^{N-1} j\ln j \\ &= (N-1)\ln N + \sum_{j=2}^{N-1}\big((j-1) - j\big)\ln j \\ &= (N-1)\ln N - \sum_{j=2}^{N-1}\ln j.\end{aligned}$$

Adding the final piece $N \ln K - N \ln N$:

$$\begin{aligned} I &= N \ln K - \ln N - \sum_{j=2}^{N-1} \ln j \\ &= N \ln K - \sum_{j=1}^{N} \ln j \\ &= N \ln K - \ln(N!), \end{aligned}$$

as required.

A slicker view

Instead of slicing the shaded region into vertical strips, slice it into horizontal layers. For $1 \le k \le N$, the layer between heights $k-1$ and k is present exactly where $[\mathrm{e}^x] \ge k$, that is, where $\mathrm{e}^x \ge k$, i.e. for $\ln k \le x \le \ln K$. That layer has area $\ln K - \ln k$, so

$$I = \sum_{k=1}^{N} (\ln K - \ln k) = N \ln K - \ln(N!)$$

at once. Formally this uses $[y] = \sum_{k=1}^{N} \mathbf{1}[y \ge k]$ for $1 \le y < N+1$, and then integrates term by term. It is the same rearrangement as the index shift above, done geometrically.

Checks

- $K = 1$: the range of integration has length zero and the formula gives $1 \cdot 0 - \ln 1 = 0$.
- $K = \mathrm{e}$: here $N = 2$, and directly $I = \ln 2 + 2(1 - \ln 2) = 2 - \ln 2 \approx 1.3069$, matching $2 \ln \mathrm{e} - \ln 2$.
- For $K = 10$ the formula gives $10 \ln 10 - \ln(10!) \approx 7.9214$, and midpoint quadrature of $[\mathrm{e}^x]$ over $[0, \ln 10]$ agrees to 5×10^{-6}. Random values of K in $[1, 40]$ also agree.

Common slips

- Running the sum up to N with the interval $[\ln N, \ln(N+1)]$. The top of the range is $\ln K$, not $\ln(N+1)$, so the last piece must be handled separately.
- Getting the end terms wrong when re-indexing: write out the sums for $N = 3$ to check.

Beyond STEP

Swapping $\sum k(a_{k+1} - a_k)$ for $-\sum a_k$ plus boundary terms is *summation by parts* (Abel summation), the discrete analogue of integration by parts. Taking $K = N$ and using $\mathrm{e}^x - 1 < [\mathrm{e}^x] \le \mathrm{e}^x$ gives $N \ln N - N + 1 \le \ln(N!) \le N \ln N - N + 1 + \ln N$, a first step

towards Stirling's approximation. The same idea, written as a Riemann–Stieltjes integral $\int f \, \mathrm{d}[t]$, turns sums into integrals throughout analytic number theory; see Apostol, *Introduction to Analytic Number Theory*.

WORKED SOLUTION

STEP II Specimen paper, Question 8

Pure • Orders in Modular Groups

STEP II Specimen paper, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q8

The key idea

Modulo N, the number $N-1$ is just -1, and $m+n=N$ says $n \equiv -m$. So every power of n is a power of m up to a sign: $n^k \equiv (-1)^k m^k$. Whether k is even or odd decides whether that sign matters, and the three parts are the three ways the parities of the orders can combine. The final part then uses these facts to find orders without computing every power.

Solution

Throughout, $\equiv$ means congruent modulo N. We take the identity of G to be 1, as the question intends. (In fact this is forced whenever $N-1 \in G$: if e is the identity then $e(N-1) \equiv N-1$, that is $-e \equiv -1$, so $e=1$.)

The order of $N-1$

We have

$$(N-1)^2 = N^2 - 2N + 1 \equiv 1,$$

and $N-1 \neq 1$ because $N \geq 3$. So $(N-1)^1$ is not the identity but $(N-1)^2$ is: the order of $N-1$ is 2.

Setting up the parts

Since $m+n=N$, we have $n \equiv -m$, and so for every positive integer k

$$n^k \equiv (-1)^k m^k, \qquad \text{equivalently} \qquad m^k \equiv (-1)^k n^k.$$

Also $N-1 \equiv -1 \equiv n\,m^{-1}$ lies in G, because G is a group containing n and m^{-1}. So -1 is an element of G, and $-1 \not\equiv 1$ since $N \geq 3$. We shall also use the standard fact that if x has order r and $x^k = 1$, then $r \mid k$ (write $k = ar + b$ with $0 \leq b < r$; then $x^b = 1$, so $b = 0$ by minimality of r).

Part (i)

Let p and q both be even. Then

$$n^p \equiv (-1)^p m^p = m^p \equiv 1,$$

so $q \mid p$. In the same way $m^q \equiv (-1)^q n^q = n^q \equiv 1$, so $p \mid q$. Two positive integers dividing each other are equal, so $p = q$.

Part (ii)

Let p be even and q odd. As p is even, $n^p \equiv m^p \equiv 1$, so $q \mid p$. Also $2q$ is even, so

$$m^{2q} \equiv n^{2q} = (n^q)^2 \equiv 1,$$

giving $p \mid 2q$. Write $p = kq$; then $kq \mid 2q$, so $k \mid 2$ and $k = 1$ or 2. But $k = 1$ would make $p = q$ odd, which it isn't. Hence $k = 2$ and $p = 2q$.

? **Ask yourself** Why can't we simply argue $m^q \equiv -n^q \equiv -1$, so "$p$ must be $2q$"?

That calculation is correct and shows $m^q \not\equiv 1$ while $m^{2q} \equiv 1$, but on its own it only tells us $p \mid 2q$ and $p \nmid q$; it does not pin p down (for instance p could a priori be 2 with $q = 3$). The extra divisibility $q \mid p$, which comes from p being even, is what closes the gap.

Part (iii)

Suppose, for contradiction, that p and q are both odd, and let $r = pq$, which is odd. Then

$$n^r = (n^q)^p \equiv 1, \qquad \text{but also} \qquad n^r \equiv (-1)^r m^r = -(m^p)^q \equiv -1.$$

So $1 \equiv -1$, that is $N \mid 2$, which is impossible as $N \geq 3$. Hence p and q cannot both be odd.

(By symmetry, part (ii) also gives $q = 2p$ when q is even and p odd. So in every case the orders of m and $N - m$ are either equal and even, or one is twice the other, which is odd.)

The group modulo 21

Successive powers of 2 and 5, reduced modulo 21, are

k	1	2	3	4	5	6
2^k	2	4	8	16	11	1
5^k	5	4	20	16	17	1

(for example $2^5 = 32 \equiv 11$, $5^2 = 25 \equiv 4$, $5^4 \equiv 4^2 = 16$, $5^5 \equiv 80 \equiv 17$). The first 1 appears at $k = 6$ in both rows, so 2 and 5 both have order 6.

Now read off orders from the table: if x has order 6, then x^k has order $6/\gcd(k, 6)$. So

- $11 = 2^5$ and $17 = 5^5$ have order 6;

- $4 = 2^2$ and $16 = 2^4$ have order 3;
- $8 = 2^3$ and $20 = 5^3$ have order 2 (as $20 = N - 1$, the first part already tells us this);
- 1 has order 1.

That leaves 10, 13 and 19, each of which pairs with a known element to make 21:

- $13 = 21 - 8$, and 8 has even order 2. By (iii) and the remark after it, the order of 13 is either 2 (equal and even) or 1 (half of 2). It isn't 1, since $13 \neq 1$, so 13 has order 2. (Check: $13^2 = 169 = 8 \cdot 21 + 1$.)
- $19 = 21 - 2$ and $10 = 21 - 11$, and 2 and 11 have order 6. So each of 19 and 10 has order 6 or 3.

So every element of G has order at most 6. The cyclic group of order 12 has a generator, an element of order 12, and an isomorphism preserves orders of elements. Since G has no element of order 12, it is not isomorphic to the cyclic group of order 12.

There is a second way to finish: G has three elements of order 2, namely 8, 13 and 20, while a cyclic group of order 12 has exactly one (if g generates it, g^k has order 2 only for $k = 6$).

Checks

A computer check finds the full list of orders in G: 1 has order 1; $8, 13, 20$ have order 2; $4, 16$ have order 3; and $2, 5, 10, 11, 17, 19$ have order 6. (Directly, $10^3 = 1000 \equiv 13$ and $19^3 \equiv (-2)^3 = -8 \equiv 13$, so neither has order 3.) The counts $1 + 3 + 2 + 6 = 12$ match the size of G. Parts (i)–(iii) were also tested on every pair m, $N - m$ in every subgroup of the units generated by at most two elements, for all $N < 60$: nearly 1900 pairs, with no exceptions.

Common slips

- Assuming $N - 1 \in G$ without saying why. It is not given, but it follows from $m, n \in G$, since $N - 1 \equiv n\,m^{-1}$.
- In (ii), proving only $p \mid 2q$; one more divisibility is needed to get equality.
- Taking the identity for granted in weird cases: a subset such as $\{3\}$ modulo 6 is closed under multiplication with identity 3, and with $m = n = 3$ it would violate (iii). The argument relies on the identity being 1, which is clearly what is meant.

Beyond STEP

G is the group of units modulo 21, and by the Chinese remainder theorem it is isomorphic to the product of the unit groups modulo 3 and modulo 7, that is $C_2 \times C_6$: this is why it has three elements of order 2 and nothing of order 12. Deciding exactly

when the units modulo N form a cyclic group (precisely for $N = 2, 4, p^k, 2p^k$ with p an odd prime, the cases with a "primitive root") is a classic result of first-year number theory. Davenport's *The Higher Arithmetic* covers primitive roots, and Allenby's *Rings, Fields and Groups* treats the group theory.

WORKED SOLUTION

STEP II Specimen paper, Question 9

Pure • Spheres, Planes and Reflections

STEP II Specimen paper, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q9

The key idea

Both parts are about one picture: a plane and the perpendicular dropped onto it. In (i), the sphere is centred at the origin, so everything is decided by the foot of the perpendicular from the origin to the plane: its distance from O settles whether the two meet, and Pythagoras then gives the circle. In (ii), split $\mathbf{x}$ into a part along $\mathbf{c}$ and a part perpendicular to $\mathbf{c}$, and watch what the formula does to each.

Solution

Part (i)

The plane $(\mathbf{x}-\mathbf{a})\cdot\mathbf{b}=0$ passes through the point $\mathbf{a}$ and has normal $\mathbf{b}$. Let N be the foot of the perpendicular from the origin O to the plane. Since ON is along the normal, $\mathbf{n}=\lambda\mathbf{b}$ for some scalar λ, and since N lies on the plane,

$$(\lambda\mathbf{b}-\mathbf{a})\cdot\mathbf{b}=0 \quad\Longrightarrow\quad \lambda=\frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{b}|^2},$$

which is allowed because $\mathbf{b}\neq\mathbf{0}$. So

$$\mathbf{n}=\frac{(\mathbf{a}\cdot\mathbf{b})\,\mathbf{b}}{|\mathbf{b}|^2}, \qquad d=|\mathbf{n}|=\frac{|\mathbf{a}\cdot\mathbf{b}|}{|\mathbf{b}|},$$

where d is the distance from O to the plane.

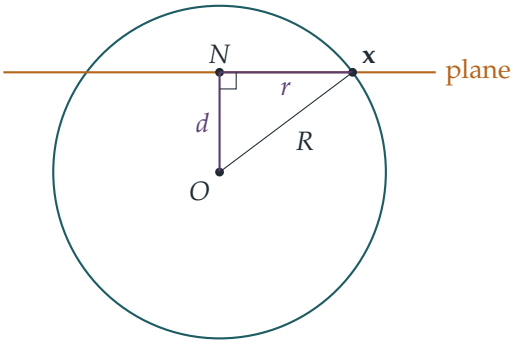
Now take any point $\mathbf{x}$ on the plane. The vector $\mathbf{x}-\mathbf{n}$ lies in the plane, so it is perpendicular to $\mathbf{b}$ and hence to $\mathbf{n}$. By Pythagoras,

$$|\mathbf{x}|^2=|\mathbf{n}|^2+|\mathbf{x}-\mathbf{n}|^2=d^2+|\mathbf{x}-\mathbf{n}|^2. \qquad (*)$$

So points of the plane have $|\mathbf{x}|\geq d$, and every value $|\mathbf{x}|\geq d$ occurs (move away from N within the plane). Hence the sphere $|\mathbf{x}|=R$ meets the plane if and only if $R\geq d$, that is,

$$\boxed{|\mathbf{a}\cdot\mathbf{b}|\leq R\,|\mathbf{b}|} \qquad \text{or equivalently} \qquad (\mathbf{a}\cdot\mathbf{b})^2\leq R^2|\mathbf{b}|^2.$$

In the equality case the plane is a tangent plane and the "circle" shrinks to the single point N.



When the condition holds, the points on both surfaces satisfy $|\mathbf{x}| = R$ and, by $(*)$,

$$|\mathbf{x} - \mathbf{n}|^2 = R^2 - d^2.$$

So they lie in the plane at a fixed distance from N, which is a point of the plane: they form a circle with centre N. Its centre and radius are

$$\mathbf{n} = \frac{(\mathbf{a}\cdot\mathbf{b})\,\mathbf{b}}{|\mathbf{b}|^2}, \qquad r = \sqrt{R^2 - \frac{(\mathbf{a}\cdot\mathbf{b})^2}{|\mathbf{b}|^2}} = \frac{\sqrt{R^2|\mathbf{b}|^2 - (\mathbf{a}\cdot\mathbf{b})^2}}{|\mathbf{b}|}.$$

Part (ii)

Write $\hat{\mathbf{c}} = \mathbf{c}/|\mathbf{c}|$, a unit vector (fine since $\mathbf{c} \neq \mathbf{0}$). Then the relation is

$$\mathbf{x}' = \mathbf{x} - 2(\mathbf{x}\cdot\hat{\mathbf{c}})\,\hat{\mathbf{c}}.$$

? **Ask yourself** What is $(\mathbf{x}\cdot\hat{\mathbf{c}})\,\hat{\mathbf{c}}$, geometrically?

It is the component of $\mathbf{x}$ along $\mathbf{c}$: the projection of $\mathbf{x}$ onto the line through O in the direction $\mathbf{c}$. So split

$$\mathbf{x} = \mathbf{x}_{\parallel} + \mathbf{x}_{\perp}, \qquad \mathbf{x}_{\parallel} = (\mathbf{x}\cdot\hat{\mathbf{c}})\,\hat{\mathbf{c}}, \qquad \mathbf{x}_{\perp} = \mathbf{x} - \mathbf{x}_{\parallel},$$

where $\mathbf{x}_{\perp}\cdot\mathbf{c} = \mathbf{x}\cdot\mathbf{c} - (\mathbf{x}\cdot\hat{\mathbf{c}})(\hat{\mathbf{c}}\cdot\mathbf{c}) = 0$. The relation becomes

$$\mathbf{x}' = \mathbf{x}_{\perp} - \mathbf{x}_{\parallel}.$$

The component in the plane Π through O perpendicular to $\mathbf{c}$ (the plane $\mathbf{x}\cdot\mathbf{c} = 0$) is unchanged, and the component normal to Π is reversed. That is exactly what a mirror does: **$\mathbf{x}'$ is the reflection of $\mathbf{x}$ in the plane $\mathbf{x}\cdot\mathbf{c} = 0$**, the plane through the origin with normal $\mathbf{c}$.

To confirm this against the definition of a reflection:

- $\mathbf{x}' - \mathbf{x} = -2\mathbf{x}_{\parallel}$ is a multiple of $\mathbf{c}$, so the segment from $\mathbf{x}$ to $\mathbf{x}'$ is perpendicular to Π;
- the midpoint $\frac{1}{2}(\mathbf{x} + \mathbf{x}') = \mathbf{x}_{\perp}$ satisfies $\mathbf{x}_{\perp} \cdot \mathbf{c} = 0$, so it lies on Π.

A few consequences make the picture concrete. Points of Π (where $\mathbf{x} \cdot \mathbf{c} = 0$) are fixed; $\mathbf{c}$ itself goes to $-\mathbf{c}$; $|\mathbf{x}'| = |\mathbf{x}|$, since $|\mathbf{x}'|^2 = |\mathbf{x}_{\perp}|^2 + |\mathbf{x}_{\parallel}|^2$; and applying the map twice gives $\mathbf{x}$ back, as reflecting twice should. The relation is unchanged if $\mathbf{c}$ is replaced by any non-zero multiple $k\mathbf{c}$, which fits: only the plane matters, not the length or sense of its normal.

Checks

In (i), take $\mathbf{a} = \mathbf{b} = (0, 0, 1)$ and $R = 2$: the plane is $z = 1$, which meets the sphere $x^2 + y^2 + z^2 = 4$ in the circle $x^2 + y^2 = 3$, $z = 1$. The formulae give centre $(0, 0, 1)$ and radius $\sqrt{4 - 1} = \sqrt{3}$, as they should. In (ii), take $\mathbf{c} = (0, 0, 1)$: the formula sends (x, y, z) to $(x, y, -z)$, the reflection in $z = 0$. We also tested both parts numerically on 20,000 random choices of $\mathbf{a}$, $\mathbf{b}$, R and $\mathbf{x}$, $\mathbf{c}$: points built on the predicted circle lay on both the sphere and the plane to within 10^{-9}, and the map in (ii) fixed vectors in Π, preserved lengths and dot products, and was its own inverse.

Common slips

- Forgetting the modulus: the condition is $|\mathbf{a} \cdot \mathbf{b}| \leq R|\mathbf{b}|$, not $\mathbf{a} \cdot \mathbf{b} \leq R|\mathbf{b}|$; a plane on the far side of O has $\mathbf{a} \cdot \mathbf{b} < 0$.
- Giving $\mathbf{a}$ as the centre of the circle. The point $\mathbf{a}$ is just some point of the plane; the centre is the foot of the perpendicular from O.
- In (ii), saying "reflection in $\mathbf{c}$" or "in the line along $\mathbf{c}$". Reflection in that line would be $\mathbf{x} \mapsto 2\mathbf{x}_{\parallel} - \mathbf{x}$, which is $-\mathbf{x}'$.

Beyond STEP

The map in (ii) is a *Householder reflection*, $\mathbf{x} \mapsto (I - 2\hat{\mathbf{c}}\hat{\mathbf{c}}^{\mathsf{T}})\mathbf{x}$, a linear map that is orthogonal and symmetric, with determinant -1. Every rotation of space is a product of two such reflections, and numerical analysts use them to factorise matrices stably (the QR decomposition). Strang's *Introduction to Linear Algebra* covers projections, reflections and orthogonal matrices from exactly this point of view.

WORKED SOLUTION

STEP II Specimen paper, Question 10

Pure • Logic Puzzles

STEP II Specimen paper, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q10

The key idea

Everything rests on one translation: a speaker is a toad exactly when what they say is true. So for each character we ask "could they be a toad?" and "could they be a frog?", and throw out any choice that makes their statement have the wrong truth value. Watch especially for statements about the speaker themselves: some of them can never be said by one kind of creature at all, and that is usually the way in.

Solution

Throughout, "toad" means a truth-teller and "frog" means a liar, so

$$\text{speaker is a toad} \iff \text{their statement is true.}$$

In each scene we find every assignment consistent with this rule, and check that exactly one survives.

Part (i)

Suppose A were a toad. Then A's statement would be true, so in particular A would be a frog, which is a contradiction. Hence A is a frog.

So A's statement is false: it is *not* the case that both A and B are frogs. Since A is a frog, the only way this can fail is for B to be a toad.

Check: with A a frog and B a toad, "both of us are frogs" is false, as a frog's statement must be. So A **is a frog and** B **is a toad**.

Part (ii)

This is the mirror image of part (i). Suppose C were a frog. Then "at least one of D and myself is a frog" would be true (because of C alone), but a frog cannot say something true. Hence C is a toad.

So C's statement is true: at least one of C, D is a frog. Since C is a toad, D must be the frog.

Check: with C a toad and D a frog, the statement is true, as a toad's must be. So C **is a toad and** D **is a frog**.

Part (iii)

Write S for E's statement, "G and H are both toads". Then G asserts that S is true, and H asserts that S is false.

Ask yourself We know nothing about anyone yet. Where can we possibly start?

Look at G and H together rather than at E. Their two statements are exact opposites, so exactly one of them is true. Hence exactly one of G, H is a toad and the other is a frog, whatever S turns out to be.

That settles S: since one of G, H is a frog, they are not both toads, so S is false. Therefore:

- E said something false, so E is a frog;
- G claimed S is true, which is false, so G is a frog;
- H claimed S is false, which is true, so H is a toad.

Check: G is a frog, so "G and H are both toads" is indeed false, consistent with every verdict. So E **and** G **are frogs and** H **is a toad**.

Part (iv)

Here "us" means the three of I, J, K.

I is a frog. If I were a toad, "all of us are frogs" would be true and so I would be a frog, a contradiction. So I is a frog, and I's statement is false: not all three are frogs, so

$$\text{at least one of } J,\ K \text{ is a toad.} \qquad (*)$$

J is a toad. Suppose instead J were a frog. Then by $(*)$, K is a toad, and the creatures are I frog, J frog, K toad: exactly one toad. That makes J's statement true, which a frog cannot say. So J is a toad.

K is a frog. Since J is a toad, J's statement is true: there is exactly one toad among the three. That toad is J, so K is a frog.

Ask yourself Does $(*)$ still hold now that K is a frog?

Yes: $(*)$ needs only one of J, K to be a toad, and J is. So the full check is: I's "all of us are frogs" is false (J is a toad), and J's "exactly one of us is a toad" is true (only J). Both statements have the right truth value, so I **and** K **are frogs and** J **is a toad**.

Checks

Each scene has only 2^2 or 2^3 possible assignments, so a complete check is a short truth table: for each assignment, test whether every speaker's statement is true exactly when that speaker is a toad. We ran this exhaustive search for all four scenes, and in each one exactly one assignment survives, the one found above. (The uniqueness matters: the question asks us to *decide*, so we need to rule the others out, not just find one that works.)

Common slips

- Finding a consistent assignment and stopping. You must also show that no other assignment works, which is what the "suppose it were…" contradictions do.
- In part (iii), treating G's "this is true" as a statement about G rather than about E's claim.
- In part (iv), forgetting that K, who says nothing, is still included in "all of us" and "exactly one of us".

Beyond STEP

Puzzles like these are exercises in propositional logic: each creature is a Boolean variable t, each statement a formula P, and every speaker imposes the constraint $t \iff P$. Deciding whether such a system of constraints has a solution is the *satisfiability* problem, which is central to theoretical computer science, and part (i)'s self-defeating statement is a gentle cousin of the liar paradox. Raymond Smullyan's *What Is the Name of This Book?* is a delightful collection of knights-and-knaves puzzles that builds up to Gödel's incompleteness theorem.

WORKED SOLUTION

STEP II Specimen paper, Question 11

Mechanics • Projectiles and Collisions

STEP II Specimen paper, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

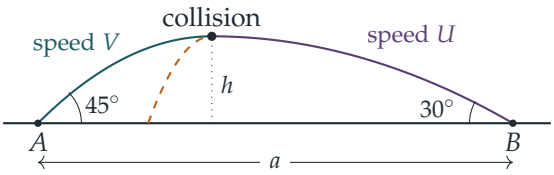
step.maths.org/questions/spec-s2-q11

The key idea

"Each particle is at its highest point when they collide" is two conditions in one: both particles must take the *same time* to reach the top, and they must be at the *same place* then. The time condition links U to V, and the horizontal distances then add up to a. After the collision, the vital observation is that both particles are moving horizontally at the top, so the combined particle starts with no vertical velocity at all.

Solution

We model both particles as projectiles moving under gravity alone (no air resistance), in the vertical plane containing AB. Take the direction from A to B as positive horizontally and upwards as positive vertically.



Showing $U^2 = 2V^2$

The particle from A has initial velocity components $\frac{V}{\sqrt{2}}$ horizontally and $\frac{V}{\sqrt{2}}$ vertically. The particle from B has components $\frac{\sqrt{3}}{2}U$ horizontally (towards A) and $\frac{1}{2}U$ vertically.

A projectile is at its highest point when its vertical velocity is zero. With initial vertical speed w this happens at time w/g. Both particles are launched at the same instant, so if they are both at their highest points at the moment of collision, these times must be equal:

$$\frac{V}{\sqrt{2}\,g} = \frac{U}{2g} \quad \Longrightarrow \quad U = \sqrt{2}\,V \quad \Longrightarrow \quad U^2 = 2V^2,$$

as required. Call this common time $T = \dfrac{V}{\sqrt{2}\,g}$.

Ask yourself Shouldn't we also check that the two highest points are at the same height?

Yes: equal times is necessary, but the particles must also meet. A projectile with initial vertical speed w rises to height $w^2/(2g)$. For A's particle this is $\frac{V^2}{2}/2g = \frac{V^2}{4g}$, and for B's it is $\frac{U^2}{4}/2g = \frac{2V^2}{8g} = \frac{V^2}{4g}$. These agree, so the heights match automatically once $U^2 = 2V^2$. (In fact this has to happen: both have zero vertical velocity at the same time T under the same acceleration, so their heights are $\frac{1}{2}gT^2$ for both.) Write

$$h = \frac{V^2}{4g}$$

for the height of the collision.

Finding a

The horizontal positions must also agree at time T. Horizontal velocities are constant, so in time T the particle from A travels $\frac{V}{\sqrt{2}}T$ towards B, and the particle from B travels $\frac{\sqrt{3}}{2}UT$ towards A. They meet, so these distances add up to a:

$$\begin{aligned} a &= \left(\frac{V}{\sqrt{2}} + \frac{\sqrt{3}}{2}\cdot\sqrt{2}\,V\right)\frac{V}{\sqrt{2}\,g} \\ &= \frac{V^2}{g}\left(\frac{1}{2} + \frac{\sqrt{3}}{2}\right) = \frac{V^2}{2g}(1+\sqrt{3}), \end{aligned}$$

which is the required result.

After the collision

At the instant of impact each particle is at its highest point, so each is moving *horizontally*: A's particle with velocity $+\frac{V}{\sqrt{2}}$ and B's with velocity $-\frac{\sqrt{3}}{2}U = -\frac{\sqrt{3}\,V}{\sqrt{2}}$.

Ask yourself Which momentum is conserved, and in which direction?

During the very short impact the only large forces are the equal and opposite forces between the particles; gravity is finite and acts for a negligible time. So total momentum is conserved. Horizontally, with v the velocity of the combined particle of mass $2m$,

$$2m\,v = m\frac{V}{\sqrt{2}} - m\frac{\sqrt{3}\,V}{\sqrt{2}} \quad\Longrightarrow\quad v = -\frac{(\sqrt{3}-1)V}{2\sqrt{2}}.$$

The negative sign says the combined particle moves back towards A, with speed

$\dfrac{(\sqrt{3}-1)V}{2\sqrt{2}}$. Vertically, both particles had zero vertical velocity, so the combined particle starts with zero vertical velocity too.

The combined particle is now a projectile launched horizontally from height $h = \frac{V^2}{4g}$. Its horizontal velocity stays $|v|$. Its vertical speed w on reaching the plane follows from $w^2 = 0 + 2gh$:

$$w^2 = 2g \cdot \frac{V^2}{4g} = \frac{V^2}{2}, \qquad w = \frac{V}{\sqrt{2}}.$$

The angle ϕ that the velocity makes with the horizontal therefore satisfies

$$\begin{aligned}\tan\phi = \frac{w}{|v|} &= \frac{V}{\sqrt{2}} \cdot \frac{2\sqrt{2}}{(\sqrt{3}-1)V} = \frac{2}{\sqrt{3}-1} \\ &= \frac{2(\sqrt{3}+1)}{(\sqrt{3}-1)(\sqrt{3}+1)} = \frac{2(\sqrt{3}+1)}{2} = 1+\sqrt{3},\end{aligned}$$

rationalising the denominator in the second line. This completes the question.

Checks

- *Units.* V^2/g has units $(\mathrm{m\,s^{-1}})^2/(\mathrm{m\,s^{-2}}) = \mathrm{m}$, as a distance should, and $\tan\phi$ is a ratio of speeds, so dimensionless. Note that $\tan\phi$ does not depend on V or g at all, which is a good sign.
- *Sanity of the geometry.* A's particle reaches its top a horizontal distance $\frac{V^2}{2g}$ from A; B's reaches its top $\sqrt{3}\,\frac{V^2}{2g}$ from B. The shallower 30° launch at the larger speed covers more ground, and the two distances add to a.
- *Numerical.* With $V = 7.3\ \mathrm{m\,s^{-1}}$ and $g = 9.8\ \mathrm{m\,s^{-2}}$ we stepped both trajectories forward in time and found them at the same point, $(2.719, 1.359)$ in metres, with both vertical velocities zero, at $t = T$. Stepping the combined particle down to the plane gave $\tan\phi \approx 2.73205$, matching $1+\sqrt{3} = 2.732050\ldots$

Common slips

- Conserving momentum in the vertical direction with non-zero vertical velocities. At the top of each trajectory the vertical velocity is zero, which is exactly what makes the collision so simple.
- Adding the horizontal momenta instead of subtracting: the particles are moving in opposite directions.
- Using $\tan\phi = |v|/w$ (the angle with the vertical) rather than $w/|v|$.

Beyond STEP

Coalescing particles are the simplest example of a perfectly inelastic collision: momentum is conserved but kinetic energy is not (here a good deal is lost, since the particles were moving in opposite directions). At university this becomes the study of collisions in the centre-of-mass frame, where the total momentum is zero and the energy lost is the whole of the kinetic energy relative to the centre of mass. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats momentum and collisions in this way with many excellent problems.

WORKED SOLUTION

STEP II Specimen paper, Question 12

Mechanics • Oscillations on a Wire

STEP II Specimen paper, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

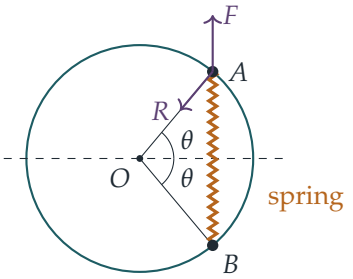
step.maths.org/questions/spec-s2-q12

The key idea

The wire is smooth, so the only forces doing work are the spring's, and mechanical energy is conserved. All we need is the length of the spring in terms of θ (a chord of the circle) and the speed of each bead. Two things need a moment's thought. Why does the line bisecting angle AOB stay fixed? And in part (ii), what happens if the beads have enough energy to reach opposite ends of a diameter?

Solution

Model the beads as particles on a smooth circular wire in a horizontal plane, joined by a light spring obeying Hooke's law. Gravity is vertical and is balanced by the vertical part of the wire's reaction, so it plays no part in the horizontal motion.



In the diagram, F is the spring's push on A (along BA, since the spring starts compressed) and R is the horizontal part of the wire's reaction on A.

Setting up the energy equation

Measure each bead's position by its angle on the circle: A at angle ϕ_A and B at angle ϕ_B.

Ask yourself Why can we assume both beads move with the same speed, symmetrically about a fixed line?

Take moments about the vertical axis through O. The spring forces on A and B are equal and opposite and act along the same line AB, so their moments cancel. The horizontal part R of each wire reaction is perpendicular to the wire, so it points along the radius through O and has no moment. The vertical forces have no moment about a vertical axis. So the angular momentum $ma^2(\dot\phi_A + \dot\phi_B)$ is conserved. It starts at zero, since the beads are released from rest, so $\phi_A + \phi_B$ stays constant: the bisector of angle AOB is fixed. Take it as the dashed line in the diagram, so that $\phi_A = \theta$ and $\phi_B = -\theta$. Then each bead moves with speed $a\dot\theta$.

The spring is the chord AB, of length $2a\sin\theta$. Its natural length is $2a\sin\alpha$, so its extension is $x = 2a(\sin\theta - \sin\alpha)$, and its elastic energy is

$$\frac{\lambda x^2}{2\cdot 2a\sin\alpha} = \frac{4\lambda a^2(\sin\theta-\sin\alpha)^2}{4a\sin\alpha} = \frac{\lambda a(\sin\theta-\sin\alpha)^2}{\sin\alpha}.$$

This formula also holds when $x < 0$ (compression), because x is squared. The kinetic energy is $2\cdot\frac{1}{2}m(a\dot\theta)^2 = ma^2\dot\theta^2$. The wire is smooth and its reaction is always perpendicular to the motion, so it does no work, and energy is conserved. At release $\theta = \beta$ and $\dot\theta = 0$, so

$$ma^2\dot\theta^2 + \frac{\lambda a(\sin\theta-\sin\alpha)^2}{\sin\alpha} = \frac{\lambda a(\sin\beta-\sin\alpha)^2}{\sin\alpha}.$$

Multiplying through by $\sin\alpha/a$ gives

$$ma\dot\theta^2\sin\alpha + \lambda(\sin\theta-\sin\alpha)^2 = \lambda(\sin\beta-\sin\alpha)^2,$$

as required. Note that the spring starts compressed, so $\beta < \alpha$.

Part (i)

Differentiate the energy equation with respect to t:

$$2ma\sin\alpha\,\dot\theta\ddot\theta + 2\lambda(\sin\theta-\sin\alpha)\cos\theta\,\dot\theta = 0.$$

Dividing by $2\dot\theta$ (valid whenever the beads are moving, and the result extends to all t by continuity) gives the equation of motion

$$ma\sin\alpha\,\ddot\theta = -\lambda(\sin\theta-\sin\alpha)\cos\theta.$$

The equilibrium is at $\theta = \alpha$, where the spring has its natural length. If $\beta - \alpha$ is small then θ stays close to α (the energy equation gives $|\sin\theta - \sin\alpha| \le |\sin\beta - \sin\alpha|$), so write $\theta = \alpha + \varepsilon$ with ε small. To first order, $\sin\theta - \sin\alpha \approx \varepsilon\cos\alpha$, and $\cos\theta \approx \cos\alpha$, which multiplies a quantity that is already small. So

$$\ddot\varepsilon \approx -\frac{\lambda\cos^2\alpha}{ma\sin\alpha}\,\varepsilon,$$

which is simple harmonic motion with $\omega^2 = \lambda\cos^2\alpha/(ma\sin\alpha)$. The period is therefore

$$T = \frac{2\pi}{\omega} \approx 2\pi\sqrt{\frac{ma\sin\alpha}{\lambda\cos^2\alpha}}.$$

Part (ii)

Write $k = \sqrt{ma\sin\alpha/\lambda}$. From the energy equation,

$$\dot\theta^2 = \frac{1}{k^2}\Big[(\sin\alpha - \sin\beta)^2 - (\sin\theta - \sin\alpha)^2\Big].$$

The beads stop when $\sin\theta - \sin\alpha = \pm(\sin\alpha - \sin\beta)$, that is when $\sin\theta = \sin\beta$ or when $\sin\theta = 2\sin\alpha - \sin\beta$. Starting from $\theta = \beta$, θ increases (the compressed spring pushes the beads apart), and $\dot\theta^2 > 0$ for as long as $\sin\beta < \sin\theta < 2\sin\alpha - \sin\beta$.

? **Ask yourself** What decides which of the two cases we are in?

Whether θ can reach a second stopping point before $2\theta = \pi$, where A and B are at opposite ends of a diameter and $\sin\theta$ reaches its largest value, 1. That depends on whether $2\sin\alpha - \sin\beta$ is less than or greater than 1.

(a) $\sin\beta > 2\sin\alpha - 1$**.** Then $2\sin\alpha - \sin\beta < 1$, so there is an acute angle γ with $\sin\gamma = 2\sin\alpha - \sin\beta$, and $\gamma > \alpha$. The beads come to rest when $\theta = \gamma$ (the spring is then stretched by as much as it was originally compressed), and θ oscillates between β and γ. The motion out and back takes equal times, so

$$T = 2\sqrt{\frac{ma\sin\alpha}{\lambda}}\int_\beta^\gamma \frac{\mathrm{d}\theta}{\sqrt{(\sin\alpha - \sin\beta)^2 - (\sin\theta - \sin\alpha)^2}}, \qquad \gamma = \arcsin(2\sin\alpha - \sin\beta).$$

(b) $\sin\beta < 2\sin\alpha - 1$**.** Now $2\sin\alpha - \sin\beta > 1 \geq \sin\theta$, so $\dot\theta$ never vanishes for θ between β and $\frac{1}{2}\pi$: the beads pass the ends of a diameter while still moving. After that the chord shortens again. The energy equation involves θ only through $\sin\theta$, which is symmetric about $\frac{1}{2}\pi$, so the motion from $\frac{1}{2}\pi$ to $\pi - \beta$ mirrors the motion from β to $\frac{1}{2}\pi$. The beads stop at $\theta = \pi - \beta$, where the spring is again compressed to its original length. It then pushes them back the way they came. (At $\theta = \pi - \beta$ the equation of motion gives $\ddot\theta < 0$, since $\sin\theta - \sin\alpha < 0$ and $\cos\theta < 0$.) A full oscillation is $\beta \to \pi - \beta \to \beta$, which is four copies of the quarter from β to $\frac{1}{2}\pi$:

$$T = 4\sqrt{\frac{ma\sin\alpha}{\lambda}}\int_\beta^{\pi/2} \frac{\mathrm{d}\theta}{\sqrt{(\sin\alpha - \sin\beta)^2 - (\sin\theta - \sin\alpha)^2}}.$$

Since $\sin\alpha < 1$ we have $2\sin\alpha - 1 < \sin\alpha$, so both cases are consistent with $\beta < \alpha$. Case (b) can only happen if $2\sin\alpha - 1 > 0$, that is if $\frac{1}{6}\pi < \alpha < \frac{1}{4}\pi$.

Checks

- In case (a), the substitution $\sin\theta = \sin\alpha - (\sin\alpha - \sin\beta)\cos u$ turns the integral into $\int_0^\pi \mathrm{d}u/\cos\theta$. When $\beta \to \alpha$ this tends to $\pi/\cos\alpha$, which gives back the period in part (i).
- We simulated the two beads moving independently on the wire with $m = 1.3$, $a = 0.7$, $\lambda = 2.1$. The energy equation held to rounding error and the bisector

stayed fixed. For $\alpha = 0.6$, $\beta = 0.59$ the period was 3.766, against 3.7657 from part (i). For $\alpha = 0.7$, $\beta = 0.4$ (case (a)) it was 4.8847, against 4.8846 from the integral. For $\alpha = 0.75$, $\beta = 0.1$ (case (b)) it was 7.0553, against 7.0553, and θ peaked at $\pi - \beta$ as predicted.

Common slips

- Using $\lambda x^2/(2a)$ for the elastic energy. The denominator is twice the *natural length*, $2 \cdot 2a \sin\alpha$.
- In case (b), stopping the integral at $\theta = \frac{1}{2}\pi$ or claiming the beads stop there. They pass the ends of a diameter with nonzero speed, so the period is $4\int_\beta^{\pi/2}$, not $2\int_\beta^{\pi/2}$.

Beyond STEP

The period integrals in part (ii) cannot be done in elementary terms. They are relatives of the *elliptic integrals* that give the exact period of a simple pendulum. Notice that the small-oscillation period in part (i) is independent of amplitude, while the exact period is not. That contrast between linearised and nonlinear oscillators is a central theme of a first course in classical mechanics. Conservation of angular momentum for a system with internal forces, which fixed the bisector here, is treated carefully in Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP II Specimen paper, Question 13

Mechanics • Equilibrium and Moments

STEP II Specimen paper, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q13

The key idea

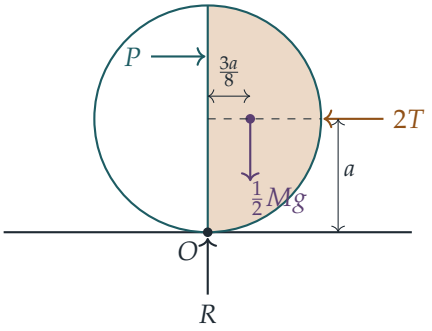
The orange would fall apart by its pieces toppling outwards, each one pivoting about the single point where the sphere touches the table. So we cut the orange in two along one of its slicing planes and take moments for one half about a horizontal line through that lowest point. The weight of the half tries to tip it over; only the ribbon, and nothing else, can hold it back. The forces from the other half can only make matters worse, and that is where the "at least" comes from.

Solution

The model

We treat the chocolate as a rigid uniform solid, the ribbon as light with the same tension T all the way round, and we take the segments to be equal and even in number (a real chocolate orange has twenty), so that one slicing plane through the axis splits the orange into two halves, each a hemisphere-shaped wedge of angle π. At the end we see that the argument works for any group of segments if the cut faces are smooth.

The sphere touches the table at just one point O, the bottom of the axis. Set up axes with O as origin, z vertically up, and x horizontal, pointing from the axis into the half we study (the right-hand half in the figure). The dividing plane is $x = 0$. Let ℓ be the horizontal line through O lying in that plane (the y-axis); we take moments about ℓ, counting a moment as positive if it tends to tip the half outwards, away from the axis.



The forces on one half

The right-hand half has mass $\frac{1}{2}M$, and the external forces on it are as follows.

- **Weight** $\frac{1}{2}Mg$, acting at the centre of mass. Putting $2\theta = \pi$ in the given result, this is at distance
$$\frac{3\pi a \sin(\pi/2)}{16(\pi/2)} = \frac{3a}{8}$$
from the axis, on the side $x > 0$. Its moment about ℓ is $\frac{1}{2}Mg \cdot \frac{3a}{8} = \frac{3}{16}Mga$, in the tipping sense.
- **The table**: a normal reaction and possibly friction, all acting at O. Since O lies on ℓ, these have no moment about ℓ.
- **The ribbon** round its half of the equator. A narrow ribbon presses on the surface normally, with force T/a per unit length (it is like a string over a smooth curve). On the arc between angles $-\frac{\pi}{2}$ and $\frac{\pi}{2}$ the x-components add up to
$$\int_{-\pi/2}^{\pi/2} \frac{T}{a} \cos\phi \; a \, \mathrm{d}\phi = 2T$$
towards the axis, and the y-components cancel by symmetry. (Equivalently, cut the ribbon where it crosses the dividing plane: each cut end pulls with tension T in the $-x$ direction, giving $2T$ again.) All of this force is horizontal and acts at height a, so its moment about ℓ is $2T \cdot a$ in the *restoring* sense.
- **The other half**, acting across the flat face $x = 0$. This is the only awkward force, and we don't need to know how it is distributed.

? **Ask yourself** Why can the force from the other half only help the half to topple?

Split the force on each small patch of the face into a part perpendicular to the face and a part along it (friction, if the cut faces are rough).

- The perpendicular part is in the x-direction. The two halves are only pressed together, never glued, so this part is a push, in the $+x$ direction. It acts at some height $z \geq 0$, so its moment about ℓ is z times its size, which is never negative: it tends to tip the half outwards.

- Any part along the face lies in the plane $x = 0$, which also contains ℓ. A force whose line of action lies in the same plane as ℓ either meets ℓ or is parallel to it, so it has no moment about ℓ.

So the other half contributes a total moment $P_{\text{mom}} \geq 0$ in the tipping sense, whatever the details of the contact.

Moments about ℓ

The half is in equilibrium, so the total moment about ℓ is zero:

$$2Ta = \tfrac{3}{16}Mga + P_{\text{mom}}.$$

Since $P_{\text{mom}} \geq 0$,

$$2Ta \;\geq\; \tfrac{3}{16}Mga, \qquad \text{so} \qquad T \;\geq\; \tfrac{3}{32}Mg,$$

as required. Equality holds exactly when the two halves press on each other with no force at all, so they only just stay together.

Without an even number of segments

If instead the cut faces are smooth, the same calculation works for any wedge of angle 2θ made of whole segments. Take x along its plane of symmetry and ℓ the horizontal line through O perpendicular to that plane. The wedge has weight $\frac{\theta}{\pi}Mg$ at distance $\frac{3\pi a \sin\theta}{16\theta}$, giving a tipping moment of $\frac{3}{16}Mga\sin\theta$. The ribbon on its arc pulls inwards with $2T\sin\theta$ at height a. Smooth faces give only horizontal pushes perpendicular to the faces, and these have a non-negative component in the $+x$ direction, so they again only add to the tipping moment. So $2Ta\sin\theta \geq \frac{3}{16}Mga\sin\theta$, and the $\sin\theta$ cancels to give $T \geq \frac{3}{32}Mg$ once more. Whichever group of segments we choose, the bound is the same.

Checks

- The given centre of mass result at $2\theta = \pi$ gives $\frac{3a}{8}$, the familiar centre of mass of a solid hemisphere measured from its flat face. A Monte Carlo estimate gives $0.376a$ for the half and agrees with the formula for other wedge angles too.
- Summing the ribbon's normal forces numerically over an arc of angle 2θ gives $2T\sin\theta$, for example $2T$ for the half.
- The bound $\frac{3}{32}Mg \approx 0.094Mg$ comes out the same for every wedge angle checked ($2\theta = 0.6,\ \pi,\ 2.4,\ 5$). A bound that depends only on the whole orange, and not on how we choose to cut it, is reassuring.
- Units: the moments are force times length, and a cancels, leaving a force.

Common slips

- Taking moments about the centre of the sphere, or about a point on the equator. Then the unknown force from the table, or from the other half, enters the equation.

The point O is chosen precisely because the table's force has no moment about a line through it.

- Saying the ribbon's force on the half is T. The ribbon pulls on both ends of the half's arc, so the resultant is $2T$.
- Setting the force between the halves equal to zero without justification. That is what gives equality. The "at least" comes from noticing that this force can only push.

Beyond STEP

This is a statics problem with an unknown internal force, handled by choosing an axis that removes the unknowns we can't find and then using an inequality (contact forces push and never pull) to deal with the rest. The same reasoning lies behind the analysis of masonry arches and domes, where a structure stands up only if a line of thrust can be found that stays inside the material. The centre of mass result itself is a standard triple integral in spherical polar coordinates, well worth deriving. For more on rigid-body equilibrium, try Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP II Specimen paper, Question 14

Mechanics • Rotation of Rigid Bodies

STEP II Specimen paper, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q14

The key idea

Both the disc and the particle turn about the same fixed vertical axis, so the natural tool is the rotational equation of motion, torque = moment of inertia × angular acceleration, applied to each body separately. The friction between them is an internal pair of forces: it speeds one up exactly as much (in angular momentum) as it slows the other down. That gives the conserved combination at once, and subtracting the two equations gives a first-order equation for the *relative* angular speed, which decays exponentially.

Solution

The model

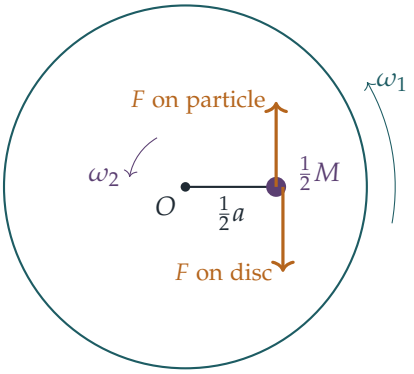
The disc is a uniform lamina turning about a smooth fixed axis through its centre O, so its moment of inertia about the axis is

$$I_1 = \tfrac{1}{2}Ma^2.$$

The particle, of mass $\tfrac{1}{2}M$, is held by a light rigid wire at a fixed distance $\tfrac{1}{2}a$ from the axis, so it moves on a circle of radius $\tfrac{1}{2}a$ and its moment of inertia about the axis is

$$I_2 = \tfrac{1}{2}M\left(\tfrac{1}{2}a\right)^2 = \tfrac{1}{8}Ma^2.$$

Initially $\omega_1 = \Omega$ and $\omega_2 = 0$, so the disc is moving faster than the particle at the point of contact. Friction therefore acts on the particle in its direction of motion (tangentially, at radius $\tfrac{1}{2}a$), and by Newton's third law an equal and opposite force acts on the disc at the same point.



(View from above; $F = Mak(\omega_1 - \omega_2)$. The tension in the wire is radial, so it has no moment about the axis.)

Equations of motion

The friction force $F = Mak(\omega_1 - \omega_2)$ acts at distance $\frac{1}{2}a$ from the axis, perpendicular to the radius, so its moment about the axis is

$$F \cdot \tfrac{1}{2}a = \tfrac{1}{2}Ma^2k(\omega_1 - \omega_2).$$

The weight of the particle and the normal reactions are vertical, so they have no moment about the vertical axis, and the axis itself is smooth. For the disc, friction opposes the rotation:

$$\tfrac{1}{2}Ma^2\frac{\mathrm{d}\omega_1}{\mathrm{d}t} = -\tfrac{1}{2}Ma^2k(\omega_1 - \omega_2) \quad \Longrightarrow \quad \frac{\mathrm{d}\omega_1}{\mathrm{d}t} = -k(\omega_1 - \omega_2).$$

For the particle, friction drives the rotation:

$$\tfrac{1}{8}Ma^2\frac{\mathrm{d}\omega_2}{\mathrm{d}t} = \tfrac{1}{2}Ma^2k(\omega_1 - \omega_2) \quad \Longrightarrow \quad \frac{\mathrm{d}\omega_2}{\mathrm{d}t} = 4k(\omega_1 - \omega_2).$$

(For the particle you can equally use Newton's second law tangentially: $\frac{1}{2}M \cdot \frac{1}{2}a\,\dot{\omega}_2 = F$, which is the same equation.)

The conserved combination

Multiplying the first equation by 4 and adding the second,

$$4\frac{\mathrm{d}\omega_1}{\mathrm{d}t} + \frac{\mathrm{d}\omega_2}{\mathrm{d}t} = -4k(\omega_1 - \omega_2) + 4k(\omega_1 - \omega_2) = 0,$$

so $4\omega_1 + \omega_2$ is constant. At $t = 0$ it equals $4\Omega + 0$, hence

$$4\omega_1 + \omega_2 = 4\Omega,$$

as required. This is conservation of angular momentum about the axis: $I_1\omega_1 + I_2\omega_2 = \frac{1}{8}Ma^2(4\omega_1 + \omega_2)$, and the only external forces (the reaction at the axis, gravity and the vertical reactions) have no moment about it.

Solving for ω_1

> **Ask yourself** We have two coupled equations. Which single quantity satisfies an equation on its own?

Both rates depend only on the difference $u = \omega_1 - \omega_2$, so look at that. Subtracting the equations,

$$\frac{\mathrm{d}u}{\mathrm{d}t} = \frac{\mathrm{d}\omega_1}{\mathrm{d}t} - \frac{\mathrm{d}\omega_2}{\mathrm{d}t} = -ku - 4ku = -5ku,$$

with $u(0) = \Omega - 0 = \Omega$. Separating variables (or recognising exponential decay),

$$\omega_1 - \omega_2 = u = \Omega\,\mathrm{e}^{-5kt}.$$

Adding this to $4\omega_1 + \omega_2 = 4\Omega$ eliminates ω_2:

$$5\omega_1 = 4\Omega + \Omega\,\mathrm{e}^{-5kt}, \qquad \text{so} \qquad \omega_1 = \frac{\Omega}{5}\left(4 + \mathrm{e}^{-5kt}\right),$$

as required. It follows that $\omega_2 = 4\Omega - 4\omega_1 = \frac{4}{5}\Omega\left(1 - \mathrm{e}^{-5kt}\right)$.

Alternatively, substitute $\omega_2 = 4\Omega - 4\omega_1$ into the disc's equation to get $\dot{\omega}_1 = -5k\omega_1 + 4k\Omega$, a linear equation whose solution with $\omega_1(0) = \Omega$ is the same.

Checks

- At $t = 0$: $\omega_1 = \frac{1}{5}\Omega(4+1) = \Omega$ and $\omega_2 = 0$. Correct.
- As $t \to \infty$: $\omega_1, \omega_2 \to \frac{4}{5}\Omega$, a common speed, as friction must eventually produce. The final kinetic energy is $\frac{1}{2} \cdot \frac{5}{8}Ma^2 \cdot \frac{16}{25}\Omega^2 = \frac{1}{5}Ma^2\Omega^2$, against $\frac{1}{4}Ma^2\Omega^2$ initially, so a fifth of the energy is lost to friction, as expected in a "perfectly inelastic" coupling.
- Numerically, with $M = a = k = \Omega = 1$, stepping the two equations of motion to $t = 1$ gives $\omega_1 = 0.80135$ and $\omega_2 = 0.79461$, matching $\frac{1}{5}(4 + \mathrm{e}^{-5})$ and $\frac{4}{5}(1 - \mathrm{e}^{-5})$, with $4\omega_1 + \omega_2$ staying at 4 throughout.

Common slips

- Taking the particle's moment of inertia as $\frac{1}{2}Ma^2$ (as if it were at the rim) instead of $\frac{1}{2}M(\frac{1}{2}a)^2$.
- Using the friction force itself, rather than its moment $F \cdot \frac{1}{2}a$, in the rotational equations.
- Giving friction the same sign in both equations: it slows the disc and speeds up the particle.

Beyond STEP

The friction here is an internal force, and the conservation of $4\omega_1 + \omega_2$ is the rotational version of momentum conservation in a collision; the exponential approach to a common speed is the same mathematics as two bodies coupled by viscous drag. More generally, a linear system $\dot{\mathbf{x}} = A\mathbf{x}$ decouples when written in terms of the eigenvectors of A: here they are the combinations $4\omega_1 + \omega_2$ (eigenvalue 0) and $\omega_1 - \omega_2$ (eigenvalue $-5k$). Kleppner and Kolenkow, *An Introduction to Mechanics*, develops rigid-body rotation and angular momentum carefully, and Boyce and DiPrima, *Elementary Differential Equations*, treats linear systems by eigenvectors.

WORKED SOLUTION

STEP II Specimen paper, Question 15

Probability and Statistics • Expectation and Sample Means

STEP II Specimen paper, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q15

The key idea

The king fines a huge number of vendors every day, so by the law of large numbers his income per vendor per day is, to all intents, the *expected* fine. We therefore need the expected fine under each scheme. The first two schemes are about the mean of a sample of normal variables, whose distribution is again normal with a smaller variance. The third is a waiting-time problem, so it calls for the geometric distribution.

Solution

Let $W_1, W_2, \ldots$ be the weights (in grams) of the kippers in the selected carton. The carton holds an essentially random selection of kippers, so the W_i are independent, each $W_i \sim \mathrm{N}(200, 10^2)$. Choosing kippers from the carton at random doesn't depend on their weights, so any kippers we pick out, in any order, are still independent $\mathrm{N}(200, 10^2)$ variables.

We use the standard fact that if $W_1, \ldots, W_m$ are independent $\mathrm{N}(\mu, \sigma^2)$, then their mean satisfies

$$\overline{W} = \frac{1}{m}\sum_{i=1}^{m} W_i \sim \mathrm{N}\left(\mu, \frac{\sigma^2}{m}\right).$$

Part (i)

Here $m = 625$, so $\overline{W} \sim \mathrm{N}\left(200, \frac{100}{625}\right)$, and the standard deviation is $\frac{10}{25} = 0.4$. Then

$$\mathrm{P}(\overline{W} < 199) = \Phi\left(\frac{199 - 200}{0.4}\right) = \Phi(-2.5) = 1 - \Phi(2.5) \approx 1 - 0.99379 = 0.00621.$$

The expected fine is

$$1500 \times 0.00621 \approx 9.3 \text{ crowns.}$$

Part (ii)

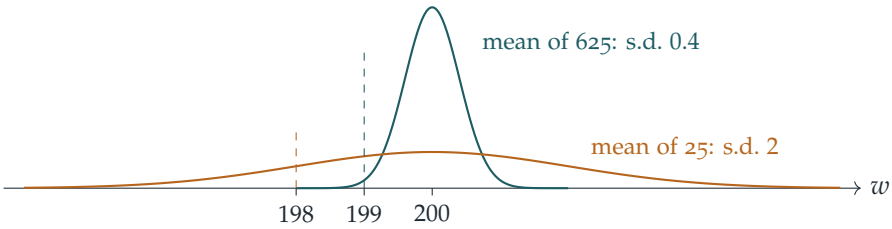
Now $m = 25$, so $\overline{W} \sim \mathrm{N}\left(200, \frac{100}{25}\right)$, with standard deviation 2. Then

$$\mathrm{P}(\overline{W} < 198) = \Phi\left(\frac{198 - 200}{2}\right) = \Phi(-1) = 1 - \Phi(1) \approx 1 - 0.84134 = 0.15866,$$

and the expected fine is

$$100 \times 0.15866 \approx 15.9 \text{ crowns.}$$

The picture shows why the smaller fine wins. A threshold one gram below the mean is 2.5 standard deviations away for the average of 625 kippers, but a threshold two grams below is only one standard deviation away for the average of 25.



Part (iii)

Let N be the number of kippers removed. Each kipper weighs more than 200 g with probability $\frac{1}{2}$, since 200 is the mean of a symmetric distribution and $\mathrm{P}(W = 200) = 0$. The kippers drawn are independent, so N is the number of trials up to and including the first success in independent trials with success probability $\frac{1}{2}$. In other words, N is geometric:

$$\mathrm{P}(N = n) = \left(\tfrac{1}{2}\right)^n, \qquad n = 1, 2, 3, \ldots$$

(Strictly, the carton runs out after 625 kippers, but the chance that none of them exceeds 200 g is 2^{-625}, which is utterly negligible.)

Ask yourself How do we find $\mathrm{E}[N(N-1)]$ without first computing $\mathrm{E}[N^2]$?

Differentiate the geometric series twice. For $|x| < 1$,

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x} \quad \Longrightarrow \quad \sum_{n=2}^{\infty} n(n-1)x^{n-2} = \frac{2}{(1-x)^3}.$$

Multiplying by x^2 and putting $x = \frac{1}{2}$ (the $n = 1$ term is zero anyway):

$$\mathrm{E}[N(N-1)] = \sum_{n=1}^{\infty} n(n-1)\left(\tfrac{1}{2}\right)^n = \left(\tfrac{1}{2}\right)^2 \cdot \frac{2}{\left(\frac{1}{2}\right)^3} = 4.$$

(Equivalently, $\mathrm{E}[N] = 2$ and $\mathrm{Var}(N) = \frac{1-p}{p^2} = 2$ with $p = \frac{1}{2}$, so $\mathrm{E}[N^2] = 6$ and $\mathrm{E}[N^2] - \mathrm{E}[N] = 4$.) So the expected fine is

$$\mathrm{E}[3N(N-1)] = 3 \times 4 = 12 \text{ crowns.}$$

Conclusion

The expected fines per vendor per day are about 9.3, 15.9 and 12 crowns for schemes (i), (ii) and (iii). Because vast numbers of cartons are inspected, the king's total income is very close to the number of inspections times the expected fine. He should therefore choose **scheme (ii)**. It raises roughly 70% more than scheme (i) and about a third more than scheme (iii).

Checks

- Scheme (iii) can't be checked against a table, but the series is easy to add up: $\frac{3\cdot2}{4} + \frac{3\cdot6}{8} + \frac{3\cdot12}{16} + \cdots = 1.5 + 2.25 + 2.25 + 1.875 + \cdots$. The partial sums climb quickly towards 12.
- A simulation of 10^5 cartons of 625 normal kippers, run through all three schemes, gave average fines of about 10.2 (from 2×10^4 cartons, so noisy), 15.9 and 11.9 crowns. These are close to 9.31, 15.87 and 12.

Common slips

- Using a standard deviation of $\frac{10}{625}$ or $\frac{100}{25}$ for the sample mean. The *variance* is divided by m, so the standard deviation is divided by $\sqrt{m}$.
- Treating N as starting from 0 (counting failures only). Here n includes the heavy kipper, so $\mathrm{P}(N = 1) = \frac{1}{2}$ and the fine is then zero.
- Comparing probabilities of being fined instead of expected fines: scheme (iii) fines a vendor (with $n \geq 2$) half the time, but usually only 6 crowns.

Beyond STEP

The justification "vast quantities, so use the expectation" is the law of large numbers, and the fact that $\overline{W}$ is normal with variance σ^2/m comes with the central limit theorem, which makes it approximately true even when the individual weights aren't normal. A cautious king might also weigh the spread: scheme (i), a rare large fine, has by far the largest variance. The idea of finding factorial moments like $\mathrm{E}[N(N-1)]$ by differentiating a series leads to probability generating functions, a central tool in first-year probability. Grimmett and Welsh, *Probability: An Introduction*, covers all of this clearly.

WORKED SOLUTION

STEP II Specimen paper, Question 16

Probability and Statistics • Knockout Tournament Probability

STEP II Specimen paper, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q16

This question has a full guided treatment in *Beyond STEP*, Section 12.7, *Who Meets Whom in a Knockout*: https://beyondstep.keithmcnulty.org/ch12/knockout.html

The key idea

Follow the two chosen players, A and B, round by round. In any round where both are still in and have not yet met, only two things matter: is A drawn against B, and if not, do both of them win? Each of these is a simple probability, and the products that appear telescope beautifully. For the last part there is also a slick symmetry argument: count the matches in the whole tournament.

Solution

Write $N = 2^n$. Call the two chosen players A and B. After $r-1$ rounds there are 2^{n-r+1} players left, so round r has $M_r = 2^{n-r+1}$ players, and round n (with $M_n = 2$) is the final.

The fact we use repeatedly: if both A and B are among M players who are paired at random, then A's opponent is equally likely to be any of the other $M-1$ players, so

$$\mathrm{P}(A \text{ is drawn against } B) = \frac{1}{M-1}.$$

(In the final there is no draw, but with $M = 2$ the formula gives 1, which is right.)

Part (i)

In the first round $M_1 = N$, so

$$\mathrm{P}(A \text{ and } B \text{ meet in the first round}) = \frac{1}{2^n - 1}.$$

Part (ii)

Let S_r be the event that A and B are both still in the tournament at the start of round r (which forces them not to have met yet, since the loser of a match is out). Given S_r, they reach round $r+1$ together exactly when they are *not* drawn together in round r and then both win their (separate) matches. The two match results are independent, each with probability $\frac{1}{2}$, so

$$\mathrm{P}(S_{r+1} \mid S_r) = \frac{M_r - 2}{M_r - 1} \cdot \frac{1}{4}.$$

Ask yourself How do we turn this into a product that we can actually evaluate?

Write $M = 2^k$, and notice that $M - 2 = 2(2^{k-1} - 1)$, so

$$\frac{M-2}{M-1} = \frac{2\,(2^{k-1}-1)}{2^k - 1}.$$

The numerator for one round is (apart from the factor 2) the denominator for the next round. Since S_1 is certain, multiplying over the rounds with $k = n, n-1, \ldots, n-r+2$ gives a telescoping product:

$$\mathrm{P}(S_r) = \frac{1}{4^{r-1}} \prod_{k=n-r+2}^{n} \frac{2\,(2^{k-1}-1)}{2^k - 1} = \frac{2^{r-1}}{4^{r-1}} \cdot \frac{2^{n-r+1}-1}{2^n - 1} = \frac{2^{n-r+1}-1}{2^{r-1}\,(2^n - 1)}.$$

Given S_r, they meet in round r with probability $1/(M_r - 1) = 1/(2^{n-r+1} - 1)$, so

$$\mathrm{P}(A \text{ and } B \text{ meet in round } r) = \frac{1}{2^{r-1}\,(2^n - 1)}.$$

This is a pleasantly simple formula: each round halves the chance of meeting there. Setting $r = 1$ recovers part (i). For the final, $r = n$:

$$\mathrm{P}(A \text{ and } B \text{ meet in the final}) = \frac{1}{2^{n-1}\,(2^n - 1)}.$$

As a sanity check by symmetry: the rules treat every player identically, so the pair of finalists is equally likely to be any of the $\binom{N}{2} = 2^{n-1}(2^n - 1)$ pairs of players, which gives the same answer.

Part (iii)

The events "meet in round r" for $r = 1, \ldots, n$ are mutually exclusive (two players can meet at most once, as the loser leaves), so we add, using the geometric series:

$$\begin{aligned}\mathrm{P}(A \text{ and } B \text{ meet}) &= \frac{1}{2^n - 1} \sum_{r=1}^{n} \frac{1}{2^{r-1}} = \frac{1}{2^n - 1} \cdot 2\left(1 - \frac{1}{2^n}\right) \\ &= \frac{2}{2^n} \cdot \frac{2^n - 1}{2^n - 1} = \frac{1}{2^{n-1}}.\end{aligned}$$

Ask yourself Is there a way to see this answer without any summing?

Yes. Every match eliminates exactly one player and all but the champion are eliminated, so there are always exactly $2^n - 1$ matches. Each match is played by a pair of players, and by symmetry every one of the $\binom{N}{2}$ pairs has the same probability p of playing. The expected number of matches is the sum of these probabilities over all pairs, so

$$\binom{2^n}{2} p = 2^n - 1 \quad \Longrightarrow \quad p = \frac{2(2^n-1)}{2^n(2^n-1)} = \frac{2}{2^n} = \frac{1}{2^{n-1}},$$

in agreement with the sum.

Checks

- $n = 1$: two players, who certainly meet in the first round, which is the final. All three formulas give 1.
- $n = 2$: four players. Part (i) gives $\frac{1}{3}$, the final gives $\frac{1}{6}$, and the total is $\frac{1}{3} + \frac{1}{6} = \frac{1}{2}$, as part (iii) says.
- $n = 3$: enumerating every draw and every result exactly gives $\frac{1}{7}$, $\frac{1}{28}$ and $\frac{1}{4}$. A simulation of 2×10^5 tournaments with 16 players gave 0.0664, 0.0085 and 0.1247, against the exact $\frac{1}{15} \approx 0.0667$, $\frac{1}{120} \approx 0.0083$ and $\frac{1}{8}$.

Common slips

- Forgetting that, to meet in a later round, the two players must first *avoid* each other in every earlier round: the factor $\frac{M-2}{M-1}$ is easy to drop.
- Answering part (ii) with $\left(\frac{1}{2^{n-1}}\right)^2$, the chance that both reach the final *if they were independent*. They are not: if they are drawn together early, at most one of them can get through.

Beyond STEP

The argument in part (iii) is *linearity of expectation* with indicator variables: the expected number of matches is the sum, over all pairs, of the probability that the pair plays, whether or not the events are independent. This is one of the most powerful tools in probability and combinatorics, and it underlies the "probabilistic method" for proving that objects with given properties exist. Blitzstein and Hwang, *Introduction to Probability*, has a full chapter of lovely examples of indicator-variable arguments.

WORKED SOLUTION

STEP III Specimen paper, Question 1

Pure • Proof by Induction

STEP III Specimen paper, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q1

The key idea

For (i), work out the first few products and look at them the right way: once each one is written with denominator $2n$, the pattern jumps out. Factorising $1 - \frac{1}{m^2} = \frac{(m-1)(m+1)}{m^2}$ explains why, and makes the inductive step a one-line calculation. For (ii), induct on k with n fixed: adding one more term to the sum is exactly Pascal's rule in disguise.

Solution

Part (i)

Write $P_n = \left(1 - \frac{1}{4}\right)\left(1 - \frac{1}{9}\right)\cdots\left(1 - \frac{1}{n^2}\right)$. The first few values are

$$P_2 = \frac{3}{4}, \qquad P_3 = \frac{3}{4}\cdot\frac{8}{9} = \frac{2}{3}, \qquad P_4 = \frac{2}{3}\cdot\frac{15}{16} = \frac{5}{8}, \qquad P_5 = \frac{5}{8}\cdot\frac{24}{25} = \frac{3}{5}.$$

The denominators 4, 8 suggest writing everything over $2n$: $P_2 = \frac{3}{4}$, $P_3 = \frac{4}{6}$, $P_4 = \frac{5}{8}$, $P_5 = \frac{6}{10}$. So we guess

$$P_n = \frac{n+1}{2n} \qquad \text{for } n \geq 2.$$

Proof by induction. Let $S(n)$ be the statement $P_n = \dfrac{n+1}{2n}$.

Base case. $P_2 = 1 - \frac{1}{4} = \frac{3}{4} = \frac{2+1}{2\cdot 2}$, so $S(2)$ is true.

Inductive step. Suppose $S(n)$ is true for some $n \geq 2$. Then, since $P_{n+1} = P_n\left(1 - \frac{1}{(n+1)^2}\right)$ and $1 - \frac{1}{(n+1)^2} = \frac{(n+1)^2-1}{(n+1)^2} = \frac{n(n+2)}{(n+1)^2}$,

$$\begin{aligned} P_{n+1} &= \frac{n+1}{2n}\cdot\frac{n(n+2)}{(n+1)^2} \\ &= \frac{n+2}{2(n+1)}, \end{aligned}$$

which is $S(n+1)$. By induction, $S(n)$ holds for all $n \geq 2$.

(The factorisation also explains the guess: $P_n = \prod_{m=2}^{n} \frac{m-1}{m} \cdot \prod_{m=2}^{n} \frac{m+1}{m} = \frac{1}{n} \cdot \frac{n+1}{2}$, both products telescoping.)

Part (ii)

Fix a positive integer n, and let $T(k)$ be the statement

$$\sum_{r=0}^{k} (-1)^r \binom{n}{r} = (-1)^k \binom{n-1}{k}.$$

We prove $T(k)$ for $k = 0, 1, \ldots, n-1$ by induction on k.

Base case. For $k = 0$ both sides equal $\binom{n}{0} = \binom{n-1}{0} = 1$.

Inductive step. Suppose $T(k)$ holds for some k with $0 \leq k \leq n-2$ (so that $k+1$ is still in the range). Adding the next term,

$$\begin{aligned} \sum_{r=0}^{k+1} (-1)^r \binom{n}{r} &= (-1)^k \binom{n-1}{k} + (-1)^{k+1} \binom{n}{k+1} \\ &= (-1)^{k+1} \left[\binom{n}{k+1} - \binom{n-1}{k} \right]. \end{aligned}$$

? **Ask yourself** What is $\binom{n}{k+1} - \binom{n-1}{k}$?

Pascal's rule says $\binom{n}{k+1} = \binom{n-1}{k} + \binom{n-1}{k+1}$: choosing $k+1$ objects from n, either a particular object is chosen (and k more come from the other $n-1$) or it isn't (and all $k+1$ come from the other $n-1$). So the bracket is $\binom{n-1}{k+1}$, and

$$\sum_{r=0}^{k+1} (-1)^r \binom{n}{r} = (-1)^{k+1} \binom{n-1}{k+1},$$

which is $T(k+1)$. By induction, $T(k)$ holds for $k = 0, 1, \ldots, n-1$, as required.

If $n = 1$ the only case is $k = 0$, covered by the base case, so the argument needs no special treatment there.

Checks

- $P_{300} = \frac{301}{600} \approx 0.5017$, and a computer multiplying the factors exactly with fractions agreed with $\frac{n+1}{2n}$ for every n from 2 to 300. As $n \to \infty$ the product tends to $\frac{1}{2}$.
- For $n = 4$: the partial sums $1, 1-4 = -3, -3+6 = 3, 3-4 = -1$ match $\binom{3}{0}, -\binom{3}{1}, \binom{3}{2}, -\binom{3}{3}$. The computer checked (ii) exactly for all $n \leq 79$ and all $k \leq n-1$.
- Taking $k = n-1$ and adding the last term $(-1)^n \binom{n}{n}$ gives $(-1)^{n-1} + (-1)^n = 0$, which is the familiar fact $\sum_{r=0}^{n} (-1)^r \binom{n}{r} = (1-1)^n = 0$.

Common slips

- Guessing from fully simplified fractions ($\frac{3}{4}, \frac{2}{3}, \frac{5}{8}, \frac{3}{5}$) and missing the pattern: uncancelling to a common form is the trick.
- In (ii), inducting on n instead of k (possible, but messier), or forgetting that the step from k to $k+1$ only needs to run while $k+1 \leq n-1$.

Beyond STEP

Part (i) is the start of the infinite product $\prod_{m\geq 2}\left(1-\frac{1}{m^2}\right) = \frac{1}{2}$. It is a special case of Euler's product $\frac{\sin \pi x}{\pi x} = \prod_{m\geq 1}\left(1-\frac{x^2}{m^2}\right)$: divide both sides by the $m=1$ factor $1-x^2$ and let $x \to 1$. Making sense of infinite products and when they converge is a topic in first-year analysis. Part (ii) is one of many binomial identities that follow from Pascal's rule or from generating functions: it is the coefficient of x^k in $(1-x)^n \cdot \frac{1}{1-x} = (1-x)^{n-1}$. Abbott's *Understanding Analysis* is a good place to read about convergence of infinite sums and products.

WORKED SOLUTION

STEP III Specimen paper, Question 2

Pure • Binomial Inequalities and Limits

STEP III Specimen paper, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q2

The key idea

Writing $y = 1 + x$ with $x > 0$ turns powers of y into binomial expansions, and every term of such an expansion is non-negative. So we may throw away all but one term and still have a lower bound. Keeping the term in x gives a bound that grows like n, which beats any fixed K; to beat K after dividing by n we need a bound that grows like n^2, so we keep the term in x^2 instead.

Solution

The inequality

Let $x \geq 0$ and let n be a positive integer. By the binomial theorem,

$$(1+x)^n = 1 + nx + \binom{n}{2}x^2 + \cdots + x^n.$$

Since $x \geq 0$, every term after the first two is non-negative, so

$$(1+x)^n \geq 1 + nx > nx.$$

The final inequality is strict because of the 1, and that is what makes the result hold even when $x = 0$ (where it reads $1 > 0$).

Deducing that y^n eventually exceeds K

Let $y > 1$ and put $x = y - 1$, so $x > 0$. The first part gives, for every positive integer n,

$$y^n = (1+x)^n > n(y-1).$$

If $K \leq 0$ there is nothing to do: $y^n > 0 \geq K$ for every n, so $N = 1$ works. If $K > 0$, choose an integer N with

$$N > \frac{K}{y-1},$$

which we can do because $y - 1 > 0$, so the right-hand side is a fixed number. Then for every integer $n \geq N$,

$$y^n > n(y-1) \geq N(y-1) > K,$$

using $y - 1 > 0$ in the middle step. This is exactly what we needed. Note that the same N works for *all* $n \geq N$, not just for $n = N$, because the lower bound $n(y-1)$ increases with n.

Showing that y^n/n eventually exceeds K

The bound $y^n > n(y-1)$ only gives $y^n/n > y - 1$, which is a fixed number and is no use against a large K. We need a lower bound for y^n that grows faster than n.

Ask yourself Which term of the binomial expansion should we keep this time?

Keep the x^2 term. With $x = y - 1 > 0$, all terms of the expansion are positive, so for every positive integer n,

$$y^n = (1+x)^n \geq 1 + nx + \frac{n(n-1)}{2}x^2 > \frac{n(n-1)}{2}x^2.$$

(For $n = 1$ there is no x^2 term, but the final inequality still holds since its right-hand side is 0.) Dividing by $n > 0$,

$$\frac{y^n}{n} > \frac{(n-1)(y-1)^2}{2}.$$

The right-hand side now grows without limit as n increases, and that is what we need. If $K \leq 0$, then $N = 1$ works since $y^n/n > 0$. If $K > 0$, choose an integer N with

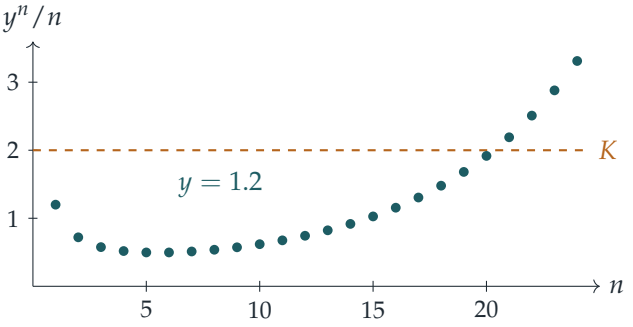
$$N > 1 + \frac{2K}{(y-1)^2}.$$

Then for every integer $n \geq N$ we have $n - 1 \geq N - 1 > 2K/(y-1)^2$, and so

$$\frac{y^n}{n} > \frac{(n-1)(y-1)^2}{2} > \frac{2K}{(y-1)^2} \cdot \frac{(y-1)^2}{2} = K,$$

which is the required result.

Again the bound increases with n, so the argument works for every $n \geq N$. This matters here: y^n/n itself is *not* always increasing. For y close to 1 it falls at first (for example, $y^2/2 < y$ whenever $y < 2$), as the sketch for $y = 1.2$ shows. So we could not simply find one n with $y^n/n > K$ and argue that later terms are larger.



Checks

Try $y = 1.01$ and $K = 1000$. Our recipes give $N = 100\,001$ for $y^n > K$ and $N = 20\,000\,002$ for $y^n/n > K$. A direct computation shows that $1.01^n/n > 1000$ already holds for every $n \geq 1424$. Our N is far from the smallest possible, but the question only asks for *some* N. We also tested both inequalities exactly with rational x for thousands of random x and n, and tested the chosen N (and many larger n) for thousands of random $y > 1$ and K.

Common slips

- Showing only that $y^N > K$ for one N, without explaining why $y^n > K$ for *every* $n \geq N$.
- Using $y^n > n(y-1)$ for the last part: dividing by n leaves a constant, which cannot beat an arbitrary K.
- Forgetting that K may be negative or zero, in which case $K/(y-1)$ is not a sensible lower bound on its own (though any N then works).

Beyond STEP

This is the first rigorous "ε–N" style argument most students meet: it proves that $y^n \to \infty$ and $y^n/n \to \infty$ when $y > 1$. The same trick, keeping the x^{k+1} term, shows that $y^n/n^k \to \infty$ for any fixed k, so exponentials beat every polynomial. Abbott's *Understanding Analysis* builds the theory of sequences and limits from exactly this kind of reasoning.

WORKED SOLUTION

STEP III Specimen paper, Question 3

Pure • Complex Triangle Inequality

STEP III Specimen paper, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q3

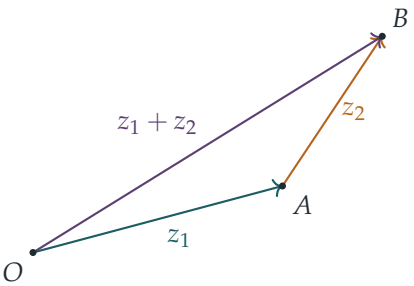
The key idea

The inequality is the triangle inequality: no side of a triangle is longer than the other two put together. Applied repeatedly, it lets us bound the modulus of a whole sum by the sum of the moduli. If the left-hand side of the equation is always strictly smaller than 1 in modulus when $|z| \leq \frac{1}{3}$, it can never equal 1, and a geometric series does the bounding.

Solution

The geometric meaning

In the Argand diagram, think of z_1 as the displacement from the origin O to a point A, and z_2 as the displacement from A to a point B. Then $z_1 + z_2$ is the displacement from O straight to B.



The lengths of the sides of triangle OAB are $|z_1|$, $|z_2|$ and $|z_1 + z_2|$. The inequality says that going straight from O to B is never longer than going via A: one side of a triangle is at most the sum of the other two. Equality holds exactly when the triangle is flat with A lying on the segment OB, that is, when z_1 and z_2 point in the same direction (one is a non-negative real multiple of the other, or one of them is zero).

By induction, the same holds for any finite number of terms:

$$|w_1 + w_2 + \cdots + w_n| \leq |w_1| + |w_2| + \cdots + |w_n|,$$

since $|w_1 + \cdots + w_{k+1}| \leq |w_1 + \cdots + w_k| + |w_{k+1}|$. Geometrically, a path made of n straight steps is at least as long as the straight line from its start to its end.

No solutions with $|z| \leq \frac{1}{3}$

Suppose $|z| \leq \frac{1}{3}$ and $|a_i| \leq 2$ for every i. Using the extended triangle inequality, then $|ab| = |a||b|$,

$$\begin{aligned}\left|a_1 z + a_2 z^2 + \cdots + a_n z^n\right| &\leq |a_1||z| + |a_2||z|^2 + \cdots + |a_n||z|^n \\ &\leq 2\left(\tfrac{1}{3} + \tfrac{1}{3^2} + \cdots + \tfrac{1}{3^n}\right),\end{aligned}$$

where the second step uses $|a_i| \leq 2$ and $|z|^k \leq \left(\frac{1}{3}\right)^k$ for each k (all the quantities involved are non-negative, so we may bound term by term).

The bracket is a finite geometric series with first term $\frac{1}{3}$ and ratio $\frac{1}{3}$:

$$2\sum_{k=1}^{n} \frac{1}{3^k} = 2 \cdot \frac{\frac{1}{3}\left(1 - 3^{-n}\right)}{1 - \frac{1}{3}} = 1 - \frac{1}{3^n} < 1.$$

Ask yourself Why is the inequality strict, when each step above used only $\leq$?

The strictness comes from the series, not from the triangle inequality. Because the sum stops at n terms, it falls short of its infinite total 1 by exactly $3^{-n} > 0$. So whatever the a_i and whatever z with $|z| \leq \frac{1}{3}$,

$$\left|a_1 z + a_2 z^2 + \cdots + a_n z^n\right| \leq 1 - 3^{-n} < 1.$$

But if z were a solution, the left-hand side would be $|1| = 1$. This is a contradiction, so the equation has no solutions with $|z| \leq \frac{1}{3}$, as required.

Checks

- The worst case is $a_1 = \cdots = a_n = 2$ and $z = \frac{1}{3}$, when every inequality becomes an equality and the left-hand side is $1 - 3^{-n}$. For $n = 10$ this is $0.99998\ldots$, just short of 1.
- The constant $\frac{1}{3}$ cannot be improved. With every $a_i = 2$, the real function $2(x + x^2 + \cdots + x^n) - 1$ is negative at $x = \frac{1}{3}$, and for large n it is positive just beyond $\frac{1}{3}$, so there are real roots as close to $\frac{1}{3}$ as we like.
- A numerical test of 200 000 random choices of $n \leq 30$, of a_i with $|a_i| \leq 2$ and of z with $|z| \leq \frac{1}{3}$ never gave a left-hand side of modulus 1 or more.

Common slips

- Bounding by the infinite series and getting ≤ 1, which does not rule out equality. You need the strict inequality, and it comes from the series being finite.
- Writing $|a_1 z + \cdots + a_n z^n| \leq |a_1 z| + \cdots$ without saying why: this is the triangle inequality extended to n terms, which is the reason the first part is there.

Beyond STEP

This is the simplest example of a *root-free region*: bounds on the coefficients of a polynomial keep its roots away from a disc. The same estimate, run the other way, gives Cauchy's bound that every root of $z^n + c_{n-1}z^{n-1} + \cdots + c_0$ has modulus less than $1 + \max |c_k|$, and in complex analysis Rouché's theorem turns arguments like this into exact counts of roots inside a circle. For a beautifully geometric account, read Needham, *Visual Complex Analysis*.

WORKED SOLUTION

STEP III Specimen paper, Question 4

Pure • Tangents to a Parabola

STEP III Specimen paper, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q4

The key idea

A line touches a parabola exactly when substituting it into the parabola's equation gives a quadratic with a repeated root. So the first part is a one-line discriminant calculation, and it hands us a family of tangents labelled by t. For the second part, think of the tangents from a given point as the values of t satisfying a quadratic: then everything about L and M can be written in terms of the sum and product of the roots, which come straight from the coefficients.

Solution

The tangent line

From the line, $x = ty - at^2$. Substituting into $y^2 = 4ax$ gives

$$y^2 = 4aty - 4a^2t^2 \iff y^2 - 4aty + 4a^2t^2 = 0 \iff (y - 2at)^2 = 0.$$

This quadratic in y has the repeated root $y = 2at$, so the line meets the parabola at exactly one point, and does so with multiplicity two: it touches the parabola. (It is not a line parallel to the axis, which would also meet the parabola only once, because it is not horizontal; for $t = 0$ it is the line $x = 0$, the tangent at the vertex.) The corresponding x is $x = t(2at) - at^2 = at^2$, so the point of contact is

$$(at^2,\ 2at).$$

As a cross-check, differentiating $y^2 = 4ax$ implicitly gives $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2a}{y} = \frac{1}{t}$ at this point, which is the gradient of $ty = x + at^2$.

The locus

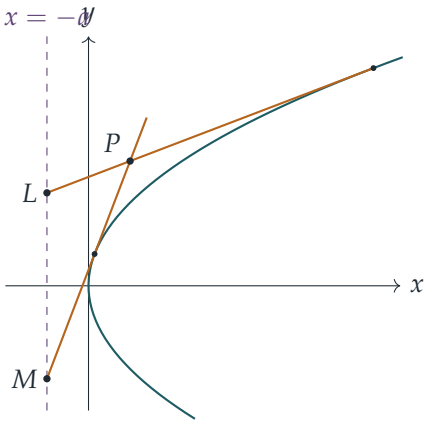
Every point of the parabola is $(at^2, 2at)$ for exactly one t, so every tangent is one of the lines $ty = x + at^2$. Let the given point be $P = (h, k)$. The tangent with parameter t

passes through P if and only if

$$at^2 - kt + h = 0.$$

For two tangents we need two distinct real roots t_1, t_2, that is $k^2 > 4ah$ (P lies outside the parabola). Then

$$t_1 + t_2 = \frac{k}{a}, \qquad t_1 t_2 = \frac{h}{a}.$$



The tangent with parameter t meets the directrix $x = -a$ where $ty = -a + at^2$. Provided $t \neq 0$,

$$y = a\left(t - \frac{1}{t}\right).$$

(If $t = 0$ the tangent is $x = 0$, parallel to the directrix, and there is no point of intersection. That happens only when $h = t_1 t_2\, a = 0$, so for L and M to exist we need $h \neq 0$, and then $t_1, t_2 \neq 0$.) So L and M are $\left(-a,\ a(t_1 - \frac{1}{t_1})\right)$ and $\left(-a,\ a(t_2 - \frac{1}{t_2})\right)$, and since both lie on the same vertical line,

$$\begin{aligned} LM &= \left|a(t_1 - t_2) - a\left(\frac{1}{t_1} - \frac{1}{t_2}\right)\right| = \left|a(t_1 - t_2) + a\,\frac{t_1 - t_2}{t_1 t_2}\right| \\ &= |a|\,|t_1 - t_2|\,\left|\frac{t_1 t_2 + 1}{t_1 t_2}\right|. \end{aligned}$$

Ask yourself How do we get $t_1 - t_2$ without solving the quadratic?

Its square is symmetric in the roots:

$$(t_1 - t_2)^2 = (t_1 + t_2)^2 - 4t_1 t_2 = \frac{k^2}{a^2} - \frac{4h}{a} = \frac{k^2 - 4ah}{a^2}.$$

Also

$$\frac{t_1 t_2 + 1}{t_1 t_2} = \frac{h/a + 1}{h/a} = \frac{h + a}{h}.$$

Squaring the expression for LM and substituting,

$$LM^2 = a^2 \cdot \frac{k^2 - 4ah}{a^2} \cdot \frac{(h+a)^2}{h^2} = \frac{(h+a)^2(k^2-4ah)}{h^2}.$$

If $LM = \ell$, multiplying through by h^2 gives

$$(h+a)^2(k^2-4ah) = \ell^2 h^2,$$

so $P = (h,k)$ lies on $(x+a)^2(y^2-4ax) = \ell^2 x^2$, as required.

Checks

- Take $a = 1$ and $P = (1,3)$, as in the diagram. The parameters satisfy $t^2 - 3t + 1 = 0$, so $t = \frac{1}{2}(3 \pm \sqrt{5})$, and $t - \frac{1}{t} = \pm\sqrt{5}$ (because $t_1 t_2 = 1$ makes $1/t_1 = t_2$). So $LM = 2\sqrt{5}$ and $LM^2 = 20$, while the formula gives $(1+1)^2(9-4)/1^2 = 20$.
- A point on the directrix, $h = -a$: the formula says $LM = 0$. Indeed the tangents from such a point meet the directrix at the point itself, so $L = M$.
- We found the tangents independently (lines $y = k + m(x-h)$ with $hm^2 - km + a = 0$) for over 14,000 random a, h, k, and the computed LM^2 matched $(h+a)^2(k^2-4ah)/h^2$ every time. For example $a = 1$, $P = (2,5)$ gives $LM^2 = 38.25$ both ways.

Common slips

- Using the tangent's parameter equation the wrong way round: P fixed and t unknown gives the quadratic $at^2 - kt + h = 0$, not one in h and k.
- Writing $\frac{1}{t_1} - \frac{1}{t_2} = \frac{t_1 - t_2}{t_1 t_2}$; it is $\frac{t_2 - t_1}{t_1 t_2}$, and the sign is what makes the two terms add rather than cancel.

Beyond STEP

The tangents from a point to a conic, and the line joining their points of contact (the *polar* of the point), are the start of projective geometry, where poles, polars and the directrix fit into one picture: the directrix is the polar of the focus. Working with roots through their symmetric functions, as here, is also the first step towards the theory of equations and Galois theory. Siklos's *Advanced Problems in Mathematics* has more coordinate geometry problems in this style.

WORKED SOLUTION

STEP III Specimen paper, Question 5

Pure • Counting Equation Roots

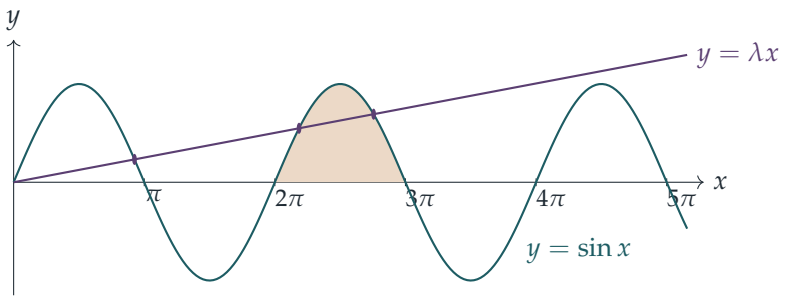
STEP III Specimen paper, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q5

The key idea

Picture the line $y = \lambda x$ against the curve $y = \sin x$: solutions are crossings. The line can only meet the curve where $\sin x > 0$, that is on the positive "arches" $\big((2n-2)\pi, (2n-1)\pi\big)$, and because $\sin x$ is concave on each arch, the line crosses an arch either twice or not at all (apart from the first arch, which starts at the origin). For the small-ϵ estimate, the Maclaurin series $\sin x \approx x - \frac{1}{6}x^3$ does the work.



(Here $\lambda = 0.08$: one crossing on the first arch, two on the shaded second arch, and the line has risen above 1 before the third arch peaks, so $N = 3$.)

Solution

Part (i)

No solutions for $\lambda > 1$. The curve $y = \sin x$ has gradient $\cos x \leq 1$ and passes through the origin, and the line $y = x$ is its tangent there. For $x > 0$ we have $\sin x < x$ (the function $x - \sin x$ is zero at 0 and has derivative $1 - \cos x \geq 0$, equal to zero only at isolated points, so it is strictly increasing). Graphically, the curve lies strictly below its tangent $y = x$ for $x > 0$, and if $\lambda > 1$ the line $y = \lambda x$ lies above $y = x$ there. Hence

$$\sin x < x < \lambda x \qquad \text{for all } x > 0,$$

and there is no non-zero solution.

The case $\lambda = 1 - \epsilon^2$. For $x > 0$ the equation is equivalent to

$$\frac{\sin x}{x} = 1 - \epsilon^2.$$

First, a solution exists and it is the only one. On $(0, \pi]$ the function $\frac{\sin x}{x}$ is continuous, tends to 1 as $x \to 0^+$ and equals 0 at $x = \pi$, so it takes the value $1 - \epsilon^2$ somewhere in $(0, \pi)$; it is strictly decreasing there (the line through the origin to a point of the concave arch gets less steep as the point moves along it), so this happens just once. For $x \geq \pi$ there is nothing more: $\lambda x \geq (1 - \epsilon^2)\pi > 1 \geq \sin x$ because ϵ is small.

Now we locate it. Using $\sin x = x - \frac{1}{6}x^3 + \frac{1}{120}x^5 - \cdots$,

$$\frac{\sin x}{x} = 1 - \frac{x^2}{6} + \frac{x^4}{120} - \cdots = 1 - \epsilon^2 \quad \Longrightarrow \quad \frac{x^2}{6} - \frac{x^4}{120} + \cdots = \epsilon^2.$$

? **Ask yourself** Why may we throw away the x^4 term? We don't yet know that x is small.

As $\epsilon \to 0$, $\frac{\sin x}{x} \to 1$, and since $\frac{\sin x}{x}$ is strictly decreasing from 1 on $(0, \pi)$, the root must tend to 0. So for small ϵ the root is small, and then x^4 is negligible beside x^2. To leading order $x^2/6 = \epsilon^2$, so

$$x \approx \epsilon\sqrt{6}.$$

(Keeping the next term, $x^2 \approx 6\epsilon^2 + \frac{6}{120}(6\epsilon^2)^2$, gives the refinement $x \approx \epsilon\sqrt{6}\left(1 + \frac{3}{20}\epsilon^2\right)$.)

Part (ii)

Let $g(x) = \sin x - \lambda x$, so the solutions are the zeros of g with $x > 0$. Since $\lambda x > 0$ for $x > 0$, every solution has $\sin x > 0$, so it lies on one of the arches

$$A_n = \big((2n-2)\pi,\, (2n-1)\pi\big), \qquad n = 1, 2, 3, \ldots$$

The first arch. At the origin the line has gradient $\lambda < 1$, smaller than the curve's, so the line starts below the curve; at $x = \pi$ it is above ($\lambda\pi > 0 = \sin\pi$). Since $\sin x$ is concave on A_1 and the line is straight, they cross exactly once there. So A_1 holds exactly one solution, which must be x_1, and $0 < x_1 < \pi$.

Later arches. On $A_{n+1} = (2n\pi, (2n+1)\pi)$ with $n \geq 1$, $g''(x) = -\sin x < 0$, so g is concave and has at most two zeros. At both ends of the arch $\sin x = 0 < \lambda x$, so $g < 0$ there. The maximum of g is where $g'(x) = \cos x - \lambda = 0$, at $c_n = 2n\pi + \arccos\lambda$, and $c_n < (2n + \frac{1}{2})\pi$ because $\arccos\lambda < \frac{\pi}{2}$ for $\lambda > 0$. Thus g increases on $(2n\pi, c_n)$ and decreases on $(c_n, (2n+1)\pi)$, so:

- if $g(c_n) < 0$ the line misses the arch;
- if $g(c_n) = 0$ the line touches it at one point;
- if $g(c_n) > 0$ there are exactly two solutions, one in $(2n\pi, c_n)$ and one in $(c_n, (2n+1)\pi)$.

Also, once the line misses an arch it misses all later ones: if $x \in A_{n+2}$ then $x - 2\pi \in A_{n+1}$, so $\sin x = \sin(x-2\pi) < \lambda(x-2\pi) < \lambda x$.

So the arches the line crosses are $A_1, A_2, \ldots$ up to some last one, contributing $1, 2, 2, \ldots$ solutions, possibly followed by a single tangency. Since $N = 2R+1$ is odd there is no tangency, and the line crosses exactly the arches $A_1, \ldots, A_{R+1}$. Listing the solutions in increasing order, arch A_{n+1} ($1 \le n \le R$) contains x_{2n} and x_{2n+1}, with

$$2n\pi < x_{2n} < c_n < (2n+\tfrac{1}{2})\pi, \qquad 2n\pi < x_{2n+1} < (2n+1)\pi.$$

The first gives the second line of the required inequalities. Writing x_{2n+1} as $x_{2(n+1)-1}$, the second, together with $0 < x_1 < \pi$, gives $(2m-2)\pi < x_{2m-1} < (2m-1)\pi$ for $m = 1, \ldots, R+1$. This is the result.

Estimating N. For small λ the line crosses many arches, so $R \ge 1$. At x_{2R} we have $\lambda x_{2R} = \sin x_{2R} \le 1$, and $x_{2R} > 2R\pi$ from above, so

$$2R\pi\lambda < 1.$$

On the other hand the line misses $A_{R+2} = \big((2R+2)\pi, (2R+3)\pi\big)$. At its midpoint $(2R+\frac{5}{2})\pi$ we have $\sin x = 1$, so we need $\lambda(2R+\frac{5}{2})\pi > 1$ (otherwise $g \ge 0$ there and the line would meet the arch). Combining,

$$\frac{1}{2\pi\lambda} - \frac{5}{4} < R < \frac{1}{2\pi\lambda}, \qquad \text{so} \qquad \frac{1}{\pi\lambda} - \frac{3}{2} < N < \frac{1}{\pi\lambda} + 1.$$

When λ is very small, $\frac{1}{\pi\lambda}$ is huge and the bounded error is negligible in comparison, so

$$N \approx \frac{1}{\pi\lambda}.$$

This makes sense: the line reaches height 1 at $x = 1/\lambda$, and before that point there are about $\frac{1}{2\pi\lambda}$ complete periods of $\sin x$, each contributing two crossings.

Checks

- With $\epsilon = 0.01$ ($\lambda = 0.9999$), bisection gives the root 0.0244953, against $\epsilon\sqrt{6} = 0.0244949$; the ratio is 1.000015, exactly the $1 + \frac{3}{20}\epsilon^2$ correction.
- For $\lambda = 0.001$ the computer finds $N = 317$ roots, against $\frac{1}{\pi\lambda} \approx 318.3$; for $\lambda = 0.01$ it finds 31 against 31.8. All roots found for several λ obey the interval inequalities of (ii), and N lies within the bounds above.
- No roots were found for $\lambda = 1.0001$, 1.1, 2 or 5.

Common slips

- Forgetting that x_1 lives on the first arch alone: the origin is a crossing too, which is why that arch contributes one non-zero solution rather than two.
- Placing x_{2n} in the first half of its arch without a reason. The point is that g only starts decreasing after $c_n = 2n\pi + \arccos\lambda$, which is before the peak of $\sin x$.

Beyond STEP

The approximation $x \approx \epsilon\sqrt{6}$ is the first step of an asymptotic expansion, and the argument that the root must be small before we truncate the series is the kind of care analysis courses insist on. Counting solutions by concavity, monotonicity and the intermediate value theorem is standard first-year analysis; see Abbott, *Understanding Analysis*. A related classic is $\tan x = x$, whose roots approach $(n + \frac{1}{2})\pi$ and which appears in heat conduction and vibration problems.

WORKED SOLUTION

STEP III Specimen paper, Question 6

Pure • Hyperbolic Reduction Formulae

STEP III Specimen paper, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q6

The key idea

The integrand $\operatorname{sech}^{n+2}u\,\sinh^2 u$ splits naturally as $\sinh u$ times $\sinh u\,\operatorname{sech}^{n+2}u$, and the second factor is, up to a constant, the derivative of $\operatorname{sech}^{n+1}u$. That makes integration by parts the obvious move. For the reduction formula, the identity $\cosh^2 u - \sinh^2 u = 1$ lets us rewrite the same integral as a difference of two of the I's.

Solution

The integral identity

First, a derivative we shall need. Since $\operatorname{sech} u = (\cosh u)^{-1}$, the chain rule gives

$$\frac{\mathrm{d}}{\mathrm{d}u}\operatorname{sech}^{n+1}u = -(n+1)\cosh^{-n-2}u\,\sinh u = -(n+1)\operatorname{sech}^{n+2}u\,\sinh u.$$

So we integrate by parts with

$$f = \sinh u, \qquad g' = \operatorname{sech}^{n+2}u\,\sinh u, \qquad \text{so} \qquad f' = \cosh u, \qquad g = -\frac{\operatorname{sech}^{n+1}u}{n+1}.$$

This gives

$$\begin{aligned}\int_0^\infty \operatorname{sech}^{n+2}u\,\sinh^2 u\,\mathrm{d}u &= \left[-\frac{\sinh u\,\operatorname{sech}^{n+1}u}{n+1}\right]_0^\infty + \frac{1}{n+1}\int_0^\infty \cosh u\,\operatorname{sech}^{n+1}u\,\mathrm{d}u \\ &= -\frac{1}{n+1}\left[\tanh u\,\operatorname{sech}^n u\right]_0^\infty + \frac{1}{n+1}\int_0^\infty \operatorname{sech}^n u\,\mathrm{d}u,\end{aligned}$$

using $\sinh u\,\operatorname{sech}^{n+1}u = \tanh u\,\operatorname{sech}^n u$ and $\cosh u\,\operatorname{sech}^{n+1}u = \operatorname{sech}^n u$.

? **Ask yourself** Why does the boundary term vanish, and where is $n > 0$ used?

At $u = 0$ we have $\tanh 0 = 0$. As $u \to \infty$, $\tanh u \to 1$ while $\operatorname{sech} u = 2/(\mathrm{e}^u + \mathrm{e}^{-u}) \to 0$, so $\operatorname{sech}^n u \to 0$ *because* $n > 0$. (For $n = 0$ the term would tend to 1, not 0.) The same

fact, $\operatorname{sech}^n u < 2^n e^{-nu}$, shows that I_n converges for $n > 0$. Hence the boundary term is zero and

$$\int_0^\infty \operatorname{sech}^{n+2} u \sinh^2 u \, du = \frac{1}{n+1} I_n,$$

as required.

The reduction formula

Now evaluate the same integral a second way. Since $\sinh^2 u = \cosh^2 u - 1$,

$$\operatorname{sech}^{n+2} u \sinh^2 u = \operatorname{sech}^{n+2} u \cosh^2 u - \operatorname{sech}^{n+2} u = \operatorname{sech}^n u - \operatorname{sech}^{n+2} u.$$

Both pieces have convergent integrals over $[0, \infty)$ (as $n > 0$), so

$$\int_0^\infty \operatorname{sech}^{n+2} u \sinh^2 u \, du = I_n - I_{n+2}.$$

Comparing with the first result,

$$I_n - I_{n+2} = \frac{1}{n+1} I_n \quad \Longrightarrow \quad I_{n+2} = \left(1 - \frac{1}{n+1}\right) I_n = \frac{n}{n+1} I_n,$$

that is, $(n+1) I_{n+2} = n I_n$, as required.

The value of I_6

The reduction formula steps n by 2, so we need a starting value with n even. The easiest is

$$I_2 = \int_0^\infty \operatorname{sech}^2 u \, du = \Big[\tanh u\Big]_0^\infty = 1 - 0 = 1.$$

(We can't start from I_0: it diverges, and the formula with $n = 0$ would give the false statement $I_2 = 0$. This is exactly where $n > 0$ matters.) Then

$$I_4 = \frac{2}{3} I_2 = \frac{2}{3},$$
$$I_6 = \frac{4}{5} I_4 = \frac{4}{5} \cdot \frac{2}{3} = \frac{8}{15}.$$

So

$$I_6 = \frac{8}{15}.$$



Checks

- The sequence $I_2, I_4, I_6 = 1, \frac{2}{3}, \frac{8}{15}$ is decreasing, as it must be: $0 < \operatorname{sech} u \leq 1$, so higher powers are smaller.
- Starting instead from $I_1 = \int_0^\infty \operatorname{sech} u \, \mathrm{d}u = \left[2 \arctan(\mathrm{e}^u)\right]_0^\infty = \frac{\pi}{2}$, the formula gives $I_3 = \frac{1}{2} I_1 = \frac{\pi}{4}$, which agrees with direct numerical integration.
- Simpson's rule on $[0, 60]$ gives $I_6 \approx 0.5333333$, matching $\frac{8}{15}$; both the identity and the reduction formula also check numerically for non-integer n, such as $n = 0.5$ and $n = 2.5$.

Common slips

- Getting the sign or the factor $n + 1$ wrong in $\frac{\mathrm{d}}{\mathrm{d}u} \operatorname{sech}^{n+1} u$: $\operatorname{sech} u$ is decreasing for $u > 0$, so the derivative must be negative.
- Not justifying why the boundary term vanishes at infinity. This is the one place the condition $n > 0$ is needed.
- Writing the answer as $\frac{4}{5} \cdot \frac{2}{3} \cdot I_2$ without evaluating I_2.

Beyond STEP

Reduction formulae like this one are the same structure as Wallis's formula: with $t = \tanh u$ (or $\sin \theta = \tanh u$, the Gudermannian substitution), I_n becomes $\int_0^{\pi/2} \cos^{n-1} \theta \, \mathrm{d}\theta$, and the recurrence is Wallis's. At university these integrals reappear as values of the Beta function, $I_n = \frac{1}{2} B\left(\frac{n}{2}, \frac{1}{2}\right)$, which extends the recurrence to all real $n > 0$, and the care we took over the boundary term is the beginning of the theory of improper integrals. For a rigorous treatment, see Abbott, *Understanding Analysis*; for many more problems in this style, Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP III Specimen paper, Question 7

Pure • Reduction of Order

STEP III Specimen paper, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q7

The key idea

When a second-order linear equation has x^2y'', xy' and y in it, a power of x is the natural thing to try: each term then becomes a multiple of x^α, and the equation turns into a condition on α. Once one solution y_1 is known, writing $y = y_1 v$ is the method of *reduction of order*: the terms in v itself cancel, leaving a first-order equation for v'.

Solution

A power solution

Try $y = x^\alpha$ (for $x > 0$, so that x^α makes sense for any real α). Then $y' = \alpha x^{\alpha-1}$ and $y'' = \alpha(\alpha-1)x^{\alpha-2}$, so

$$xy' - y = (\alpha-1)x^\alpha, \qquad x^2y'' = \alpha(\alpha-1)x^\alpha.$$

Substituting,

$$x^2y'' + (x-2)(xy'-y) = \big[\alpha(\alpha-1) + (x-2)(\alpha-1)\big]x^\alpha = (\alpha-1)(\alpha+x-2)\,x^\alpha.$$

For this to vanish for *all* x we need the bracket to be zero identically. The factor $\alpha + x - 2$ depends on x, so it can't be identically zero; hence $\alpha = 1$.

So $y = x$ is a solution, and you can see it directly: $y'' = 0$ and $xy' - y = x - x = 0$. Any multiple $y = Ax$ is then also a solution, which is the sense in which the solution is "proportional to x^α".

The general solution

Following the hint, put $y = xv$, where v is a function of x. Then

$$y' = v + xv', \qquad y'' = 2v' + xv'',$$

and the combination that appears in the equation simplifies nicely:

$$xy' - y = xv + x^2v' - xv = x^2v'.$$

Substituting into the equation,

$$\begin{aligned} 0 &= x^2(2v' + xv'') + (x-2)x^2v' \\ &= x^2\left(xv'' + 2v' + xv' - 2v'\right) \\ &= x^3\left(v'' + v'\right). \end{aligned}$$

Ask yourself Why has every term in v itself (with no derivative) disappeared?

Because $y = x$ is a solution: if v were constant, $y = xv$ would be a multiple of x and the left-hand side would be zero. So only v' and v'' can survive. This is what makes reduction of order work, and it is a useful check on the algebra.

On any interval not containing 0 we may divide by x^3, giving

$$v'' + v' = 0.$$

This is first order in $w = v'$: $w' = -w$, so $w = Be^{-x}$ for a constant B. Integrating once more (and absorbing the sign into the constant),

$$v = A + Be^{-x}.$$

Therefore the general solution is

$$\boxed{y = Ax + Bxe^{-x}}$$

with A and B arbitrary constants. It contains two arbitrary constants, as the general solution of a second-order linear equation must, and the two pieces x and xe^{-x} are not multiples of each other.

As a direct check of the new piece, $y = xe^{-x}$ gives $y' = (1-x)e^{-x}$ and $y'' = (x-2)e^{-x}$, so

$$x^2y'' + (x-2)(xy' - y) = (x-2)e^{-x}\left[x^2 + x(1-x) - x\right] = 0.$$

A remark on $x = 0$. The coefficient of y'' vanishes at $x = 0$, which is why we divided by x^3 only away from 0. In principle one could take different constants on $x > 0$ and $x < 0$; but every such function has $y(0) = 0$, $y'(0^{\pm}) = A_{\pm} + B_{\pm}$ and $y''(0^{\pm}) = -2B_{\pm}$, so for y to be twice differentiable at 0 we need $B_+ = B_-$ and then $A_+ = A_-$. So $y = Ax + Bxe^{-x}$ is the general solution on the whole real line too.

Checks

- Substitute both $y = x$ and $y = xe^{-x}$ back into the equation, as above: each gives zero.
- Numerically: solving the equation step by step from $x = 1$ with $y(1) = y_0$, $y'(1) = y_1$, the formula predicts $A = y_1$, $B = (y_0 - y_1)e$. With $y_0 = 2$, $y_1 = 1$ this gives $y(4) = 4 + 4e^{-3} \approx 4.199$, and a fourth-order Runge–Kutta integration agrees to better than 10^{-8} for several random starting values.

Common slips

- Concluding $\alpha = 2 - x$ from the factor $(\alpha + x - 2)$: α must be a constant, so this root is not allowed.
- Making an error in y'' for $y = xv$ (it is $2v' + xv''$, not $v' + xv''$). If the terms in v don't cancel, the differentiation has gone wrong.
- Stopping at $v' = Be^{-x}$ and forgetting the second constant when integrating to get v.

Beyond STEP

Reduction of order works for any second-order linear equation $y'' + p(x)y' + q(x)y = 0$: given one solution y_1, the substitution $y = y_1 v$ always gives a first-order linear equation for v', and the second solution can be written as an integral involving y_1 and p. It is closely tied to the Wronskian and Abel's formula, and the trial of x^α at a point like $x = 0$ is the first step of the Frobenius method for series solutions. All of this is covered in Boyce and DiPrima, *Elementary Differential Equations*.

WORKED SOLUTION

STEP III Specimen paper, Question 8

Pure • Vectors and Cauchy-Schwarz

STEP III Specimen paper, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q8

The key idea

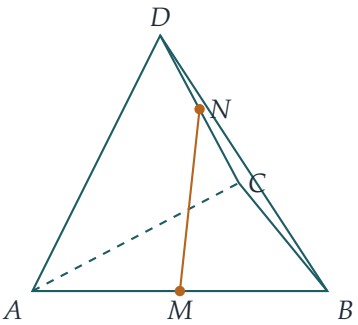
In part (i), give the four vertices position vectors. Every length we need is the modulus of a simple combination of them, and squared lengths are dot products we can expand. Both sides then come out as the same expression, and symmetry does most of the bookkeeping. In part (ii), the left-hand side looks like a dot product with the fixed vector $(3, 2, 6)$, whose length is a whole number. That is exactly what the Cauchy–Schwarz inequality is made for.

Solution

Part (i)

Let the tetrahedron be $ABCD$, with position vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}$ relative to any origin. Its six edges come in three pairs of opposite edges (edges with no common vertex):

$$AB \text{ and } CD, \qquad AC \text{ and } BD, \qquad AD \text{ and } BC.$$



For the pair AB, CD: the midpoints are $M = \frac{1}{2}(\mathbf{a} + \mathbf{b})$ and $N = \frac{1}{2}(\mathbf{c} + \mathbf{d})$, so

$$MN^2 = \tfrac{1}{4}\,|\mathbf{c} + \mathbf{d} - \mathbf{a} - \mathbf{b}|^2.$$

The other two midlines are found in the same way, so four times the sum of the squares of the three midlines is

$$S = |-\mathbf{a}-\mathbf{b}+\mathbf{c}+\mathbf{d}|^2 + |-\mathbf{a}+\mathbf{b}-\mathbf{c}+\mathbf{d}|^2 + |-\mathbf{a}+\mathbf{b}+\mathbf{c}-\mathbf{d}|^2.$$

Expand each term using $|\mathbf{u}|^2 = \mathbf{u}\cdot\mathbf{u}$. Each term contributes $|\mathbf{a}|^2+|\mathbf{b}|^2+|\mathbf{c}|^2+|\mathbf{d}|^2$, plus cross terms $\pm 2\,\mathbf{u}\cdot\mathbf{v}$, with a plus sign when $\mathbf{u}$ and $\mathbf{v}$ have the same sign in that term.

? **Ask yourself** How do the cross terms add up across the three terms?

Take any pair of vertices, say $\mathbf{a}$ and $\mathbf{b}$. They have the same sign in the first term and opposite signs in the other two, so their cross terms total $2\,\mathbf{a}\cdot\mathbf{b} - 2\,\mathbf{a}\cdot\mathbf{b} - 2\,\mathbf{a}\cdot\mathbf{b} = -2\,\mathbf{a}\cdot\mathbf{b}$. The same holds for every pair: each of the three terms splits the four vertices into two pairs (a vertex pair together with its opposite pair), and every vertex pair has matching signs in exactly one of the three terms. Hence

$$S = 3\big(|\mathbf{a}|^2+|\mathbf{b}|^2+|\mathbf{c}|^2+|\mathbf{d}|^2\big) - 2\big(\mathbf{a}\cdot\mathbf{b}+\mathbf{a}\cdot\mathbf{c}+\mathbf{a}\cdot\mathbf{d}+\mathbf{b}\cdot\mathbf{c}+\mathbf{b}\cdot\mathbf{d}+\mathbf{c}\cdot\mathbf{d}\big).$$

Now the edges. For example, $AB^2 = |\mathbf{b}-\mathbf{a}|^2 = |\mathbf{a}|^2+|\mathbf{b}|^2-2\,\mathbf{a}\cdot\mathbf{b}$. Summing over all six edges, each vertex lies on three edges, so each $|\mathbf{a}|^2$ appears three times, and each pair gives one $-2\,\mathbf{u}\cdot\mathbf{v}$:

$$AB^2+AC^2+AD^2+BC^2+BD^2+CD^2 = 3\sum|\mathbf{a}|^2 - 2\sum_{\text{pairs}}\mathbf{a}\cdot\mathbf{b} = S.$$

So the sum of the squares of the edges equals four times the sum of the squares of the three lines joining midpoints of opposite edges, as required.

Part (ii)

Take

$$\mathbf{a} = \begin{pmatrix}3\\2\\6\end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix}x\\y\\1\end{pmatrix}, \qquad \text{so} \qquad \mathbf{a}\cdot\mathbf{b} = 3x+2y+6,$$

with $|\mathbf{a}| = \sqrt{9+4+36} = 7$ and $|\mathbf{b}| = \sqrt{x^2+y^2+1}$. Since any real number is at most its modulus,

$$3x+2y+6 = \mathbf{a}\cdot\mathbf{b} \le |\mathbf{a}\cdot\mathbf{b}| \le |\mathbf{a}|\,|\mathbf{b}| = 7\sqrt{x^2+y^2+1},$$

which is the required inequality.

Equality. We need equality at both steps.

- $|\mathbf{a}\cdot\mathbf{b}| = |\mathbf{a}||\mathbf{b}|$ holds exactly when $\mathbf{a}$ and $\mathbf{b}$ are parallel, since $\mathbf{a}\cdot\mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta$ and we need $|\cos\theta| = 1$. As $\mathbf{a}\neq\mathbf{0}$, this means $\mathbf{b} = \lambda\mathbf{a}$ for some λ. The third components give $1 = 6\lambda$, so $\lambda = \frac{1}{6}$ and $x = \frac{1}{2}$, $y = \frac{1}{3}$.
- $\mathbf{a}\cdot\mathbf{b} = |\mathbf{a}\cdot\mathbf{b}|$ needs $\mathbf{a}\cdot\mathbf{b} \ge 0$. Here $\mathbf{a}\cdot\mathbf{b} = \lambda|\mathbf{a}|^2 = \frac{49}{6} > 0$, so this holds too.

Directly: at $x = \frac{1}{2}$, $y = \frac{1}{3}$ the left side is $\frac{3}{2} + \frac{2}{3} + 6 = \frac{49}{6}$ and the right side is $7\sqrt{\frac{1}{4} + \frac{1}{9} + 1} = 7 \cdot \frac{7}{6} = \frac{49}{6}$. So equality holds if and only if

$$x = \tfrac{1}{2}, \qquad y = \tfrac{1}{3}.$$

Checks

- For a regular tetrahedron with edge 1, the six squared edges sum to 6. Placing it at alternate corners of a cube of side $\frac{1}{\sqrt{2}}$, each midline joins centres of opposite faces, so it has length $\frac{1}{\sqrt{2}}$, and $4 \times 3 \times \frac{1}{2} = 6$.
- A numerical test on 10^4 random tetrahedra agreed with the identity to rounding error. For (ii), $7\sqrt{x^2 + y^2 + 1} - (3x + 2y + 6)$ was positive at 2×10^5 random points, and its smallest value on a fine grid was at $(0.5, 0.335)$, close to $(\frac{1}{2}, \frac{1}{3})$, where it is 0.

Common slips

- Forgetting the step $\mathbf{a} \cdot \mathbf{b} \leq |\mathbf{a} \cdot \mathbf{b}|$, or ignoring it in the equality case. Here it causes no trouble because $\lambda > 0$, but it has to be checked: with $\lambda < 0$ the two sides would differ in sign.
- Using $(3, 2, 6)$ with $(x, y, 1)$ but taking $|\mathbf{a}|$ to be anything other than 7. The question is built so that $3^2 + 2^2 + 6^2$ is a perfect square.

Beyond STEP

Part (i) is a three-dimensional relative of the parallelogram law $|\mathbf{u} + \mathbf{v}|^2 + |\mathbf{u} - \mathbf{v}|^2 = 2|\mathbf{u}|^2 + 2|\mathbf{v}|^2$. In fact the three midlines meet at the centroid and bisect each other. Part (ii) is the Cauchy–Schwarz inequality, which holds in any inner-product space, including spaces of functions, where it bounds integrals. Strang's *Introduction to Linear Algebra* is a good place to see both ideas in a wider setting.

WORKED SOLUTION

STEP III Specimen paper, Question 9

Pure • Groups of Matrices

STEP III Specimen paper, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q9

The key idea

Everything follows from one calculation: multiply two general matrices of this shape and see what comes out. The product turns out to have the same shape, with entries given by a simple formula, and that formula hands us closure, the identity, the inverse, the integer subgroup and the commuting condition almost for free. Write

$$M(x,y,z) = \begin{pmatrix} 1 & 0 & 0 \\ x & 1 & 0 \\ y & z & 1 \end{pmatrix}.$$

Solution

G is a group

The product formula. Multiplying row by column,

$$M(a,b,c)\,M(x,y,z) = \begin{pmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 \\ x & 1 & 0 \\ y & z & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ a+x & 1 & 0 \\ b+cx+y & c+z & 1 \end{pmatrix},$$

that is,

$$M(a,b,c)\,M(x,y,z) = M(a+x,\ b+cx+y,\ c+z). \qquad (*)$$

Closure. The entries $a+x$, $b+cx+y$ and $c+z$ are real, so the product lies in G.

Associativity. We are allowed to assume this for matrix multiplication.

Identity. The identity matrix is $I = M(0,0,0)$, which is in G, and $IM = MI = M$ for every matrix M.

Inverses. Given $M(a,b,c)$, we look for $M(x,y,z)$ with product $M(0,0,0)$. By $(*)$ we need $a+x=0$, $c+z=0$ and $b+cx+y=0$, so $x=-a$, $z=-c$ and $y=-b-cx=ac-b$. Check the other order too, because matrix multiplication isn't commutative:

$$M(-a,\ ac-b,\ -c)\,M(a,b,c) = M\big(0,\ (ac-b)+(-c)a+b,\ 0\big) = M(0,0,0).$$

So $M(a,b,c)^{-1} = M(-a,\ ac-b,\ -c)$, which is in G.

All four group axioms hold, so G is a group under matrix multiplication.

The integer matrices

Let H be the set of $M(x,y,z)$ with $x,y,z \in \mathbb{Z}$. It is a subset of G and contains $I = M(0,0,0)$. If a,b,c,x,y,z are integers then so are $a+x$, $b+cx+y$ and $c+z$, so by $(*)$ H is closed under multiplication. The inverse $M(-a,\ ac-b,\ -c)$ also has integer entries, so H is closed under inverses. Associativity is inherited from G. So **yes**, H is a subgroup of G.

? **Ask yourself** Why isn't closure under multiplication enough on its own?

For an infinite subset it can fail: the matrices $M(x,y,z)$ with x,y,z *non-negative* integers are closed under multiplication, but $M(1,0,0)$ has inverse $M(-1,0,0)$, which is not in that set. Here the inverse formula keeps us inside the integers, and that is what makes H a subgroup.

An element commuting with everything

Take $A = M(a,b,c)$ and a general $B = M(x,y,z)$. By $(*)$, in both orders,

$$AB = M(a+x,\ b+cx+y,\ c+z), \qquad BA = M(x+a,\ y+za+b,\ z+c).$$

The first and third entries always agree, so

$$AB = BA \iff cx = az.$$

(In particular G is not abelian: $M(1,0,0)$ and $M(0,0,1)$ don't commute.)

For A to commute with *every* B we need $cx = az$ for all real x and z. Choosing $x=1$, $z=0$ gives $c=0$; choosing $x=0$, $z=1$ gives $a=0$. Conversely, if $a=c=0$ then $cx = 0 = az$ for all x,z. So the elements that commute with all of G are exactly

$$A = M(0,b,0) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ b & 0 & 1 \end{pmatrix}, \qquad b \in \mathbb{R}.$$

Yes, such an A exists: any $b \neq 0$ gives $A \neq I$, for example $A = M(0,1,0)$. By $(*)$, $M(0,1,0)\,M(x,y,z) = M(x,\ 1+y,\ z) = M(x,y,z)\,M(0,1,0)$ for every $B = M(x,y,z)$ in G.

Checks

- Each matrix in G is lower triangular with ones on the diagonal, so it has determinant 1 and is certainly invertible: our inverse must exist, and it is again of the same shape.
- Numerically: with $A = M(2,3,5)$ and $B = M(1,0,0)$, $(*)$ gives $AB = M(3,8,5)$ but $BA = M(3,3,5)$, so A does not commute with B, as $cx = 5 \neq 0 = az$ predicts.

We also checked the product and inverse formulas, and the commuting condition, exactly with fractions for thousands of random matrices.

Common slips

- Checking the inverse on one side only. In a non-abelian setting it is worth confirming both products (or noting that for square matrices a one-sided inverse is automatically two-sided).
- Answering the final part with "G is not abelian, so no". A non-abelian group can still have non-identity elements commuting with everything, and this one does.
- Forgetting the converse: after showing $a = c = 0$ is necessary, we must also check that such A really does commute with every B.

Beyond STEP

This is the *Heisenberg group*, and the elements commuting with everything form its *centre* $Z(G)$, here the matrices $M(0, b, 0)$, a copy of the real numbers under addition. The integer version H is the discrete Heisenberg group, a favourite example in group theory and geometry, and the Heisenberg group itself appears in quantum mechanics through the commutation relations of position and momentum. For groups, subgroups and centres, see Allenby, *Rings, Fields and Groups*.

WORKED SOLUTION

STEP III Specimen paper, Question 10

Pure • Modular Arithmetic

STEP III Specimen paper, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q10

The key idea

Both parts are about remainders, so work modulo a well-chosen number. In (i), squares can only take a few values modulo 8, and three of those values can never add up to 7. In (ii), rewrite the powers so that both terms involve the same base modulo 17; the two coefficients then add up to a multiple of 17.

Solution

Part (i)

Odd squares. Write an odd number as $2k+1$, where k is an integer. Then

$$(2k+1)^2 = 4k^2 + 4k + 1 = 4k(k+1) + 1.$$

Of the two consecutive integers k and $k+1$, one is even, so $k(k+1)$ is even and $4k(k+1)$ is a multiple of 8. Hence every odd square leaves remainder 1 on division by 8.

Even squares. Write an even number as $2k$. Then $(2k)^2 = 4k^2$.

- If k is even, $k = 2j$, then $4k^2 = 16j^2$, a multiple of 8: remainder 0.
- If k is odd, then k^2 is odd, say $k^2 = 2j+1$, so $4k^2 = 8j+4$: remainder 4.

So every even square leaves remainder 0 or 4 on division by 8.

The deduction. Every square is congruent to 0, 1 or 4 modulo 8 (and this includes $0^2 = 0$, so we are allowing some of the three squares to be zero). Suppose $N = a^2 + b^2 + c^2$. Then N mod 8 is the remainder of a sum of three numbers each chosen from $\{0, 1, 4\}$. Sorting by how many of them equal 4:

number of 4s	possible sums of the rest, and the total mod 8
0	$0, 1, 2, 3$
1	$4 + \{0, 1, 2\} = 4, 5, 6$
2	$8 + \{0, 1\} \equiv 0, 1$
3	$12 \equiv 4$

The possible remainders are $0, 1, 2, 3, 4, 5, 6$, and 7 never appears. A number of the form $8n + 7$ leaves remainder 7 on division by 8, so it cannot be a sum of three squares.

Ask yourself Why can't we just check that 7 itself isn't a sum of three squares?

Because the claim is about infinitely many numbers $7, 15, 23, \ldots$, and checking them one by one never finishes. Working modulo 8 handles them all at once: whatever a, b, c are, only their remainders matter, and the table covers every case.

Part (ii)

Split off the powers so that each term is a constant times an nth power:

$$2^{3n+1} = 2 \cdot 8^n, \qquad 3\left(5^{2n+1}\right) = 15 \cdot 25^n.$$

Now $25 = 17 + 8$, so $25 \equiv 8 \pmod{17}$, and therefore $25^n \equiv 8^n \pmod{17}$ for every $n \geq 0$ (congruences can be multiplied). Hence

$$\begin{aligned} 2^{3n+1} + 3\left(5^{2n+1}\right) &= 2 \cdot 8^n + 15 \cdot 25^n \\ &\equiv 2 \cdot 8^n + 15 \cdot 8^n \\ &= 17 \cdot 8^n \equiv 0 \pmod{17}. \end{aligned}$$

So 17 divides $2^{3n+1} + 3(5^{2n+1})$ for all integers $n \geq 0$, as required.

Ask yourself What if I haven't met congruences?

The same idea works by induction. Let $f(n) = 2 \cdot 8^n + 15 \cdot 25^n$. Then $f(0) = 17$, and

$$f(n+1) - 8f(n) = 15 \cdot 25^{n+1} - 15 \cdot 8 \cdot 25^n = 15 \cdot 25^n(25 - 8) = 17 \cdot 15 \cdot 25^n,$$

so $f(n+1) = 8f(n) + 17 \cdot 15 \cdot 25^n$. If $17 \mid f(n)$ then $17 \mid f(n+1)$, and the result follows by induction.

Checks

Try small cases. $7 = 4 + 1 + 1 + 1$ needs four squares, and $15 = 9 + 4 + 1 + 1$ also needs four. On the other hand $6 = 4 + 1 + 1$ and $14 = 9 + 4 + 1$ are fine. A computer search confirms that no number $8n + 7$ up to 5000 is a sum of three squares. For (ii): $n = 0$ gives $2 + 15 = 17$, and $n = 1$ gives $16 + 375 = 391 = 17 \times 23$. The expression was checked to be divisible by 17 for every n from 0 to 299.

Common slips

- Forgetting that 0 is a square, so a "sum of three squares" may really use only one or two non-zero squares. The argument above includes this case automatically, since 0 is one of the residues.
- Saying odd squares are 1 mod 8 because "$4k^2 + 4k$ is divisible by 4": that only gives 1 mod 4. You need the factor of 2 from $k(k+1)$.

Beyond STEP

Part (i) is the easy half of Legendre's three-square theorem: a positive integer is a sum of three squares *exactly* when it is not of the form $4^a(8n+7)$. The converse direction is much harder, and needs the theory of quadratic forms. By contrast Lagrange proved that *every* positive integer is a sum of four squares. Part (ii) is the everyday arithmetic of $\mathbb{Z}/17\mathbb{Z}$, which first-year courses develop into the study of groups and rings. Davenport's *The Higher Arithmetic* covers congruences and sums of squares beautifully.

WORKED SOLUTION

STEP III Specimen paper, Question 11

Mechanics • First-Order Differential Equations

STEP III Specimen paper, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q11

This question has a full guided treatment in *Beyond STEP*, Section 11.13, *The Snowplough Problem*: https://beyondstep.keithmcnulty.org/ch11/snowplough.html

The key idea

The speed of each plough is given in terms of the snow depth, and the depth depends on *when* the plough reaches a point. So it is natural to treat time as a function of position and write down $\mathrm{d}t/\mathrm{d}x$, the reciprocal of the speed. For the first plough the depth is just the original depth plus what has fallen; for the second, the key observation is that the snow in front of it has only been falling since the first plough cleared that spot.

Solution

Model each plough as a point moving along the road, and measure distance from the common starting point. Throughout, depths are in metres, times in seconds and distances in metres.

Part (i)

At time t, the snow that the first plough meets has been falling undisturbed since $t = 0$, so its depth is

$$z = h + kt.$$

The speed is therefore

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{k}{\alpha(h+kt)}, \qquad \text{so} \qquad \frac{\mathrm{d}t}{\mathrm{d}x} = \frac{\alpha(h+kt)}{k} = \alpha t + \frac{\alpha h}{k}.$$

(Inverting is fine because the speed is positive, so x is a strictly increasing function of t.) This is linear in t, and it is neatest to write it as

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(t + \frac{h}{k}\right) = \alpha\left(t + \frac{h}{k}\right).$$

Hence $t + h/k = Ce^{\alpha x}$ for a constant C. At $x = 0$ we have $t = 0$, so $C = h/k$, and the time taken to travel x metres is

$$t_1(x) = \left(e^{\alpha x} - 1\right)\frac{h}{k} \text{ seconds},$$

as required.

Part (ii)

Setting up the equation. Suppose that at time $t > T$ the second plough is at position y. The first plough passed this point at time $t_1(y)$ from part (i) and cleared all the snow there.

Ask yourself What is the depth of snow in front of the second plough?

Not $h + kt$: the first plough left bare road behind it. At position y the snow has been building up from zero since time $t_1(y)$, at rate k, so the depth is

$$z = k\big(t - t_1(y)\big).$$

(This needs $t > t_1(y)$, i.e. the second plough is still behind the first, which is true until they meet.) The speed of the second plough is therefore

$$\frac{dy}{dt} = \frac{k}{\alpha k\big(t - t_1(y)\big)} \qquad \Longrightarrow \qquad \frac{1}{\alpha}\frac{dt}{dy} = t - \left(e^{\alpha y} - 1\right)\frac{h}{k},$$

which is the given equation. The second plough starts from the origin at time T, so the initial condition is $t = T$ when $y = 0$.

Verifying the solution. Write the proposed solution as

$$t = t_1(y) + g(y), \qquad g(y) = \left(T - \frac{\alpha h y}{k}\right)e^{\alpha y}.$$

Then $t - t_1(y) = g(y)$, so we need $\frac{1}{\alpha}\,dt/dy = g(y)$. Differentiating, with the product rule for g,

$$\begin{aligned}\frac{dt}{dy} &= \frac{\alpha h}{k}e^{\alpha y} - \frac{\alpha h}{k}e^{\alpha y} + \alpha\left(T - \frac{\alpha h y}{k}\right)e^{\alpha y} \\ &= \alpha\, g(y),\end{aligned}$$

where the first term is $t_1'(y)$ and the next two are $g'(y)$. So the differential equation holds. Also, at $y = 0$, $t = 0 + T \cdot 1 = T$, so the initial condition holds too.

Why is this *the* required solution? The equation is linear and first order: if t and $\tilde{t}$ both satisfy it, their difference u satisfies $du/dy = \alpha u$, so $u = u(0)e^{\alpha y}$. Both satisfy $t(0) = T$, so $u(0) = 0$ and $u \equiv 0$. Hence there is exactly one solution through $(y, t) = (0, T)$, and we have found it.

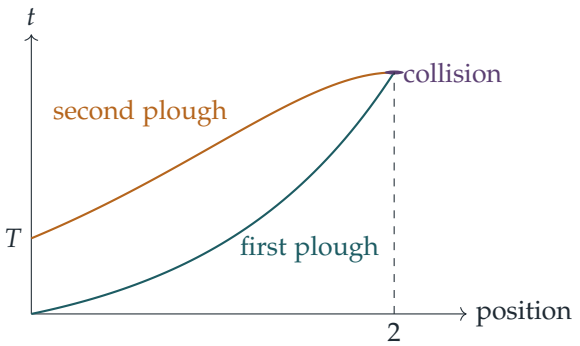
(If you prefer to derive it: multiplying by the integrating factor $e^{-\alpha y}$ gives $\frac{d}{dy}\left(te^{-\alpha y}\right) = -\frac{\alpha h}{k}\left(1 - e^{-\alpha y}\right)$, and integrating from 0 to y leads to the same formula.)

The collision. The ploughs collide at a point y if they reach it at the same time, that is, if $t = t_1(y)$, which means $g(y) = 0$. Since $e^{\alpha y} > 0$, this happens exactly when

$$T - \frac{\alpha h y}{k} = 0 \qquad \Longleftrightarrow \qquad y = \frac{kT}{\alpha h}.$$

For $0 \leq y < kT/(\alpha h)$ we have $g(y) > 0$ (taking $\alpha, h, k > 0$, as the speeds must be positive), so the second plough reaches each such point strictly after the first and they do not meet earlier. Hence the snowploughs collide when they have moved $kT/(\alpha h)$ metres. ■

The picture below shows the two paths in the (y, t) plane for $h = k = \alpha = 1$ and $T = 2$; the time gap $g(y)$ between the curves shrinks to zero at $y = 2$.



Checks

- Units: α has units m^{-1} (since $k/(\alpha z)$ is a speed), so αx is dimensionless, h/k is a time, and $kT/(\alpha h)$ is a length.
- If $T = 0$ the ploughs start together and the formula gives collision at distance 0; a bigger head start T pushes the collision further down the road, as it should.
- Numerically, for a range of random values of h, k, α and T, stepping each plough's motion forward in time (for the second, $\mathrm{d}y/\mathrm{d}t = 1/\big(\alpha(t - t_1(y))\big)$ from $y = 0$ at $t = T$) agrees with the formulae to better than 10^{-8}. With $h = k = \alpha = 1$ and $T = 2$, as in the picture, both ploughs reach $y = 2$ at time $e^2 - 1 \approx 6.39\,\mathrm{s}$.

Common slips

- Using depth $h + kt$ for the second plough too. It is ploughing a road that was cleared at time $t_1(y)$, so the depth is $k(t - t_1(y))$.
- Forgetting the initial condition $t = T$ at $y = 0$ when checking the solution; a solution of the differential equation alone is not enough.
- Writing $\mathrm{d}y/\mathrm{d}t$ instead of $\mathrm{d}t/\mathrm{d}y$: with time as the dependent variable the equations become linear and easy.

Beyond STEP

This is the classic "snowplough problem" family, and the trick of swapping the roles of dependent and independent variable to make an equation linear is used constantly in first-year differential equations. The uniqueness argument above is a baby case of the existence and uniqueness theorem for initial value problems. Notice too that the second plough's speed blows up as the snow in front of it thins to nothing, which is a sign of the model's limits rather than of reality. Boyce and DiPrima, *Elementary Differential Equations*, covers linear first-order equations, integrating factors and uniqueness.

WORKED SOLUTION

STEP III Specimen paper, Question 12

Mechanics • Equilibrium of a Rod

STEP III Specimen paper, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

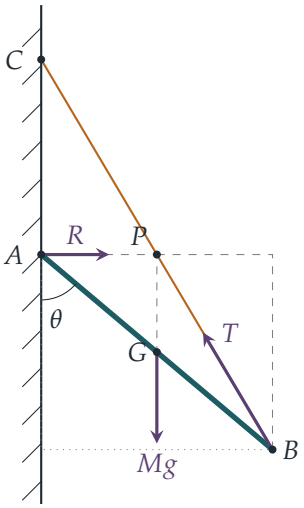
step.maths.org/questions/spec-s3-q12

The key idea

The rod is acted on by exactly three forces: its weight, the reaction from the smooth wall and the tension in the string. Three non-parallel forces holding a body in equilibrium must pass through a single point, and here that point is easy to locate. It pins down the geometry completely, and a single resolution then gives the tension.

Solution

Model. The rod is uniform, so its weight Mg acts at its midpoint G. The wall is smooth, so the reaction R at A is horizontal (perpendicular to the wall). The string is light and inextensible, so it pulls on B with a tension T directed along BC.



The inclination of the rod

The reaction R acts along the horizontal line through A, and the weight acts along the vertical line through G. These meet at the point P level with A and directly above (or below) G. Since the three forces are in equilibrium, the moments about P must balance; R and Mg have no moment about P, so T has none either, and the line of the string must pass through P.

Ask yourself Does the rod go up or down from A?

Measure horizontal distances from the wall. B is at distance $L\sin\theta$ from the wall, where θ is the angle between the rod and the vertical, and G (hence P) is at half that distance, $\frac{1}{2}L\sin\theta$. The string runs in a straight line from B to C, which is on the wall, so a point halfway across horizontally is halfway along the string: P is the *midpoint* of BC. So the height of P is the average of the heights of B and C. But P is level with A and C is above A, so B must be *below* A, by exactly the distance AC.

So the rod hangs downwards from A. Let θ be its angle with the downward vertical. Then B is a depth $L\cos\theta$ below A, and so $AC = L\cos\theta$. Relative to B, the point C is a horizontal distance $L\sin\theta$ away and a height $L\cos\theta + L\cos\theta = 2L\cos\theta$ above. Since the string has length αL, Pythagoras gives

$$\alpha^2 L^2 = L^2\sin^2\theta + 4L^2\cos^2\theta = L^2\left(1 + 3\cos^2\theta\right),$$

so

$$\cos^2\theta = \frac{\alpha^2 - 1}{3}, \qquad \text{i.e.} \qquad \theta = \cos^{-1}\sqrt{\frac{\alpha^2-1}{3}},$$

with the rod pointing *downwards* from A at this angle to the vertical.

The possible values of α

We need C strictly above A, so $\cos\theta > 0$. (If $\cos\theta = 0$ then C coincides with A and the rod is horizontal; then A, B, C are collinear.) We also need A, B, C not collinear, so the rod is not vertical: $\sin\theta \neq 0$, i.e. $\cos^2\theta < 1$. Hence

$$0 < \frac{\alpha^2 - 1}{3} < 1 \iff 1 < \alpha^2 < 4 \iff 1 < \alpha < 2,$$

using $\alpha > 0$. Conversely, for any α in this range the angle above exists and the construction gives a configuration in which the three forces are concurrent, so moments balance, and the resolutions below give a positive T and R. The set of possible values is $1 < \alpha < 2$.

The tension

Resolve vertically. The string makes an angle with the vertical whose cosine is (vertical rise)/(length) $= 2L\cos\theta/(\alpha L)$, and R is horizontal, so

$$T \cdot \frac{2\cos\theta}{\alpha} = Mg \quad \Longrightarrow \quad T = \frac{Mg\alpha}{2\cos\theta} = \frac{Mg\alpha}{2}\left(\frac{3}{\alpha^2-1}\right)^{\frac{1}{2}},$$

substituting $\cos\theta = \sqrt{(\alpha^2-1)/3}$ (the positive root, since $\cos\theta > 0$). This is the required result.

For completeness, resolving horizontally gives $R = T\cdot\dfrac{L\sin\theta}{\alpha L} = \tfrac{1}{2}Mg\tan\theta > 0$, so the wall does push outwards on the rod, as a wall must.

Checks

- As $\alpha \to 2^-$, $\cos\theta \to 1$: the rod hangs vertically below A with the string doubled back along it, and $T \to Mg$, since the string then carries the whole weight. As $\alpha \to 1^+$, $\theta \to 90^\circ$ and $T \to \infty$: a nearly horizontal rod needs a huge tension to hold it up.
- With $\alpha = 1.5$, we get $\cos\theta \approx 0.6455$ ($\theta \approx 49.8^\circ$) and $T \approx 1.162\,Mg$. We checked numerically, for thousands of random α in $(1,2)$, that the moment about A vanishes at exactly one angle, which matches the formula, along with the tension; and that no equilibrium exists for α outside $(1,2)$.

Common slips

- Drawing the rod sloping *upwards* from A, as in a ladder problem. The concurrency argument forces B below A, and with the wrong picture the equations have no consistent solution.
- Forgetting the reaction at a smooth wall is horizontal, and so introducing an unnecessary friction or vertical component.
- Including $\alpha = 1$ or $\alpha = 2$: these give collinear (degenerate) configurations, which the question excludes.

Beyond STEP

The three-force principle is a special case of a general fact in statics: a system of forces is in equilibrium exactly when its resultant force and its resultant moment about any one point both vanish. Choosing the point cleverly, as we did with P, is often the quickest route. The same idea underlies the analysis of trusses, cables and frameworks in engineering mechanics. For a thorough treatment of forces and torques, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP III Specimen paper, Question 13

Mechanics • Rolling Hoop

STEP III Specimen paper, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

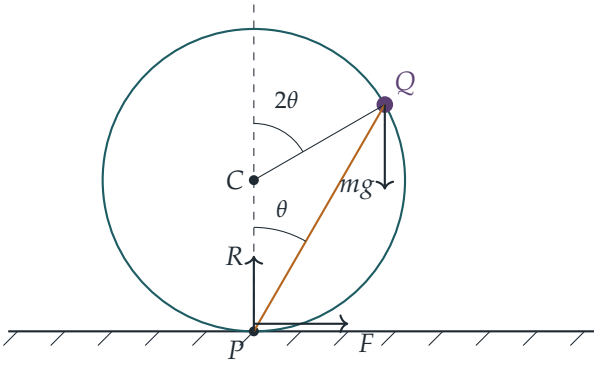
step.maths.org/questions/spec-s3-q13

The key idea

Because the hoop is light, the only kinetic energy in the system is the particle's, and because the hoop rolls, friction does no work. So energy conservation gives the speed at once. Rolling also fixes the geometry completely: once the hoop has turned through 2θ, its centre has moved a distance $2a\theta$, which gives x and y in terms of θ alone. Finally, a light hoop can carry no net force, so the table's reaction is simply what the particle's acceleration demands, and it cannot be negative.

Solution

Model the hoop as a light rigid circle rolling without slipping, and the particle as a point mass fixed to the rim. Let P be the point of contact with the table, C the centre and Q the particle. Take the hoop to roll in the positive x direction, so that θ increases from 0.



Here R and F are the normal reaction and friction of the table on the hoop at P, and the chord PQ makes angle θ with the vertical (half the angle 2θ at the centre).

The speed

Both R and F act at P, which is instantaneously at rest because the hoop rolls, so neither does any work. The hoop is light, so the only kinetic energy is the particle's and the only potential energy change is the particle's. The particle starts at height $2a$, essentially at rest, and when the radius CQ makes angle 2θ with the upward vertical its height is $a + a\cos 2\theta = 2a\cos^2\theta$. Energy conservation gives

$$\tfrac{1}{2}mv^2 = mg\big(2a - 2a\cos^2\theta\big) = 2mga\sin^2\theta,$$

so $v^2 = 4ga\sin^2\theta$ and, since $0 \le \theta < \frac{\pi}{2}$ gives $\sin\theta \ge 0$,

$$v = 2(ga)^{1/2}\sin\theta,$$

as required.

The position and $\dot\theta$

The hoop has turned through angle 2θ, and rolling without slipping means the centre has moved forward by the arc length $a \cdot 2\theta$. Relative to C, the particle is $a\sin 2\theta$ ahead and $a\cos 2\theta$ above. So

$$x = 2a\theta + a\sin 2\theta, \qquad y = a + a\cos 2\theta = 2a\cos^2\theta.$$

Differentiating with respect to time,

$$\dot x = 2a\dot\theta\,(1 + \cos 2\theta) = 4a\dot\theta\cos^2\theta,$$
$$\dot y = -2a\dot\theta\sin 2\theta = -4a\dot\theta\sin\theta\cos\theta,$$

and therefore

$$v^2 = \dot x^2 + \dot y^2 = 16a^2\dot\theta^2\cos^2\theta\big(\cos^2\theta + \sin^2\theta\big) = 16a^2\dot\theta^2\cos^2\theta.$$

? **Ask yourself** Is there a quicker way to see that $v = 4a\dot\theta\cos\theta$?

Yes: P is the instantaneous centre of rotation, and the hoop turns at rate $2\dot\theta$. The chord PQ subtends the angle $\pi - 2\theta$ at C, so $PQ = 2a\cos\theta$, and $v = PQ \cdot 2\dot\theta = 4a\dot\theta\cos\theta$, matching the calculation.

With $\dot\theta > 0$ and $\cos\theta > 0$, comparing with the speed found from energy,

$$4a\dot\theta\cos\theta = 2(ga)^{1/2}\sin\theta \quad\Longrightarrow\quad \dot\theta = \tfrac{1}{2}(g/a)^{1/2}\tan\theta,$$

which is the first required result. (The left-hand side must be the rate $\dot\theta$, not θ: the right-hand side has units of s^{-1}.)

The vertical acceleration

Substituting $\dot\theta$ into $\dot y$,

$$\dot y = -4a\sin\theta\cos\theta\cdot\tfrac{1}{2}(g/a)^{1/2}\frac{\sin\theta}{\cos\theta} = -2(ga)^{1/2}\sin^2\theta.$$

Differentiating again and using $\dot\theta$ once more,

$$\ddot y = -4(ga)^{1/2}\sin\theta\cos\theta\,\dot\theta = -4(ga)^{1/2}\sin\theta\cos\theta\cdot\tfrac12(g/a)^{1/2}\frac{\sin\theta}{\cos\theta} = -2g\sin^2\theta,$$

as required. In the same way $\dot x = 4a\cos^2\theta\,\dot\theta = (ga)^{1/2}\sin 2\theta$, and

$$\ddot x = 2(ga)^{1/2}\cos 2\theta\,\dot\theta = g\cos 2\theta\tan\theta.$$

What stops the rolling

The hoop is light, so the total force on the hoop-and-particle system equals m times the particle's acceleration. The external forces are mg down, R up and F horizontally. Resolving vertically,

$$R - mg = m\ddot y \quad\Longrightarrow\quad R = mg\left(1-2\sin^2\theta\right) = mg\cos 2\theta,$$

and horizontally $F = m\ddot x = mg\cos 2\theta\tan\theta$, so that $F/R = \tan\theta$ while $R > 0$.

Ask yourself What goes wrong when θ passes $\pi/4$?

The table can push but not pull, so $R \geq 0$. But $R = mg\cos 2\theta$ falls to zero at $\theta = \pi/4$, when the particle is level with the centre, and rolling beyond that would need $R < 0$. Equivalently, continued rolling would need $\ddot y = -2g\sin^2\theta < -g$: the particle would have to accelerate downwards faster than free fall, which gravity alone cannot provide. So at $\theta = \pi/4$ the contact force vanishes and the hoop leaves the table. The particle then moves as a projectile, starting with velocity components $(ga)^{1/2}$ forwards and $(ga)^{1/2}$ downwards, and the light hoop is simply carried along with it, its lowest point rising off the table.

Friction may intervene earlier. Rolling needs $|F| \leq \mu R$, that is $\tan\theta \leq \mu$, and $\tan\theta$ increases to 1 as $\theta \to \pi/4$. So if the coefficient of friction satisfies $\mu < 1$, the hoop starts to slip when $\tan\theta = \mu$, before reaching $\theta = \pi/4$. Only if $\mu \geq 1$ does it roll all the way to $\theta = \pi/4$ and then jump.

Checks

- At $\theta = 0$, $\dot\theta = 0$: the top position is an equilibrium, which is why the hoop has to be displaced slightly.
- At $\theta = \pi/4$: $v = (2ga)^{1/2}$, which is the speed from a fall of a, and indeed the particle has dropped from $2a$ to a.
- The computer integrated the equation of motion for $\phi = 2\theta$ independently (from the Lagrangian $ma^2(1+\cos\phi)\dot\phi^2 - mga(1+\cos\phi)$) and compared finite-difference velocities and accelerations with the formulae. The speed, $\dot\theta$, $\ddot y$, $R = mg\cos 2\theta$ and $F/R = \tan\theta$ all agreed to within 10^{-6}; for example at $\theta = \pi/4$ it gives $R = 0$ and $F/R = 1$.

Common slips

- Taking the centre to move a distance $a\theta$ rather than $2a\theta$: the hoop has turned through 2θ, not θ.
- Including friction in the energy equation. The contact point is at rest, so friction does no work while the hoop rolls.
- Saying only that the particle "falls off". The particle is attached; it is the hoop that leaves the table, because the reaction would have to become negative.

Beyond STEP

The contact point of a rolling body is its instantaneous centre of rotation, and rolling without slipping is a constraint that removes a degree of freedom without doing work. Such constraints are handled systematically by Lagrangian mechanics, where one writes the energy in a single coordinate such as θ and the equation of motion follows automatically; the check above did exactly this. Questions of when a constraint force vanishes (a bead leaving a sphere, a car leaving a hump) are typical of first-year dynamics. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rolling motion and constraint forces in detail.

WORKED SOLUTION

STEP III Specimen paper, Question 14

Mechanics • Rotational Collisions

STEP III Specimen paper, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q14

The key idea

The pivot exerts an unknown impulse on the rod, so linear momentum is *not* conserved, but that impulse has no moment about the pivot: angular momentum about the midpoint is conserved. Together with Newton's law of restitution (coefficient 1, since the collision is perfectly elastic) applied at the point of contact, that fixes everything. For the second part, find what the particle does next and follow both bodies: the far half of the rod comes round to the particle's line.

Solution

Model. The particle is a point mass; the table is smooth, so there's no friction on either body; the rod is uniform and pivoted smoothly at its midpoint O. The moment of inertia of the rod about O is

$$I = \tfrac{1}{12}m(4a)^2 = \tfrac{4}{3}ma^2.$$

During the collision the impulses act along the particle's line of motion (the rod is smooth and the particle hits it at right angles), so the particle keeps moving along the same line. Let its velocity after the collision be v, measured in its original direction of motion.

The angular speed

Angular momentum about O. The only external impulse on the system (particle and rod) comes from the pivot at O, and it has zero moment about O. Before the collision the particle's angular momentum about O is mua and the rod is at rest, so

$$mua = mva + \tfrac{4}{3}ma^2\omega \qquad \Longrightarrow \qquad u = v + \tfrac{4}{3}a\omega. \tag{1}$$

Restitution. The point of the rod that is struck is at distance a from O, so just after the collision it moves at speed $a\omega$ in the particle's original direction. With coefficient of

restitution 1, the speed of separation equals the speed of approach:

$$a\omega - v = u. \tag{2}$$

Ask yourself Can we use Newton's law of restitution for a rotating body?

Yes: it applies to the velocities of the two points in contact, along the common normal. If you prefer, use energy instead: "perfectly elastic" means kinetic energy is conserved, $\frac{1}{2}mu^2 = \frac{1}{2}mv^2 + \frac{1}{2}\cdot\frac{4}{3}ma^2\omega^2$. Writing this as $m(u-v)(u+v) = \frac{4}{3}ma^2\omega^2$ and using (1), $u - v = \frac{4}{3}a\omega$, we get $u + v = a\omega$ (dividing by $\omega \neq 0$), which is exactly (2).

Adding (1) and (2) eliminates v:

$$2u = \tfrac{7}{3}a\omega \qquad \Longrightarrow \qquad a\omega = \frac{6u}{7},$$

as required. Then from (2),

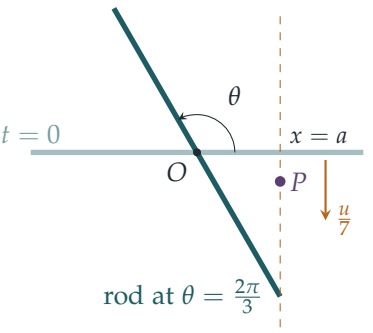
$$v = a\omega - u = -\frac{u}{7},$$

so the particle bounces back along its original line with speed $u/7$.

The second collision

Take axes on the table with origin O, the x-axis along the rod's initial position and the particle originally moving in the positive y-direction along the line $x = a$, so that it strikes the rod at $(a, 0)$ at time $t = 0$. Then for $t > 0$:

- the particle is at $\left(a,\ -\frac{u}{7}t\right)$, moving steadily in the negative y-direction;
- the struck point moved in the positive y-direction, so the rod turns anticlockwise, and at time t it makes an angle $\theta = \omega t$ with the x-axis. No further force acts on either body until they collide again, so both motions are uniform.



The rod has half-length $2a$, so it reaches the line $x = a$ only when the point of the rod on that line, at distance $a/|\cos\theta|$ from O, is within $2a$ of O; that is, when $|\cos\theta| \geq \frac{1}{2}$. Where it does, the rod crosses the line at height $y = a\tan\theta$.

- For $0 < \theta \le \frac{\pi}{3}$ the rod meets the line at $y = a\tan\theta > 0$, above the x-axis, while the particle is below it: no contact.
- For $\frac{\pi}{3} < \theta < \frac{2\pi}{3}$ the rod doesn't reach the line at all.
- For $\frac{2\pi}{3} \le \theta < \pi$ the *other* half of the rod crosses the line, at $y = a\tan\theta < 0$, on the same side as the particle.

Now compare positions. Since $t = \theta/\omega = \frac{7a\theta}{6u}$, the particle's y-coordinate at angle θ is $-\frac{u}{7}t = -\frac{a\theta}{6}$. Consider

$$f(\theta) = (\text{rod's } y \text{ on } x = a) - (\text{particle's } y) = a\tan\theta + \frac{a\theta}{6}, \qquad \tfrac{2\pi}{3} \le \theta < \pi.$$

- At $\theta = \frac{2\pi}{3}$ (the end of the rod reaches the line): $f = a\left(-\sqrt{3} + \frac{\pi}{9}\right) < 0$, since $\frac{\pi}{9} < 1 < \sqrt{3}$. The particle is at $y = -\frac{\pi}{9}a \approx -0.35a$, between the x-axis and the end of the rod at $y = -\sqrt{3}a$: the rod's end has come round *beyond* the particle.
- As $\theta \to \pi^-$: $\tan\theta \to 0$, so $f \to \frac{\pi}{6}a > 0$. The rod is sweeping back up to the x-axis, while the particle is at about $-0.52a$.

f is continuous on $[\frac{2\pi}{3}, \pi)$, so by the intermediate value theorem $f(\theta) = 0$ for some θ in between: at that instant the rod is at the particle's position, and the two collide again. (No earlier contact is possible, by the first two bullet points above.) This completes the proof.

Numerically, the collision happens where $\tan\theta = -\theta/6$, at $\theta \approx 2.716$ (about $155.6°$), at time $t \approx 3.17\,a/u$, with the particle at $y \approx -0.45a$, about $1.10a$ from O along the rod.

Checks

- Energy: with $a\omega = \frac{6}{7}u$ and $v = -\frac{1}{7}u$, the final kinetic energy is $\frac{1}{2}m\left(\frac{1}{49} + \frac{4}{3}\cdot\frac{36}{49}\right)u^2 = \frac{1}{2}m\cdot\frac{49}{49}u^2$, exactly the initial energy.
- Repeating the argument for a strike at distance d on a body of moment of inertia I gives $d\omega = 2u/(1 + I/md^2)$. As $I \to 0$ the struck point moves off at $2u$, as for an elastic collision with a very light body; as $I \to \infty$ the rod hardly turns and the particle bounces back with speed u. Here $I/ma^2 = \frac{4}{3}$ gives $\frac{6}{7}u$.
- We checked with exact fractions that $a\omega = \frac{6}{7}u$, $v = -\frac{1}{7}u$ conserve both angular momentum and energy, then stepped both motions forward in time: the first contact after the collision occurs at $\theta \approx 2.7165$, agreeing with a bisection solution of $\tan\theta + \theta/6 = 0$.

Common slips

- Conserving *linear* momentum. The pivot supplies an impulse, so only angular momentum about the pivot is conserved.

- Using $I = \frac{1}{3}m\ell^2$ (the value about an end) or forgetting that the rod's length is $4a$, not $2a$.
- Looking for the second collision on the half of the rod that was struck. That half moves away from the particle's line; it's the other half, coming round, that meets it.

Beyond STEP

Pairing angular momentum about a fixed point with an energy or restitution condition is the standard way to handle impulsive motion of rigid bodies, and leads on to the idea of the *centre of percussion*: the point at which a pivoted body can be struck with no impulse at the pivot. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body collisions and angular momentum carefully and has many problems of exactly this kind.

WORKED SOLUTION

STEP III Specimen paper, Question 15

Probability and Statistics • Random Network Connectivity

STEP III Specimen paper, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q15

The key idea

Forget the geometry: all that matters is which of the six roads are open. Every village is reachable from every other exactly when the open roads form a *connected* network on the four villages. So we sort the possible sets of open roads by how many roads they contain, count how many of each size are connected, and weight each by its probability. With only four villages, the counting is short and clean.

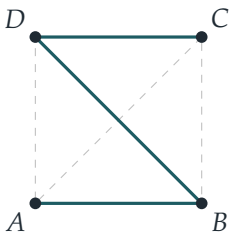
Solution

Write $q = 1 - p$ for the probability that a given road is open. Let K be the event that every village can be reached from every other after the snowfall.

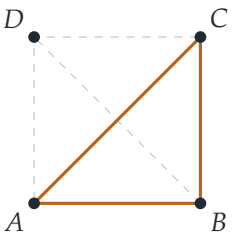
Since the six roads behave independently, any particular set of k open roads (with the other $6 - k$ blocked) has probability $q^k p^{6-k}$. So

$$\mathrm{P}(K) = \sum_{k=0}^{6} N_k \, q^k p^{6-k},$$

where N_k is the number of sets of k roads that connect all four villages.



three roads: a connecting tree



three roads: a triangle, D cut off

Counting the connected sets of roads

At most two open roads. Each open road joins two villages, so starting from one village, each road can bring in at most one new village. Reaching all three others needs at least three roads. So $N_0 = N_1 = N_2 = 0$.

Three open roads. There are $\binom{6}{3} = 20$ sets of three roads. Suppose such a set fails to connect the villages. If some village is cut off, all three roads lie among the other three villages. But only three roads join those three villages, so the set is exactly a triangle. If no village is cut off, the villages split into two pairs, and at most two roads (one inside each pair) can be open, a contradiction. So the only failures are the 4 triangles, one for each choice of the cut-off village, and

$$N_3 = 20 - 4 = 16.$$

? **Ask yourself** Why can't four open roads ever leave the network disconnected?

Four or more open roads. Use the same two ways of failing. A cut-off village would force all the open roads to lie among the other three villages, where there are only three roads. A split into two pairs allows at most two open roads. With four or more open roads, neither can happen, so every such set is connected:

$$N_4 = \binom{6}{4} = 15, \qquad N_5 = \binom{6}{5} = 6, \qquad N_6 = 1.$$

The probability

Putting the counts together,

$$\mathrm{P}(K) = q^6 + 6q^5p + 15q^4p^2 + 16q^3p^3, \qquad q = 1 - p.$$

Expanding in powers of p gives the equivalent form

$$\mathrm{P}(K) = 1 - 4p^3 - 3p^4 + 12p^5 - 6p^6.$$

The case $p = \frac{1}{2}$

Now $p = q = \frac{1}{2}$, so every one of the $2^6 = 64$ patterns of open and blocked roads is equally likely, with probability $\frac{1}{64}$. Hence

$$\mathrm{P}(K) = \frac{1 + 6 + 15 + 16}{64} = \frac{38}{64} = \frac{19}{32},$$

as required.

Checks

- *Extreme values.* At $p = 0$ every road is open and the formula gives 1. At $p = 1$ every road is blocked and it gives $1 - 4 - 3 + 12 - 6 = 0$.

- *The complement.* The network is disconnected exactly when some village is cut off, or the villages split into two pairs with no road between the pairs. A village is cut off with probability p^3. By inclusion–exclusion over the four villages (two particular villages cut off needs five blocked roads, and three cut off means all four are) this gives $4p^3 - 6p^5 + 3p^6$. Each of the 3 splits into pairs, with both inside roads open, adds p^4q^2. One minus the total agrees with the formula above: at $p = \frac{1}{2}$ the disconnected probability is $\frac{32-12+3+3}{64} = \frac{13}{32}$.
- *Brute force.* Listing all 64 sets of roads by computer confirms the counts $16, 15, 6, 1$. A simulation with $p = 0.3$ gave 0.893, against the exact value 0.8925.

Common slips

- Counting all 20 sets of three roads as connected, forgetting the four triangles.
- Assuming any set of four roads can still leave a village isolated. A village is cut off only if all three of its roads are blocked.
- Mixing up p (blocked) and $1 - p$ (open) when writing $q^k p^{6-k}$.

Beyond STEP

This is a tiny case of a *random graph*: each edge of a network survives independently, and we ask whether the result is connected. The polynomial in p is called the *reliability polynomial* of the network, and the number 16 is an instance of Cayley's formula: there are n^{n-2} trees joining n labelled points. For large networks, Erdős and Rényi found that connectivity switches on remarkably suddenly as the open-road probability passes a threshold. For more on random structures built from independent events, see Grimmett and Welsh, *Probability: An Introduction*.

WORKED SOLUTION

STEP III Specimen paper, Question 16

Probability and Statistics • Negative Binomial Distribution

STEP III Specimen paper, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q16

The key idea

Waiting for r sons is the same as waiting for one son, then starting again and waiting for the next, r times over. So C splits into a sum of r identical waiting times, and linearity of expectation does the rest. For part (ii), the trick is to imagine the births carrying on past the point where the president stops: whether he has stopped by a given time is then a question about a *fixed* number of children, and a symmetry between boys and girls settles it.

Solution

Throughout, each birth is a boy with probability $\frac{1}{2}$, independently of all other births.

Part (i)

For $i = 1, 2, \ldots, r$, let X_i be the number of children born after the $(i-1)$th son, up to and including the ith son. (For $i = 1$, count from the start.) Then

$$C = X_1 + X_2 + \cdots + X_r.$$

Each X_i counts births up to the first boy in a fresh run of independent fair births, so every X_i has the same distribution as X, where

$$\mathrm{P}(X = k) = \left(\tfrac{1}{2}\right)^{k-1} \cdot \tfrac{1}{2} = \left(\tfrac{1}{2}\right)^{k}, \qquad k = 1, 2, 3, \ldots,$$

since $X = k$ means $k - 1$ girls followed by a boy.

To find $\mathrm{E}(X)$, differentiate the geometric series $\sum_{k\geq 0} x^k = \dfrac{1}{1-x}$ for $|x| < 1$ to get $\sum_{k\geq 1} kx^{k-1} = \dfrac{1}{(1-x)^2}$. Multiplying by x and putting $x = \frac{1}{2}$,

$$\mathrm{E}(X) = \sum_{k=1}^{\infty} k\left(\tfrac{1}{2}\right)^{k} = \frac{\frac{1}{2}}{\left(1-\frac{1}{2}\right)^2} = 2.$$

This makes sense: if half of all babies are boys, you expect to wait two births for one.

By linearity of expectation (which holds whether or not the X_i are independent, though here they are),

$$\mathrm{E}(C) = \mathrm{E}(X_1) + \cdots + \mathrm{E}(X_r) = 2r.$$

Part (ii)

Imagine that the sexes of an endless sequence of babies are decided in advance, independently and fairly, and that the president has babies from this sequence until he reaches his rth son. This changes nothing about C, but it lets us talk about "the first $2r-1$ babies" even when he stops sooner.

Ask yourself How can the event $C < 2r$ be described using only the first $2r-1$ babies of the sequence?

Since C is a whole number, $C < 2r$ means $C \leq 2r-1$, that is, the rth son arrives at or before position $2r-1$. That happens exactly when there are at least r boys among the first $2r-1$ babies. So

$$\mathrm{P}(C < 2r) = \mathrm{P}(\text{at least } r \text{ boys among the first } 2r-1 \text{ babies}).$$

Now let B and G be the numbers of boys and girls among these $2r-1$ babies, so $B+G = 2r-1$. Because $2r-1$ is odd, B and G cannot be equal, and exactly one of them is at least r:

- if $B \geq r$ then $G = 2r-1-B \leq r-1$;
- if $B \leq r-1$ then $G \geq r$.

So the events $\{B \geq r\}$ and $\{G \geq r\}$ are mutually exclusive and between them cover every possibility, which gives

$$\mathrm{P}(B \geq r) + \mathrm{P}(G \geq r) = 1.$$

But boys and girls are equally likely at every birth, so swapping the labels "boy" and "girl" doesn't change any probability, and $\mathrm{P}(B \geq r) = \mathrm{P}(G \geq r)$. Hence each is $\frac{1}{2}$, and

$$\mathrm{P}(C < 2r) = \tfrac{1}{2},$$

as required.

So the expected family size is $2r$, yet the family is smaller than $2r$ exactly half the time. The family sizes are skewed: values just below $2r$ are common, and the occasional very long run of girls pulls the mean up.

Checks

- With $r = 1$: $\mathrm{E}(C) = 2$, and $C < 2$ means the first baby is a boy, which has probability $\frac{1}{2}$.

- With $r = 2$: $C < 4$ means $C = 2$ or 3, with probabilities $\frac{1}{4}$ and $2 \cdot \frac{1}{8} = \frac{1}{4}$, total $\frac{1}{2}$.
- Summing $c\binom{c-1}{r-1}2^{-c}$ exactly for $r = 1, \ldots, 8$ gives $\mathrm{E}(C) = 2r$ and $\mathrm{P}(C < 2r) = \frac{1}{2}$ every time; a simulation with $r = 3$ and 200 000 families gave a mean of 6.00 and $\mathrm{P}(C < 6) \approx 0.499$.

Common slips

- Arguing "half the children are boys, so $C = 2r$". That is true on average, but C itself is random; part (i) needs the decomposition into waiting times, not just a guess.
- Considering only the president's actual children in part (ii). When $C < 2r - 1$ there *are* no $2r - 1$ children; extending the sequence of births past the stopping point is what makes the counting argument valid.
- Forgetting that $2r - 1$ is odd. That is exactly what rules out a tie between boys and girls and makes the two events complementary.

Beyond STEP

The distribution of C is the *negative binomial* distribution: $\mathrm{P}(C = c) = \binom{c-1}{r-1}2^{-c}$ for $c \geq r$. Writing a random variable as a sum of simpler pieces and using linearity of expectation is one of the most powerful tools in first-year probability, and the trick in part (ii) of linking a "waiting time" to a "count in a fixed number of trials" is the same duality that connects the negative binomial and binomial distributions (and, in continuous time, the gamma and Poisson distributions). Blitzstein and Hwang, *Introduction to Probability*, treats both ideas with many examples.

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